

RECOVERY OF MATES ON MARTÍN NORMS AND THE RESONANCE OBSTRUCTION

ABSTRACT. For Martín’s renormings p of c_0 (canonical base, finitely or infinitely many blocks) we study the density of norm-attaining operators into ℓ_2^2 through the mate fibres $\mathcal{C}(f) = \{g : (f, g) \text{ is contractive}\}$. Density is equivalent to the statement that every pair $(f, \rho g)$ with $g \in \mathcal{C}(f)$ and $\rho < 1$ is a limit of norm-attaining pairs, and we reduce this to lower limits of the fibres along norm-attaining approximants; finitely many blocks suffice. We introduce finite certificates with coordinatewise radii, prove that the closed set they generate is transported along *every* sequence $f_n \rightarrow f$, and prove an averaging theorem over scales which covers weighted directions, locally split mates and cross mates carried by off-peak coordinates with non-summable gaps. At tame supports we identify the sharp second-order invariant, the mass-weighted coefficient Γ_w , and realize it along canonical truncations by means of transfer peaks. We then construct an admissible operator T and a non-attaining first row at which all intrinsically recoverable mates lie on one line while $\mathcal{C}(f)$ spans an infinite-dimensional space (the resonance obstruction); the canonical truncations do not recover these mates.

This is a defect of the intrinsic mechanisms only. Rebalancing through transfer peaks at first order makes *engineered* norm-attaining approximants work without any steering: at first rows with finite base support, every mate with one-sided (two-piece) linear decompositions of weighted coefficient at most one and non-negative d -mismatch is recovered, for arbitrary contact sets and any finite set of active blocks (infinite block sets are then handled through the reduction to finitely many blocks), and the case of negative mismatch is reduced to a scrambling condition. By Fenchel duality we compute the exact one-sided second-order coefficients at block-tame first rows, and deduce that every block-tame first row is recoverable, whatever its contact set; in particular every mate of the resonant example is recovered. Finally we design an admissible operator (private signatures, coarse-to-fine targets, a super-fast ladder of weights) for which every first row with finite base support and signature room is recoverable, and for which density is equivalent to a single approximation statement, Lemma Z. The density problem itself remains open; we formulate precisely what remains.

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1. SETTING AND STANDING FACTS

All spaces are real. We write $s(t) := \sqrt{1+t^2}$ throughout.

1.1. The base, the blocks and Martín's lemmas. We use the canonical base on $X = c_0$, as in the accompanying note [7]: H is a Hilbert space, $U : H \rightarrow c_0$ is compact with dense range, and

$$B_q = B_{c_0} + U(B_H), \quad q^*(a) = \|a\|_1 + \|U^*a\|_H \quad (a \in \ell_1 = X^*).$$

Since U has dense range, U^* is injective; it is compact, so $\|U^*e_j^*\|_H \rightarrow 0$. The bidual ball is $B_{q^{**}} = B_{\ell_\infty} + U(B_H)$ (the sum of a weak*-compact set and a norm compact set is weak* compact, contains B_q , and is contained in its weak* closure).

The norm q is a smoothing of the supremum norm in the sense of Debs, Godefroy and Saint-Raymond [1].

Martín's construction [6] rests on two lemmas, which we recall in the form stated there.

Remark 1.1 (Martín's Lemma A). Let Φ be a sequence of positive numbers with $\|\Phi\|_1 < 1$ and let $D_\Phi : \ell_2 \rightarrow \ell_1$ be the diagonal operator with entries $\Phi(n)$. The norm $|\cdot|_\Phi$ on ℓ_1 with unit ball $B_{\ell_1} + D_\Phi(B_{\ell_2})$ satisfies: (a) $|f|_\Phi^* = \|f\|_\infty + \|D_\Phi f\|_2$ for $f \in \ell_\infty$; (b) $|f|_\Phi^* \geq \|f\|_\infty \geq (1 - \|\Phi\|_1)|f|_\Phi^*$; (c) $|x|_\Phi \leq \|x\|_1 \leq (1 + \|\Phi\|_1)|x|_\Phi$; (d) the dual norm is strictly convex, so $|\cdot|_\Phi$ is smooth; (e) if $|f|_\Phi^* = 1$, $\langle f, x \rangle = |x|_\Phi$ and $|x(n)| > \Phi(n)|x|_\Phi$, then $f(n) = \text{sign}(x(n))\|f\|_\infty$.

Remark 1.2 (Martín's Lemma B and the choice of T). Lemma B of [6] reads: *Let Y be a separable operator range. Then there is a norm-one injective operator $T : \ell_1(\mathbb{N} \times \mathbb{N}) \rightarrow Y$ such that for every m the set $\{T(e_{n,m})/\|T(e_{n,m})\| : n \in \mathbb{N}\}$ is dense in S_Y .* Its proof takes a bijection $\sigma : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, a sequence (w'_n) in S_Y such that every point of S_Y is an accumulation point of (w'_n) , and repeats the proof of [4, Proposition 2.8] with the sequence $w_{\sigma(n,m)} = w'_n$. Two features matter below. First, the sequence (w'_n) is *arbitrary* subject to the accumulation property, so “Martín's space” is a family of spaces indexed by the admissible choices of T . Second, every block m follows the same sequence (w'_n) , so that near duplicates $T e_{n,m}/\|T e_{n,m}\| \approx T e_{n,m'}/\|T e_{n,m'}\|$ across blocks are built into the construction (up to the perturbations produced in [4]). No other property of T is used in [6], and no other property will be used here unless it is stated explicitly.

Definition 1.3 (Admissible operators). An operator $T : \ell_1(\mathbb{N} \times \mathbb{N}) \rightarrow X^* = \ell_1$ is *admissible* if

- (T-a) $q^*(T e_{n,m}) \leq 1$ for all (n, m) , with equality for some (n, m) ;
- (T-b) T is injective;
- (T-c) $Y := T(\ell_1(\mathbb{N} \times \mathbb{N}))$ satisfies $Y \cap c_{00} = \{0\}$;
- (T-d) for every m , the set $\{u_{n,m} : n \in \mathbb{N}\}$, where $u_{n,m} := T e_{n,m}/q^*(T e_{n,m})$, is norm dense in S_{q^*} .

Since $\text{NA}((X, q), \mathbb{R}) = c_{00}$ (Proposition 1.10(c) below), (T-c) is the condition $Y \cap \text{NA}(q) = \{0\}$ of [6]. As S_{q^*} has no isolated points, (T-d) implies that every tail $\{u_{n,m} : n \geq n_0\}$ is dense in S_{q^*} .

1.2. The norm p . Fix an admissible T . For $m \in \mathbb{N}$ put

$$\begin{aligned} \Phi_m(k) &:= 2^{-m-k} q^*(T e_{k,m}) \leq 2^{-m-k}, & \lambda_{k,m} &:= m \Phi_m(k), \\ (R_m x)(k) &:= \lambda_{k,m} u_{k,m}(x) & (x \in X). \end{aligned}$$

Let D_m be the diagonal operator with entries $\Phi_m(k)$, let $|\cdot|_m$ be the norm of Remark 1.1 for $\Phi = \Phi_m$ on $V_m = \ell_1$, and $N_m(w) := |w|_m^* = \|w\|_\infty + \|D_m w\|_2$. For a nonempty set $I \subseteq \mathbb{N}$ of *blocks* put

$$\begin{aligned} V &:= \left(\bigoplus_{m \in I} V_m \right)_{\ell_1}, & V^* &= \left(\bigoplus_{m \in I} (\ell_\infty, N_m) \right)_{\ell_\infty}, \\ Lx &:= (R_m x)_{m \in I}, & p(x) &:= q(x) + \|Lx\|_V. \end{aligned}$$

For $I = \mathbb{N}$ this is Martín's norm (with the canonical base); for $I = \{1, \dots, N\}$ we obtain the finite-block norms p_N of [8]. Since $|(R_m x)(k)| \leq m 2^{-m-k} q(x)$,

$\|R_m\| \leq m2^{-m}$; each R_m is a norm limit of finite-rank operators, hence compact, and L is compact. For $w \in V^*$,

$$(1) \quad L^*w = \sum_m R_m^* w_m = T \left(\sum_{m \in I} \sum_k m2^{-m-k} w_m(k) e_{k,m} \right) \in Y,$$

an absolutely convergent series; in particular L^* is injective on V^* . For $\theta \in X^{**} = \ell_\infty$, $(R_m^{**}\theta)(k) = \lambda_{k,m}\theta(u_{k,m})$, so $R_m^{**}(X^{**}) \subseteq \ell_1$.

Remark 1.4 (Finite and infinite block sets). By Lemma 1.5 below (Martín tails; cf. [8, Remark 6.9]), if $\text{NA}((c_0, p_N), \mathcal{F})$ is dense in $\mathcal{L}((c_0, p_N), \mathcal{F})$ for infinitely many N , then $\text{NA}((c_0, p), \mathcal{F})$ is dense for Martín's p ($I = \mathbb{N}$). Hence finite block sets suffice for the density problem. Sections 1–4 are valid for every nonempty I (certificates involve finitely many blocks), with two exceptions. Corollary 4.3 needs an extra hypothesis when I is infinite (Remark 4.4). Theorem 4.10 and Corollary 4.11 are stated and proved only for finite I : their proof truncates every block, and the admissible range of the block truncations shrinks with the block index, so for infinitely many blocks it is not uniform. By the first part of this remark, finite I is enough for the density problem, so both results are covered by the reduction to p_N ; we make no claim about locally split mates for $I = \mathbb{N}$. Sections 5–8 assume that I is finite.

Lemma 1.5 (Martín tails). *Fix an admissible T , let p be the norm with $I = \mathbb{N}$ and p_N the norm with $I = \{1, \dots, N\}$, and let $\mathcal{F}$ be a Banach space. If $\text{NA}((c_0, p_N), \mathcal{F})$ is dense in $\mathcal{L}((c_0, p_N), \mathcal{F})$ for infinitely many N , then $\text{NA}((c_0, p), \mathcal{F})$ is dense in $\mathcal{L}((c_0, p), \mathcal{F})$.*

Proof. Put $s_N(x) := \sum_{m > N} |R_m x|_m$, so that $p = p_N + s_N$. Since $|R_m x|_m \leq \|R_m x\|_1 \leq m2^{-m}q(x)$, we have $s_N \leq \eta_N q \leq \eta_N p_N$ with $\eta_N := \sum_{m > N} m2^{-m} = (N+2)2^{-N}$, hence $\|A\|_p \leq \|A\|_{p_N} \leq (1 + \eta_N)\|A\|_p$ for operators A . Let $S \in \text{NA}((c_0, p_N), \mathcal{F})$ with $\|S\|_{p_N} = 1 = \|Sx_0\|$, $p_N(x_0) = 1$. Let J be a norm-one functional on $(\bigoplus_{m > N} V_m)_{\ell_1}$ norming $(R_m x_0)_{m > N}$ ($J := 0$ if this vector is 0), and $\varphi(x) := J((R_m x)_{m > N})$; then $\varphi \in X^*$, $|\varphi| \leq s_N$ and $\varphi(x_0) = s_N(x_0)$. Put $S'x := Sx + \varphi(x)Sx_0$. Then $\|S'x\| \leq p_N(x) + s_N(x) = p(x)$ and $\|S'x_0\| = 1 + s_N(x_0) = p(x_0)$, so $S' \in \text{NA}((c_0, p), \mathcal{F})$, and $\|S' - S\|_p \leq \sup_x |\varphi(x)|/p(x) \leq \eta_N$. Given $A \neq 0$ and $\varepsilon > 0$, density for p_N gives such an S with $\|S - A/\|A\|_{p_N}\|_{p_N} < \varepsilon$; then $\|A\|_{p_N} S'$ attains its p -norm and $\|A\|_{p_N} S' - A\|_p \leq (1 + \eta_N)\|A\|_p(\eta_N + \varepsilon)$. Let $\varepsilon \rightarrow 0$ and $N \rightarrow \infty$ along the given N . $\square$

1.3. Elementary structure.

Lemma 1.6 (Dual ball). $B_{p^*} = B_{q^*} + L^*(B_{V^*})$, and for every $h \in X^*$

$$p^*(h) = \min\{\max(q^*(A), \|W\|_{V^*}) : A + L^*W = h\},$$

the minimum being attained. Moreover $p^{**}(\theta) = q^{**}(\theta) + \|L^{**}\theta\|_V$ for $\theta \in X^{**}$, and $L^{**}(X^{**}) \subseteq V$.

Proof. If $q^*(A) \leq 1$ and $\|W\| \leq 1$ then $(A + L^*W)(x) \leq q(x) + \|Lx\| = p(x)$. The set $B_{q^*} + L^*(B_{V^*})$ is convex and weak* compact (L^* is weak*-continuous), and its support function on X is $q(x) + \|Lx\|_V = p(x)$, which is also the support function of B_{p^*} ; two weak*-closed convex sets with the same support function on X coincide. The formula for p^* follows by homogeneity and compactness. Taking suprema over B_{p^*} gives the bidual formula; L^{**} maps into V because L is compact. $\square$

Lemma 1.7 (Block threshold; cf. [8, Lemma 6.5]). *Fix a block and drop the index m . Let $\zeta \in \ell_1 \setminus \{0\}$ and $w = J(\zeta)$ its norming functional ($N(w) = 1$, $w(\zeta) = |\zeta|$). Put $M = \|w\|_\infty$, $C = \|Dw\|_2$ and $P = \{k : |w(k)| = M\}$. Then $M > 0$, $C > 0$, $M + C = 1$, and*

$$\begin{aligned} \zeta/|\zeta| &= \alpha + D^2w/C, & \|\alpha\|_1 &= 1, \\ \text{supp } \alpha &\subseteq P, & \text{sign } \alpha(k) &= \text{sign } w(k) \quad (k \in \text{supp } \alpha). \end{aligned}$$

In particular $P \neq \emptyset$; off P one has $w(k) = C\zeta(k)/(\Phi(k)^2|\zeta|)$, and $|\zeta(k)| > |\zeta|\Phi(k)^2M/C$ implies $k \in P$ and $w(k) = M \text{sign } \zeta(k)$.

Proof. $w \neq 0$ and D is injective, so $M, C > 0$. The ball $B_{\ell_1} + D(B_{\ell_2})$ is closed (D is compact), so $\zeta/|\zeta| = \alpha + D\beta$ with $\|\alpha\|_1, \|\beta\|_2 \leq 1$. Then $1 = \langle w, \alpha \rangle + \langle Dw, \beta \rangle \leq M\|\alpha\|_1 + C\|\beta\|_2 \leq M + C = 1$, so $\langle w, \alpha \rangle = M = M\|\alpha\|_1$ and $\beta = Dw/C$; the first equality forces $\|\alpha\|_1 = 1$ and α to live on P with the signs of w . The remaining claims follow by reading the identity coordinatewise. $\square$

For a block vector $\zeta = R_m^{**}\theta$ the condition of the last sentence reads $|\theta(u_{k,m})| > \vartheta_m \Phi_m(k)$ with the *threshold constant* $\vartheta_m := |\zeta|_m M_m / (m C_m)$. For $k \in P$ we define the *margin* $\mu_{k,m} := |\theta(u_{k,m})| - \vartheta_m \Phi_m(k) \geq 0$; reading Lemma 1.7 at $k \in P$ gives the identity

$$(2) \quad |\zeta|_m |\alpha_m(k)| = \lambda_{k,m} \mu_{k,m} \quad (k \in P_m).$$

Peaks with $\alpha_m(k) = 0$ (equivalently $\mu_{k,m} = 0$) are called *degenerate*. We write $Q_m := \mathbb{N} \setminus P_m$ (strict non-peaks) and $\text{gap}_m(k) := M_m - |w_m(k)|$.

Lemma 1.8 (Rigidity and independence). (a) *If $b_+ + L^*W_+ = b_- + L^*W_-$ with $b_+ - b_- \in c_{00}$ and $W_\pm \in V^*$, then $b_+ = b_-$ and $W_+ = W_-$.* (b) *Let $J \subseteq \mathbb{N}$ be cofinite, I finite, and for each $m \in I$ let $\Lambda_m \subseteq \mathbb{N}$ be finite and $P'_m \subseteq \mathbb{N}$ be infinite, $s \in \{\pm 1\}^{P'_m}$. Put $\pi_m := \sum_{k \in P'_m} s_k \lambda_{k,m} u_{k,m}$. Then the restrictions to $c_{00}(J)$ of the functionals $u_{k,m}$ ($k \in \Lambda_m$, $m \in I$) and π_m ($m \in I$) are linearly independent; consequently $\nu \mapsto (u_{k,m}(\nu), \pi_m(\nu))$ maps $c_{00}(J)$ onto the corresponding finite-dimensional space.*

Proof. (a) $b_+ - b_- = L^*(W_- - W_+) \in Y \cap c_{00} = \{0\}$, and L^* is injective. (b) A vanishing combination $\sum c_{k,m} u_{k,m} + \sum_m d_m \pi_m$ on $c_{00}(J)$ is an element of Y supported in the finite set $\mathbb{N} \setminus J$, hence zero by (T-c). By (T-b) its coefficient at every $e_{k,m}$ vanishes: at the infinitely many $k \in P'_m \setminus \Lambda_m$ this coefficient is $d_m s_k m 2^{-m-k}$, so $d_m = 0$, and then $c_{k,m} = 0$. $\square$

Proposition 1.9 (Strict convexity and forced decomposition). (a) *Each R_m^{**} is injective, and p^{**} is strictly convex.*

(b) *Every $f \in S_{p^*}$ has a unique normer $\xi \in S_{p^{**}}$. Put $q_0 := q^{**}(\xi)$. Then $0 < q_0 < 1$, $\|L^{**}\xi\|_V = 1 - q_0$, every $R_m^{**}\xi \neq 0$, and there is exactly one decomposition $f = a + L^*w$ with $q^*(a) \leq 1$ and $\|w\|_{V^*} \leq 1$; it satisfies*

$$q^*(a) = 1, \quad a(\xi) = q_0, \quad w_m = J_m(R_m^{**}\xi), \quad N_m(w_m) = 1 \quad (m \in I).$$

(c) *Put $\nu := \|U^*a\|_H > 0$ and $e := U^*a/\nu$. Then $\hat{z} := \xi/q_0 = z + Ue$ with $z \in B_{\ell_\infty}$ and $z_j = \text{sign } a_j$ on $F := \text{supp } a$; and $a(\hat{z}) = 1$, $b(\hat{z}) = b(z) + \langle U^*b, e \rangle$ for $b \in \ell_1$.*

Proof. (a) If $\theta(u_{k,m}) = 0$ for all k , then θ vanishes on a dense subset of S_{q^*} , so $\theta = 0$. Let $p^{**}(\xi) = p^{**}(\eta) = p^{**}((\xi + \eta)/2) = 1$. Since q^{**} and each $|R_m^{**} \cdot|_m$ are seminorms whose sum is p^{**} (Lemma 1.6), we get $|R_m^{**}(\xi + \eta)|_m = |R_m^{**}\xi|_m + |R_m^{**}\eta|_m$ for every m . Fix m ; $\zeta_1 = R_m^{**}\xi$ and $\zeta_2 = R_m^{**}\eta$ are nonzero, and a norming functional w of

$\zeta_1 + \zeta_2$ norms both. If ξ, η were not positively proportional there would be $v \in S_{q^*}$ with $\xi(v) > 0 > \eta(v)$ (use surjectivity of $v \mapsto (\xi(v), \eta(v))$ if they are independent). Choose $n_j \rightarrow \infty$ with $u_{n_j, m} \rightarrow v$. Since $\Phi_m(n_j) \rightarrow 0$, Lemma 1.7 applied to ζ_1 and to ζ_2 gives $w(n_j) = \|w\|_\infty$ and $w(n_j) = -\|w\|_\infty$ for large j , which is absurd. Hence $\eta = \lambda\xi$ with $\lambda > 0$, and $\lambda = 1$.

(b) The normers of f form a convex subset of $S_{p^{**}}$, hence a point. For any decomposition $f = a + L^*w$ with $q^*(a), \|w\| \leq 1$, $1 = f(\xi) = a(\xi) + w(L^{**}\xi) \leq q^{**}(\xi) + \|L^{**}\xi\| = 1$, so $a(\xi) = q_0$ and $w_m(R_m^{**}\xi) = |R_m^{**}\xi|_m$ for each m . As $R_m^{**}\xi \neq 0$ and $|\cdot|_m$ is smooth, $w_m = J_m(R_m^{**}\xi)$ is determined, hence so is $a = f - L^*w$. Finally $q_0 > 0$ because q^{**} is a norm and $\xi \neq 0$, $q_0 < 1$ because $L^{**}\xi \neq 0$, and $q_0 = a(\xi) \leq q^*(a)q_0$ gives $q^*(a) = 1$.

(c) $\xi/q_0 \in B_{q^{**}} = B_{\ell_\infty} + U(B_H)$, so $\xi/q_0 = z + Uh$ with $\|z\|_\infty, \|h\| \leq 1$; then $1 = a(z) + \langle U^*a, h \rangle \leq \|a\|_1 + \|U^*a\| = 1$ forces $a(z) = \|a\|_1$ and $h = e$. $\square$

Throughout, the *forced data* of $f \in S_{p^*}$ are $(\xi, q_0, a, w, F, z, \hat{z}, e, \nu)$ and, for each block, $\zeta_m := R_m^{**}\xi$, $\sigma_m := |\zeta_m|_m$, $M_m, C_m, P_m, \alpha_m, Q_m, \text{gap}_m, \vartheta_m, \mu_{k,m}$. Note $q_0 + \sum_m \sigma_m = 1$. The set $K := \{j \notin F : |z_j| = 1\}$ is the set of *contacts* (kinks) and $J := \mathbb{N} \setminus (F \cup K)$ the set of *free coordinates*; the *room* of $j \in J$ is $1 - |z_j| > 0$.

1.4. Smoothness and norm-attaining functionals.

Proposition 1.10 (Fréchet smoothness). (a) *The norm p is Fréchet differentiable at every $x \neq 0$, and $\nabla p(x) = \nabla q(x) + L^*J_V(Lx)$, where $J_V(Lx) = (J_m(R_mx))_m$.*

(b) $\text{NA}((X, p), \mathbb{R}) \cap S_{p^*} = \nabla p(S_p)$, and the duality map $\nabla p : S_p \rightarrow S_{p^*}$ is norm-to-norm continuous.

(c) $\text{NA}((X, q), \mathbb{R}) = c_{00}$. More precisely, if $a' \in c_{00}$, $q^*(a') = 1$, and $z' \in c_0$ satisfies $\|z'\|_\infty \leq 1$ and $z'_j = \text{sign } a'_j$ on $\text{supp } a'$, then $y' := z' + U(U^*a'/\|U^*a'\|)$ satisfies $q(y') = 1$, $\nabla q(y') = a'$ and $\nabla p(y') = a' + L^*J_V(Ly')$. Conversely every $f' \in \text{NA}((X, p), \mathbb{R}) \cap S_{p^*}$ equals $\nabla p(y')$ for such a pair (a', z') , with $a' = \nabla q(y')$.

Proof. Gateaux smoothness. The norm q^* is strictly convex: if $q^*(a) = q^*(b) = 1 = q^*(\frac{a+b}{2})$, then $\|U^*a + U^*b\| = \|U^*a\| + \|U^*b\|$, so $U^*b = \lambda U^*a$ with $\lambda \geq 0$ (both are nonzero), hence $b = \lambda a$ by injectivity of U^* , and $\lambda = 1$. Thus q is Gateaux smooth. In the same way N_m is strictly convex (D_m is injective), so $|\cdot|_m$ is smooth. Let $x \neq 0$ and let $f, f' \in S_{p^*}$ both norm x . Write $f = A + L^*W$ with $q^*(A), \|W\| \leq 1$ (Lemma 1.6). Then $p(x) = A(x) + W(Lx) \leq q(x) + \|Lx\| = p(x)$ forces $A(x) = q(x)$ and $W_m(R_mx) = |R_mx|_m$ for every m . Since R_m is injective (by the argument of Proposition 1.9(a)), $R_mx \neq 0$, so $A = \nabla q(x)$ and $W_m = J_m(R_mx)$. The same holds for f' , so $f = f'$.

Fréchet smoothness. By Šmulyan's lemma it suffices to show: if $x \in S_p$, $f_n \in B_{p^*}$ and $f_n(x) \rightarrow 1$, then $f_n \rightarrow \nabla p(x)$ in norm. Write $f_n = a_n + L^*w_n$ with $q^*(a_n), \|w_n\| \leq 1$. As $a_n(x) \leq q(x)$, $w_n(Lx) \leq \|Lx\|$ and the sum tends to $p(x)$, we get $a_n(x) \rightarrow q(x)$. Write $x/q(x) = z + Uh$ with $z \in B_{c_0}$, $\|h\| \leq 1$. Then

$$\frac{a_n(x)}{q(x)} \leq \sum_j |a_n(j)| |z_j| + \|U^*a_n\| \leq 1 - \sum_j |a_n(j)| (1 - |z_j|),$$

so $\sum_j |a_n(j)| (1 - |z_j|) \rightarrow 0$. The set $G = \{j : |z_j| \geq \frac{1}{2}\}$ is finite, $\sum_{j \notin G} |a_n(j)| \rightarrow 0$, and the coordinates in G are bounded; hence every subsequence of (a_n) has a norm convergent subsequence, whose limit a' satisfies $q^*(a') \leq 1$, $a'(x) = q(x)$, so $a' = \nabla q(x)$. Every subsequence of (w_n) has a weak* convergent subsequence

$w_n \rightarrow w$, and $L^*w_n \rightarrow L^*w$ in norm because L is compact. Then $\nabla q(x) + L^*w \in B_{p^*}$ norms x , so it equals $\nabla p(x)$. This proves (a), and the continuity in (b).

(b) If $f \in S_{p^*}$ attains its norm at $x \in S_p$, then $f = \nabla p(x)$.

(c) If $a \in S_{q^*}$ attains its norm at $y \in S_q$, write $y = z + Uh$, $z \in B_{c_0}$, $\|h\| \leq 1$; then $1 = a(z) + \langle U^*a, h \rangle$ forces $|z_j| = 1$ on $\text{supp } a$, which is finite because $z \in c_0$. For the pair (a', z') : $y' \in B_q$ and $a'(y') = \|a'\|_1 + \|U^*a'\| = 1 = q^*(a')$, hence $q(y') = 1$ and $\nabla q(y') = a'$; then (a) gives $\nabla p(y')$. Conversely, if $f' = \nabla p(x')$ with $x' \in S_p$, then $a' := \nabla q(x') \in c_{00}$ and $y' := x'/q(x') = z' + Uh'$ with $z' \in B_{c_0}$; the equality $1 = a'(z') + \langle U^*a', h' \rangle \leq \|a'\|_1 + \|U^*a'\| = 1$ forces $z' = \text{sign } a'$ on $\text{supp } a'$ and $h' = U^*a'/\|U^*a'\|$; finally $\nabla p(y') = \nabla p(x')$. $\square$

Proposition 1.11 (Continuity of the forced data). *Let $f_n \rightarrow f$ in S_{p^*} (not necessarily norm attaining) with forced data indexed by n . Then $\xi_n \rightarrow \xi$ weak*; $R_m^{**}\xi_n \rightarrow R_m^{**}\xi$ in ℓ_1 ; $w_{n,m} \rightarrow w_m$ weak* and $D_m w_{n,m} \rightarrow D_m w_m$ in ℓ_2 , so $M_{n,m} \rightarrow M_m$, $C_{n,m} \rightarrow C_m$, $\sigma_{n,m} \rightarrow \sigma_m$ and $\vartheta_{n,m} \rightarrow \vartheta_m$; $R_m^* w_{n,m} \rightarrow R_m^* w_m$ and $L^* w_n \rightarrow L^* w$ in norm; $a_n \rightarrow a$ in ℓ_1 ; $e_n \rightarrow e$ in H ; $q_0^{(n)} \rightarrow q_0$; $\hat{z}_n \rightarrow \hat{z}$ and $z_n \rightarrow z$ weak*.*

Proof. A weak* cluster point η of (ξ_n) lies in $B_{p^{**}}$ and $f(\eta) = \lim f_n(\xi_n) = 1$, so $\eta = \xi$; $B_{p^{**}}$ is weak* metrizable. R_m is compact, so R_m^{**} maps bounded weak* convergent sequences to norm convergent ones. A weak* cluster point w' of $(w_{n,m})_n$ has $N_m(w') \leq 1$ and $w'(\zeta_m) = \lim w_{n,m}(\zeta_{n,m}) = \sigma_m$, so $w' = w_m$. Since $\Phi_m \in \ell_2$, D_m maps bounded weak* convergent sequences to norm convergent ones; $M = 1 - C$. L^* is weak*-to-norm continuous on bounded sets, and $a_n = f_n - L^*w_n$. Finally $q_0^{(n)} = a_n(\xi_n) \rightarrow a(\xi)$, $e_n = U^*a_n/\|U^*a_n\|$, and $z_n = \xi_n/q_0^{(n)} - Ue_n$. $\square$

Proposition 1.12 (Norm-attaining approximants). *Let $f \in S_{p^*}$ have forced data as above. For a sequence $f'_n \in \text{NA}((X, p), \mathbb{R}) \cap S_{p^*}$ the following are equivalent:*

- (i) $f'_n \rightarrow f$ in norm;
- (ii) $f'_n = \nabla p(y'_n)$ with $y'_n = z'_n + U(U^*a'_n/\|U^*a'_n\|)$, where $a'_n \in c_{00} \cap S_{q^*}$, $a'_n \rightarrow a$ in ℓ_1 , $z'_n \in c_0$, $\|z'_n\|_\infty \leq 1$, $z'_n = \text{sign } a'_n$ on $\text{supp } a'_n$, and $z'_n \rightarrow z$ coordinatewise.

In particular, for every n the coordinates of z'_n outside a finite window $W_n \supseteq \text{supp } a'_n$ may be chosen freely (moduli ≤ 1 , membership in c_0) as long as the windows exhaust $\mathbb{N}$; the window must contain $\text{supp } a'_n$, where $z'_n = \text{sign } a'_n$ is forced.

Proof. (ii) $\Rightarrow$ (i): $y'_n \rightarrow \hat{z}$ weak* (bounded coordinatewise convergence plus norm convergence of the U -part, as $U^*a'_n/\|U^*a'_n\| \rightarrow e$). By compactness of L , $p(y'_n) = 1 + \|Ly'_n\| \rightarrow 1 + \|L^*\hat{z}\| = 1/q_0$, so $x_n := y'_n/p(y'_n) \rightarrow \xi$ weak*. Then $\nabla q(x_n) = a'_n \rightarrow a$, and $L^*J_V(Lx_n) \rightarrow L^*w$ in norm by the argument of Proposition 1.11 (a weak* cluster point of $J_V(Lx_n)$ norms $L^{**}\xi$, whose components are nonzero). (i) $\Rightarrow$ (ii): write $f'_n = \nabla p(x'_n)$ (Proposition 1.10); by Proposition 1.11, $x'_n \rightarrow \xi$ weak*, $a'_n := \nabla q(x'_n) \rightarrow a$, and Proposition 1.10(c) gives the form of $y'_n = x'_n/q(x'_n)$; finally $z'_n = y'_n - U(\cdot) \rightarrow z$ weak*. $\square$

1.5. Mates and the reduction. For $f \in S_{p^*}$ we use, as in [7],

$$\mathcal{C}(f) := \{g \in X^* : p^*(f + tg) \leq s(t) \text{ for all } t \in \mathbb{R}\}.$$

Proposition 1.13 (Reduction). (a) *For $f \in S_{p^*}$ and $g \in X^*$ the operator $(f, g) : (X, p) \rightarrow \ell_2^2$ is contractive iff $g \in \mathcal{C}(f)$ iff $f(y)^2 + g(y)^2 \leq p(y)^2$ for all $y \in X$.* (b) *Every $S \in \mathcal{L}((X, p), \ell_2^2)$ of norm one is, after a rotation of ℓ_2^2 , of the form (f, g) with*

$f \in S_{p^*}$ and $g \in \mathcal{C}(f)$. Hence $\text{NA}((X, p), \ell_2^2)$ is dense iff $(f, \rho g) \in \text{cl NA}((X, p), \ell_2^2)$ for all $f \in S_{p^*}$, $g \in \mathcal{C}(f)$ and $\rho \in (0, 1)$. (c) If $g \in \mathcal{C}(f)$ then $p^*(g) \leq 1$, $g(\xi) = 0$, and $p^*(f + t\rho g) \leq s(\rho t) < s(t)$ for $t \neq 0$, $\rho \in (0, 1)$. (d) If f attains its norm at x and $g \in \mathcal{C}(f)$, then (f, g) attains its norm at x .

Proof. (a) $\|(f, g)\| = \sup_{\theta} p^*(\cos \theta f + \sin \theta g)$ and $p^*(\cos \theta f + \sin \theta g) = |\cos \theta| p^*(f + \tan \theta g)$ with $|\cos \theta| = s(\tan \theta)^{-1}$; the case $\cos \theta = 0$ is the limit $t \rightarrow \infty$. The primal form is $\|(f, g)\|^2 = \sup_{p(y) \leq 1} f(y)^2 + g(y)^2$. (b) $\theta \mapsto p^*(S^*(\cos \theta, \sin \theta))$ attains its maximum 1; rotate that direction to e_1 . Since NA is a cone and $(f, g) = \lim_{\rho \rightarrow 1} (f, \rho g)$, the reduction follows. (c) Divide $p^*(f + tg) \leq s(t)$ by $|t| \rightarrow \infty$; and $1 + tg(\xi) = (f + tg)(\xi) \leq s(t)$ for all t forces $g(\xi) = 0$. (d) $g(x)^2 \leq p(x)^2 - f(x)^2 = 0$ (this observation is from [3, Section 13]). $\square$

Proposition 1.14 (Primal description of the fibre). *For $f \in S_{p^*}$ put $r_f := \sqrt{p^2 - f^2}$ on X and let $\tilde{r}_f$ be its largest sublinear minorant, $\tilde{r}_f(x) = \inf \{ \sum_j r_f(y_j) : \sum_j y_j = x \}$ (finite families). Then $\tilde{r}_f$ is a seminorm, $\mathcal{C}(f) = \{g \in X^* : g \leq \tilde{r}_f\}$, $\mathcal{C}(f)$ is convex, symmetric and weak* compact, and $\max_{g \in \mathcal{C}(f)} g(x) = \tilde{r}_f(x)$ for $x \in X$.*

Proof. The infimum φ is subadditive, positively homogeneous, even and $\leq r_f$, and every sublinear minorant s of r_f satisfies $s(x) \leq \sum s(y_j) \leq \sum r_f(y_j)$, so $\varphi = \tilde{r}_f$. A linear g satisfies $g \leq r_f$ iff $g \leq \tilde{r}_f$, and as r_f is even this is $|g| \leq r_f$, i.e. $g \in \mathcal{C}(f)$ by Proposition 1.13(a). Given x , the Hahn–Banach theorem gives a linear $g \leq \tilde{r}_f$ with $g(x) = \tilde{r}_f(x)$, continuous because $\tilde{r}_f \leq p$. $\square$

Definition 1.15 (Recovery). For subsets $C_n \subseteq X^*$ let $\text{Li}_n C_n := \{g : \text{dist}(g, C_n) \rightarrow 0\}$ (Kuratowski lower limit; it is closed, and convex if the C_n are convex). A mate $g \in \mathcal{C}(f)$ is *recovered along* a sequence $f_n \rightarrow f$ in S_{p^*} if $g \in \text{Li}_n \mathcal{C}(f_n)$. We write

$$\mathcal{R} := \{f \in S_{p^*} : (f, g) \in \text{cl NA}((X, p), \ell_2^2) \text{ for every } g \in \mathcal{C}(f)\}$$

for the *recoverable set*. By Proposition 1.13, $\text{NA}((X, p), \ell_2^2)$ is dense iff $\mathcal{R} = S_{p^*}$.

Lemma 1.16 (Recoverability of a single pair). *Let $f \in S_{p^*}$, $g \in \mathcal{C}(f)$ and $\rho \in (0, 1)$. Then $(f, \rho g) \in \text{cl NA}((X, p), \ell_2^2)$ iff there are $f_n \in \text{NA}((X, p), \mathbb{R}) \cap S_{p^*}$ with $f_n \rightarrow f$ and $\rho g \in \text{Li}_n \mathcal{C}(f_n)$.*

Proof. Suppose $g_n \in \mathcal{C}(f_n)$ and $g_n \rightarrow \rho g$. Then (f_n, g_n) attains its norm (Proposition 1.13(d)) and converges to $(f, \rho g)$. Conversely let $S_k \in \text{NA}$ converge to $T := (f, \rho g)$; $\|T\| = 1$. For a unit vector $y = (\cos \theta, \sin \theta)$ with $\sin \theta \neq 0$ we have $p^*(T^*y) < 1$: if $\cos \theta \neq 0$,

$$p^*(T^*y) = |\cos \theta| p^*(f + \tan \theta \rho g) \leq |\cos \theta| s(\rho \tan \theta) < 1,$$

and $p^*(\rho g) \leq \rho < 1$ otherwise. Let S_k attain its norm at $x_k \in S_p$ and $y_k := S_k x_k / \|S_k x_k\|$. Then $p^*(T^*y_k) \geq \|S_k\| - \|S_k - T\| \rightarrow 1$, so every limit point of (y_k) is $\pm e_1$; replacing x_k by $-x_k$ we may assume $y_k \rightarrow e_1$. Let Q_k be the rotation with $Q_k y_k = e_1$; then $Q_k \rightarrow I$, and $Q_k S_k = (\tilde{f}_k, \tilde{g}_k)$ attains its norm at x_k with $\tilde{f}_k(x_k) = \|S_k\|$, $\tilde{g}_k(x_k) = 0$. The functionals $f_k := \tilde{f}_k / \|S_k\|$ and $g_k := \tilde{g}_k / \|S_k\|$ satisfy $\|(f_k, g_k)\| = 1 = f_k(x_k)$, so $f_k \in \text{NA} \cap S_{p^*}$, $g_k \in \mathcal{C}(f_k)$, and $(f_k, g_k) \rightarrow (f, \rho g)$. $\square$

2. THE LOWER-LIMIT THEOREM

Theorem 2.1 (Global compactness; [7]). *If $f_n \rightarrow f$ in S_{p^*} and $g_n \in \mathcal{C}(f_n)$, then (g_n) has a norm convergent subsequence and every limit of such a subsequence lies in $\mathcal{C}(f)$. Consequently every $\mathcal{C}(f)$ is norm compact, $f \mapsto \mathcal{C}(f)$ is upper semicontinuous, and $\mathcal{C}(f) \cup \bigcup_n \mathcal{C}(f_n)$ is norm compact.*

Proof. $p^*(g_n) \leq 1$; pass to a weak* convergent subsequence $g_n \rightarrow g$. Weak* lower semicontinuity of p^* gives $g \in \mathcal{C}(f)$. Let $\Delta := \limsup_n \|g_n - g\|_1$, realized along a further subsequence. Fix $t > 0$ and write $f_n + tg_n = A_n + L^*W_n$ with $q^*(A_n), \|W_n\| \leq s(t)$ (Lemma 1.6). Passing to a subsequence, $L^*W_n \rightarrow k$ in norm, $W_n \rightarrow W$ weak*, and $A_n \rightarrow A := f + tg - k$ weak*. Since $k(\xi) = W(L^{**}\xi) \leq s(t)(1 - q_0)$ we get $A(\xi) \geq 1 - s(t)(1 - q_0)$ and $q^*(A) \geq A(\xi)/q_0$. Moreover $U^*A_n \rightarrow U^*A$ in norm, and for bounded coordinatewise convergent sequences in ℓ_1 one has $\|A_n\|_1 - \|A_n - A\|_1 \rightarrow \|A\|_1$. Since $t(g_n - g) = (A_n - A) + (L^*W_n - k) - (f_n - f)$,

$$\begin{aligned} t\Delta &= \limsup_n \|A_n - A\|_1 = \limsup_n (q^*(A_n) - q^*(A)) \\ &\leq s(t) - \frac{1 - s(t)(1 - q_0)}{q_0} = \frac{s(t) - 1}{q_0}. \end{aligned}$$

Letting $t \rightarrow 0$ gives $\Delta = 0$. The remaining assertions follow (for the union, a sequence either has infinitely many terms in one compact fibre or a subsequence to which the first part applies). $\square$

Theorem 2.2 (Lower-limit theorem). *Let $f_n \rightarrow f$ in S_{p^*} (not necessarily norm attaining) and $\rho \in [0, 1]$. The following are equivalent:*

- (i) $\rho\mathcal{C}(f) \subseteq \text{Li}_n \mathcal{C}(f_n)$;
- (ii) $\liminf_n \tilde{r}_{f_n}(x) \geq \rho \tilde{r}_f(x)$ for every $x \in X$;
- (iii) (ii) holds for all x in a dense subset of X .

Proof. (i) $\Rightarrow$ (ii): choose $g \in \mathcal{C}(f)$ with $g(x) = \tilde{r}_f(x)$ (Proposition 1.14) and $g_n \in \mathcal{C}(f_n)$ with $g_n \rightarrow \rho g$; then $\tilde{r}_{f_n}(x) \geq g_n(x) \rightarrow \rho \tilde{r}_f(x)$. (ii) $\Leftrightarrow$ (iii): each $\tilde{r}_h$ is a seminorm bounded by p , so $|\tilde{r}_h(x) - \tilde{r}_h(y)| \leq p(x - y)$ uniformly in h . (ii) $\Rightarrow$ (i): suppose $g \in \rho\mathcal{C}(f)$ and $\text{dist}(g, \mathcal{C}(f_{n_j})) \geq \delta > 0$. All $\mathcal{C}(f_{n_j})$ lie in the compact set of Theorem 2.1; by Blaschke's selection theorem we may assume $\mathcal{C}(f_{n_j}) \rightarrow A$ in the Hausdorff metric. Then A is compact and convex and $\text{dist}(g, A) \geq \delta$. For $x \in X$ the support functions satisfy $|h_C(x) - h_{C'}(x)| \leq d_H(C, C')\|x\|_\infty$, so $h_A(x) = \lim_j \tilde{r}_{f_{n_j}}(x) \geq \rho \tilde{r}_f(x) \geq g(x)$. As A is weak* compact and convex, $g \notin A$ would give $x \in c_0$ with $g(x) > h_A(x)$. Contradiction. $\square$

Corollary 2.3 (Finite, diagonal and mate versions). *For $f \in S_{p^*}$ the following are equivalent:*

- (a) *there are $f_n \in \text{NA} \cap S_{p^*}$ with $f_n \rightarrow f$ and $\mathcal{C}(f) \subseteq \text{Li}_n \mathcal{C}(f_n)$;*
- (b) *for every $\varepsilon > 0$, $\rho < 1$ and $x_1, \dots, x_k \in X$ there is $f' \in \text{NA} \cap S_{p^*}$ with $p^*(f' - f) < \varepsilon$ and $\tilde{r}_{f'}(x_i) \geq \rho \tilde{r}_f(x_i) - \varepsilon$ for $i \leq k$;*
- (c) *for every $\varepsilon > 0$, $\rho < 1$ and $g_1, \dots, g_k \in \mathcal{C}(f)$ there is $f' \in \text{NA} \cap S_{p^*}$ with $p^*(f' - f) < \varepsilon$ and $\text{dist}(\rho g_i, \mathcal{C}(f')) < \varepsilon$ for $i \leq k$;*
- (d) *as (c), uniformly: $\sup_{g \in \mathcal{C}(f)} \text{dist}(\rho g, \mathcal{C}(f')) < \varepsilon$.*

If they hold, then $f \in \mathcal{R}$.

Proof. (a) $\Rightarrow$ (d): $\text{dist}(g, \mathcal{C}(f_n)) \rightarrow 0$ for each g , uniformly on the compact set $\mathcal{C}(f)$ (finite nets and 1-Lipschitz dependence on g); and $\text{dist}(\rho g, \mathcal{C}(f_n)) \leq \rho \text{dist}(g, \mathcal{C}(f_n))$

because $\mathcal{C}(f_n)$ is convex and contains 0. (d) $\Rightarrow$ (c) is trivial. (c) $\Rightarrow$ (b): pick $g_i \in \mathcal{C}(f)$ with $g_i(x_i) = \tilde{r}_f(x_i)$ and apply (c) with $\varepsilon/\max(1, \max_i \|x_i\|)$. (b) $\Rightarrow$ (a): let (x_i) be dense in X ; for each n apply (b) with $\varepsilon = 1/n$, $\rho = 1 - 1/n$ and $x_1, \dots, x_n$ to obtain f_n . Then $\liminf_n \tilde{r}_{f_n}(x_i) \geq \tilde{r}_f(x_i)$ for every i , and Theorem 2.2 with $\rho = 1$ gives (a). The last claim is Lemma 1.16. $\square$

Theorem 2.4 (Transitivity). $\text{NA} \cap S_{p^*} \subseteq \mathcal{R}$. Let $f \in S_{p^*}$ and suppose that for every $\rho < 1$ there is a sequence $f_n^\rho \in \mathcal{R}$ with $f_n^\rho \rightarrow f$ and $\liminf_n \tilde{r}_{f_n^\rho}(x) \geq \rho \tilde{r}_f(x)$ on a dense set of $x \in X$. Then $f \in \mathcal{R}$.

Proof. The first claim is Proposition 1.13(d). By Theorem 2.2, $\rho \mathcal{C}(f) \subseteq \text{Li}_n \mathcal{C}(f_n^\rho)$. Given $g \in \mathcal{C}(f)$ choose $g_n \in \mathcal{C}(f_n^\rho)$ with $g_n \rightarrow \rho g$; each (f_n^ρ, g_n) lies in clNA , hence so does $(f, \rho g)$; let $\rho \rightarrow 1$. $\square$

Remark 2.5 (Several rows and the residual set). For $d \geq 1$ let $\mathcal{C}_d(f) := \{(g_1, \dots, g_d) : f(y)^2 + \sum_i g_i(y)^2 \leq p(y)^2 \ (y \in X)\}$; equivalently $p^*(f + \sum_i t_i g_i) \leq s(|t|)$ for $t \in \mathbb{R}^d$. Theorem 2.1, Theorem 2.2, Corollary 2.3 and Theorem 2.4 hold verbatim for $\mathcal{C}_d$ (support functions on X^d , separation by $(c_0)^d$), with ℓ_2^{d+1} in place of ℓ_2^2 . By the residual uniform recovery theorem of [7] there is a dense G_δ set $\mathcal{O} \subseteq S_{p^*}$ at which every $\mathcal{C}_d$ is Hausdorff continuous; in particular $\mathcal{O} \subseteq \mathcal{R}$. This does not decide density: Hausdorff continuity may fail on a meagre set of first rows, and the remaining sections study what can be recovered there.

3. CERTIFICATES AND THE TRANSPORT THEOREM

Throughout this section $f \in S_{p^*}$ has forced data as in Section 1; block indices are dropped when a single block is involved. For $b \in \ell_1$ we write $\|b/a\|_\infty := \sup_{j \in F} |b_j|/|a_j|$ (when $\text{supp } b \subseteq F$), and $P^\perp$ denotes the orthogonal projection of H onto $e^\perp$ (throughout the paper; it is not used for other projections).

3.1. Definitions and exact expansions.

Definition 3.1 (Finite certificates). A *finite certificate* at f is a pair $c = (b, \omega)$, $\omega = (\omega_m)_{m \in I}$, such that

- (C1) $b \in \ell_1$, $\text{supp } b \subseteq F$, $\|b/a\|_\infty < \infty$ and $b(\hat{z}) = 0$;
- (C2) each ω_m is finitely supported in Q_m , and $\omega_m = 0$ for all but finitely many m (the *active blocks*).

Put $d_m := \langle D_m w_m, D_m \omega_m \rangle / C_m$ and define the *direction* $g_c := b + \sum_m R_m^*(\omega_m - d_m w_m)$, the coefficients

$$h(b) := \frac{\|P^\perp U^* b\|^2}{\nu}, \quad H_m(\omega_m) := \frac{\|D_m \omega_m\|_2^2 - d_m^2}{C_m},$$

$$H(c) := \max(h(b), \max_m H_m(\omega_m)),$$

the constant $\kappa(c) := \max\{2\beta/\nu, \max_m 2|d_m|M_m/C_m\}$ with $\beta := \|U^* b\|$, and the *coordinatewise radius*

$$r(c) := \min \left\{ \frac{1}{\|b/a\|_\infty}, \frac{\nu}{2\beta}, \min_m \min \left(\min_{k \in \text{supp } \omega_m} \frac{\text{gap}_m(k)}{2|\omega_m(k)|}, \frac{1}{2|d_m|}, \frac{C_m}{2|d_m|M_m} \right) \right\}$$

(with $1/0 = \infty$). Finally

$$\text{Cert}(f) := \{g_c : c \text{ a finite certificate at } f, H(c) \leq 1, g_c \in \mathcal{C}(f)\}.$$

Let $P_m^\perp$ denote the orthogonal projection of ℓ_2 onto $(D_m w_m)^\perp$. Then

$$\|P_m^\perp D_m \omega_m\|^2 = \|D_m \omega_m\|^2 - d_m^2,$$

so $H_m \geq 0$. The radius is the *coordinatewise* one: each off-peak coordinate is compared with its own gap. (A cruder radius $\min_k \text{gap}(k)/(2 \max_k |\omega(k)|)$ would not suffice for Theorem 4.6 below.)

Lemma 3.2 (Algebra). *Finite certificates form a vector space and $c \mapsto g_c$ is linear; H is convex and $H(\lambda c) = \lambda^2 H(c)$; $r(\lambda c) = r(c)/|\lambda|$, $\kappa(\lambda c) = |\lambda| \kappa(c)$, and $\kappa(c)r(c) \leq 1$. Every direction satisfies $g_c(\xi) = 0$, and $\text{Cert}(f)$ is convex and symmetric.*

Proof. d_m is linear in ω_m , and H is a maximum of positive semidefinite quadratic forms. The inequality $\kappa r \leq 1$ follows from the terms $\nu/(2\beta)$ and $C_m/(2|d_m|M_m)$ of the radius. We have $b(\xi) = q_0 b(\hat{z}) = 0$ and, by Lemma 1.7,

$$\begin{aligned} \frac{\langle \omega_m - d_m w_m, \zeta_m \rangle}{\sigma_m} &= \langle \omega_m, \alpha_m \rangle + \frac{\langle D_m \omega_m, D_m w_m \rangle}{C_m} - d_m \left(\langle w_m, \alpha_m \rangle + \frac{\|D_m w_m\|^2}{C_m} \right) \\ &= 0 + d_m - d_m(M_m + C_m) = 0, \end{aligned}$$

because α_m lives on P_m and ω_m vanishes there. Convexity of $\text{Cert}(f)$ follows from linearity, convexity of H and of $\mathcal{C}(f)$. $\square$

Lemma 3.3 (Exact base identity). *Put $\Psi(h) := \|e + h\| - 1 - \langle e, h \rangle$ for $h \in H$. For $B \in \ell_1$ and $t \in \mathbb{R}$,*

$$(3) \quad q^*(a + tB) = 1 + tB(\hat{z}) + \text{Fl}_B(t) + \sum_{j \notin F} (|tB_j| - z_j tB_j) + \nu \Psi(tU^*B/\nu),$$

where $\text{Fl}_B(t) := \sum_{j \in F} 2(-\text{sign}(a_j)tB_j - |a_j|)_+$; all terms after $tB(\hat{z})$ are nonnegative. Moreover $\Psi(h) = \|h_\perp\|^2/(\|e + h\| + 1 + \langle e, h \rangle)$ with $h_\perp = P^\perp h$, and for $\|h\| \leq \frac{1}{2}$: $\|h_\perp\|^2/(2(1 + \|h\|)) \leq \Psi(h) \leq \|h_\perp\|^2/(2(1 - \|h\|))$. Consequently:

- (a) if $B(\hat{z}) = 0$ and $|t| \|U^*B\| \leq \nu/2$, then $q^*(a + tB) - 1 \geq \frac{t^2}{2} h(B)/(1 + |t| \|U^*B\|/\nu)$;
- (b) if b satisfies (C1) and $|t| \leq \min(1/\|b/a\|_\infty, \nu/(2\beta))$, then $q^*(a + tb) \leq 1 + \frac{t^2}{2} h(b)(1 + 2|t|\beta/\nu)$.

Proof. For real $x \neq 0$ and y : $|x + y| = |x| + \text{sign}(x)y + 2(-\text{sign}(x)y - |x|)_+$. Summing over $j \in F$ (where $z_j = \text{sign } a_j$) gives $\sum_{j \in F} |a_j + tB_j| = \|a\|_1 + \sum_{j \in F} z_j tB_j + \text{Fl}_B(t)$, and $\sum_{j \notin F} |tB_j| = \sum_{j \notin F} z_j tB_j + \sum_{j \notin F} (|tB_j| - z_j tB_j)$. Next $\|U^*a + tU^*B\| = \nu\|e + h\|$ with $h = tU^*B/\nu$ and $\nu\langle e, h \rangle = t\langle U^*B, e \rangle$. Adding and using $\|a\|_1 + \nu = 1$ and $B(\hat{z}) + \langle U^*B, e \rangle = B(\hat{z})$ gives (3). Nonnegativity of Ψ : $\|e + h\| \geq \langle e, e + h \rangle$. The formula for Ψ follows from $\|e + h\|^2 - (1 + \langle e, h \rangle)^2 = \|h\|^2 - \langle e, h \rangle^2$; for $\|h\| \leq \frac{1}{2}$ the denominator lies in $[2(1 - \|h\|), 2(1 + \|h\|)]$. With $h = tU^*B/\nu$ one has $\nu\|h_\perp\|^2 = t^2 h(B)$; (a) follows by dropping the nonnegative terms, and (b) because no flip occurs for $|t| \leq 1/\|b/a\|_\infty$, the sum over $j \notin F$ vanishes, and $1/(1 - x) \leq 1 + 2x$ for $x \leq \frac{1}{2}$. $\square$

Lemma 3.4 (Block expansion). *Fix a block. Let $\omega : \mathbb{N} \rightarrow \mathbb{R}$ vanish on P with $D\omega \in \ell_2$, $d := \langle D\omega, D\omega \rangle/C$, $h_\perp := D\omega - (d/C)D\omega$ (so $\|h_\perp\|^2 = \|D\omega\|^2 - d^2$), $W(\sigma) := (1 - d\sigma)w + \sigma\omega$ and $Y := C + d\sigma M$. Then:*

$$(a) \quad \|DW(\sigma)\|_2 = \sqrt{Y^2 + \sigma^2 \|h_\perp\|^2};$$

- (b) if $1 - d\sigma \geq 0$ and $Y > 0$, then $N(W(\sigma)) \geq 1 + \sigma^2 \|h_\perp\|^2 / (\sqrt{Y^2 + \sigma^2 \|h_\perp\|^2} + Y)$;
- (c) if $|d\sigma| \leq \frac{1}{2}$ and $|\sigma\omega(k)| \leq \text{gap}(k)/2$ for all k with $\omega(k) \neq 0$, then $\|W(\sigma)\|_\infty$ equals $(1 - d\sigma)M$; if moreover $Y > 0$, equality holds in (b);
- (d) if $\omega \in c_{00}$ and $|\sigma| \leq \min(\min_k \text{gap}(k)/(2|\omega(k)|), 1/(2|d|), C/(2|d|M))$, then $N(W(\sigma)) \leq 1 + \frac{\sigma^2}{2} H(\omega)(1 + 2|d\sigma|M/C)$.

Proof. (a) Using $1/C - 1 = M/C$, $(1 - d\sigma)Dw + \sigma D\omega = (1 + d\sigma M/C)Dw + \sigma h_\perp$, and $h_\perp \perp Dw$. (b) At a peak k , $|W(\sigma)(k)| = (1 - d\sigma)M$, and $(1 - d\sigma)M + Y = M + C = 1$. (c) Off $\text{supp } \omega$, $|W(k)| = (1 - d\sigma)|w(k)|$; on $\text{supp } \omega$, $|W(k)| \leq (1 - d\sigma)(M - \text{gap}(k)) + \text{gap}(k)/2 \leq (1 - d\sigma)M$. (d) Here $Y \geq C/2$, the bracket in (c) is at most $\sigma^2 \|h_\perp\|^2 / (2Y)$, and $C/Y \leq 1/(1 - |d\sigma|M/C) \leq 1 + 2|d\sigma|M/C$. $\square$

Proposition 3.5 (Certificate expansion). *For a finite certificate c at f and $|\sigma| \leq r(c)$,*

$$p^*(f + \sigma g_c) \leq 1 + \frac{\sigma^2}{2} H(c)(1 + \kappa(c)|\sigma|).$$

Proof. $f + \sigma g_c = (a + \sigma b) + \sum_m R_m^* W_m(\sigma)$ with $W_m(\sigma) = (1 - d_m \sigma)w_m + \sigma \omega_m (= w_m$ on inactive blocks). Apply Lemma 1.6, Lemma 3.3(b) and Lemma 3.4(d). $\square$

Lemma 3.6 (Slack and validity window). (a) *For $\rho \in (0, 1)$ and real t , $s(t) - s(\rho t) \geq (1 - \rho^2) \min(t^2, |t|)/3$.* (b) *If $H(c) \leq 1 - \delta$ with $0 < \delta \leq 1$ and $|\sigma| \leq r_*(c, \delta) := \min(r(c), \delta/(2\kappa(c) + 2), \sqrt{\delta})$, then $p^*(f + \sigma g_c) \leq s(\sigma)$.*

Proof. (a) $s(t) - s(\rho t) = (1 - \rho^2)t^2/(s(t) + s(\rho t)) \geq (1 - \rho^2)t^2/(2s(t))$ and $2s(t) \leq 3 \max(1, |t|)$. (b) $(1 - \delta)(1 + \kappa|\sigma|) \leq (1 - \delta)(1 + \delta/2) \leq 1 - \delta/2$, and $1 + \frac{\sigma^2}{2}(1 - \frac{\delta}{2}) \leq 1 + \frac{\sigma^2}{2} - \frac{\sigma^4}{8} \leq s(\sigma)$ since $\sigma^2 \leq \delta$ and $\sqrt{1 + x} \geq 1 + \frac{x}{2} - \frac{x^2}{8}$ for $x \geq 0$. $\square$

Proposition 3.7 (Scale criterion). *Let $f, f' \in S_{p^*}$, $g \in \mathcal{C}(f)$, $\rho \in (0, 1)$, and let c' be a finite certificate at f' with $H(c') \leq 1 - \delta$, $0 < \delta \leq 1$. Put $r_* := r_*(c', \delta)$, $\varepsilon := p^*(f' - f)$ and $\eta := p^*(g_{c'} - \rho g)$. If $\varepsilon \leq (1 - \rho^2)r_*^2/6$ and $\eta \leq (1 - \rho^2)r_*/6$, then $g_{c'} \in \text{Cert}(f')$ and $\text{dist}_{p^*}(\rho g, \mathcal{C}(f')) \leq \eta$.*

Proof. For $|t| \leq r_*$ use Lemma 3.6(b) at f' . For $|t| \geq r_*$, $p^*(f' + t g_{c'}) \leq p^*(f + t \rho g) + \varepsilon + |t|\eta \leq s(\rho t) + \varepsilon + |t|\eta$; for $r_* \leq |t| \leq 1$ both ε and $|t|\eta$ are at most $(1 - \rho^2)t^2/6$, and for $|t| \geq 1$ their sum is at most $(1 - \rho^2)|t|/3$; conclude with Lemma 3.6(a). $\square$

Thus the slack alone certifies the scales $|t| \gtrsim \sqrt{\varepsilon/(1 - \rho^2)}$, and the smaller scales must be certified by the local structure of f' ; this is the multi-scale nature of the problem.

3.2. Transport along every sequence.

Theorem 3.8 (Transport). *Let $f_n \rightarrow f$ in S_{p^*} be an arbitrary sequence (norm attaining or not) and let $c = (b, \omega)$ be a finite certificate at f . Put*

$$G_n := \{j \in F : \text{sign } a_n(j) = \text{sign } a_j, |a_n(j)| \geq |a_j|/2\},$$

$$b'_n := b \mathbf{1}_{G_n}, \quad \tau_n := b'_n(\hat{z}_n), \quad b_n := b'_n - \tau_n a_n,$$

and $c_n := (b_n, \omega)$ (the same block coefficients; $d_{n,m}$ is computed at f_n). Then for n large c_n is a finite certificate at f_n , $\|g_{c_n} - g_c\| \rightarrow 0$, $H(c_n) \rightarrow H(c)$, $\kappa(c_n) \rightarrow \kappa(c)$ and $\liminf_n r(c_n) \geq r(c)/2$. If moreover $g_c \in \mathcal{C}(f)$ and $H(c) \leq 1$, then for every $\rho \in (0, 1)$, $\rho g_{c_n} \in \text{Cert}(f_n)$ for all large n .

Proof. Base. By Proposition 1.11, $a_n \rightarrow a$ in ℓ_1 , $\nu_n \rightarrow \nu$, $e_n \rightarrow e$ and $z_n \rightarrow z$ weak*; also $z_n = \text{sign } a_n$ on $\text{supp } a_n$. Every $j \in F$ eventually lies in G_n , so $b'_n \rightarrow b$ in ℓ_1 by dominated convergence. On G_n , $z_n(j) = \text{sign } a_n(j) = \text{sign } a_j = z_j$, hence $\tau_n = \sum_{j \in G_n} b_j z_j + \langle U^* b'_n, e_n \rangle \rightarrow b(z) + \langle U^* b, e \rangle = b(\hat{z}) = 0$, and $b_n \rightarrow b$. Further $\text{supp } b_n \subseteq \text{supp } a_n$, $\|b_n/a_n\|_\infty \leq 2\|b/a\|_\infty + |\tau_n|$, and $b_n(\hat{z}_n) = \tau_n - \tau_n a_n(\hat{z}_n) = 0$. Thus (C1) holds at f_n , $h_n(b_n) \rightarrow h(b)$ and $\|U^* b_n\| \rightarrow \beta$.

Blocks. For the finitely many active m and $k \in \text{supp } \omega_m$, $w_{n,m}(k) \rightarrow w_m(k)$ and $M_{n,m} \rightarrow M_m$, so $M_{n,m} - |w_{n,m}(k)| \rightarrow \text{gap}_m(k) > 0$: eventually $\text{supp } \omega_m \subseteq Q_{n,m}$ and the gaps converge. Also $d_{n,m} \rightarrow d_m$, $C_{n,m} \rightarrow C_m$, $D_m w_{n,m} \rightarrow D_m w_m$, so $H(c_n) \rightarrow H(c)$, $\kappa(c_n) \rightarrow \kappa(c)$ and $\liminf r(c_n) \geq r(c)/2$ (the base ratio is at most halved).

Directions. $g_{c_n} - g_c = (b_n - b) + \sum_m (d_m R_m^* w_m - d_{n,m} R_m^* w_{n,m}) \rightarrow 0$, since $R_m^* w_{n,m} \rightarrow R_m^* w_m$ in norm.

Mates. Let $\delta := (1 - \rho^2)/2$. Then $H(\rho c_n) = \rho^2 H(c_n) \rightarrow \rho^2 H(c) \leq 1 - 2\delta$, so $H(\rho c_n) \leq 1 - \delta$ for large n , and $r_*(\rho c_n, \delta)$ is bounded below by some $r_0 > 0$. Apply Proposition 3.7 with $f' = f_n$, $c' = \rho c_n$ and $g = g_c$: $p^*(f_n - f) \rightarrow 0$ and $p^*(g_{\rho c_n} - \rho g_c) = \rho p^*(g_{c_n} - g_c) \rightarrow 0$. $\square$

Corollary 3.9 (Lower semicontinuity of the certified fibre). *For every sequence $f_n \rightarrow f$ in S_{p^*} ,*

$$\text{cl Cert}(f) \subseteq \text{Li}_n \text{cl Cert}(f_n) \subseteq \text{Li}_n \mathcal{C}(f_n).$$

Consequently: (a) every $g \in \text{cl Cert}(f)$ is recovered along every norm-attaining sequence $f_n \rightarrow f$, finitely many such mates simultaneously, and (f, g) lies in $\text{cl NA}((X, p), \ell_2^2)$; (b) if $\mathcal{C}(f) = \text{cl Cert}(f)$, then $f \in \mathcal{R}$ and $\mathcal{C}$ is Hausdorff continuous at f ; (c) if for every f the set $\mathcal{C}(f) \setminus \text{cl Cert}(f)$ is contained in $\text{Li}_n \mathcal{C}(f_n)$ for some norm-attaining sequence $f_n \rightarrow f$, then $\text{NA}((X, p), \ell_2^2)$ is dense.

Proof. By Theorem 3.8, $\rho \text{Cert}(f) \subseteq \text{Li}_n \text{Cert}(f_n)$ for each $\rho < 1$; lower limits are closed. (a) follows from Lemma 1.16 and Corollary 2.3 (applied to finitely many mates). (b) Upper semicontinuity (Theorem 2.1) and $\mathcal{C}(f) \subseteq \text{Li}_n \mathcal{C}(f_n)$ give Kuratowski, hence Hausdorff, convergence inside a compact set. (c) Then $\mathcal{C}(f) \subseteq \text{Li}_n \mathcal{C}(f_n)$ for that sequence (the lower limit is closed and convex); apply Corollary 2.3. $\square$

Corollary 3.10 (Base and block directions). (i) *If ω is finitely supported strictly below the peak of w_m , $d = \langle D_m w_m, D_m \omega \rangle / C_m$, $g = R_m^*(\omega - d w_m) \in \mathcal{C}(f)$ and $H_m(\omega) \leq 1$, then $g \in \text{Cert}(f)$; in particular (f, g) is recovered along every sequence. This is the finite local-certificate theorem quoted from the unavailable note [9] in [7].* (ii) *If b satisfies (C1), then $b/\Lambda \in \text{Cert}(f)$ for every*

$$\Lambda \geq \max(1, \sqrt{2h(b)}, p^*(b)r_0/(s(r_0)-1)), \quad r_0 := \min(r(c), 1/(4\kappa(c)+4), 1/\sqrt{2}),$$

where $c = (b, 0)$.

Proof. (i) is the definition of $\text{Cert}(f)$. (ii) $H(c/\Lambda) = h(b)/\Lambda^2 \leq \frac{1}{2}$, $\kappa(c/\Lambda) \leq \kappa(c)$ and $r(c/\Lambda) \geq r(c)$, so $r_*(c/\Lambda, \frac{1}{2}) \geq r_0$ and Lemma 3.6(b) gives $p^*(f + \sigma b/\Lambda) \leq s(\sigma)$ for $|\sigma| \leq r_0$. For $|\sigma| \geq r_0$, $p^*(f + \sigma b/\Lambda) \leq 1 + |\sigma|p^*(b)/\Lambda \leq s(\sigma)$, since $(s(\sigma) - 1)/|\sigma|$ is increasing in $|\sigma|$. $\square$

3.3. Shifted certificates. Since $f = a + L^*w$, one may move a second-order amount of mass between the base and the blocks: $f = \lambda(a + L^*w) - (\lambda - 1)f$ with $\lambda = 1 + \theta\tau^2$.

Definition 3.11 (Shifted certificates). A *shifted certificate* is $c = (b, \omega, \theta)$, where (b, ω) is a finite certificate and $\theta = (\theta_m)$ is a finitely supported real family. Put $v_\theta := \sum_m \theta_m R_m^* w_m \in \ell_1$, $\kappa_q(v) := \sum_{j \notin F} (|v_j| + z_j v_j) \in [0, 2\|v\|_1]$ and

$$H^{\text{sh}}(c) := \max \left(h(b) - \frac{2}{q_0} \sum_m \theta_m \sigma_m + 2\kappa_q(v_\theta), \max_m (H_m(\omega_m) + 2\theta_m) \right).$$

The direction is $g_c := g_{(b, \omega)}$, and $\text{Cert}^{\text{sh}}(f) := \{g_c \in \mathcal{C}(f) : H^{\text{sh}}(c) \leq 1\}$.

Note $v_\theta(\hat{z}) = \sum_m \theta_m \langle w_m, \zeta_m \rangle / q_0 = \sum_m \theta_m \sigma_m / q_0$. The associated decomposition is

$$(4) \quad \begin{aligned} A(\tau) &:= a + \tau b - \tau^2 v_\theta, & W_m(\tau) &:= (1 + \theta_m \tau^2) w_m + \tau(\omega_m - d_m w_m), \\ A(\tau) + L^* W(\tau) &= f + \tau g_c. \end{aligned}$$

For $\rho > 0$ put $\rho.c := (\rho b, \rho \omega, \rho^2 \theta)$; then $g_{\rho.c} = \rho g_c$ and $H^{\text{sh}}(\rho.c) = \rho^2 H^{\text{sh}}(c)$. H^{sh} is jointly convex, so $\text{Cert}^{\text{sh}}(f)$ is convex, symmetric and contains $\text{Cert}(f)$ ($\theta = 0$).

Proposition 3.12 (Shifted expansion). *For a shifted certificate c at f there is $\varepsilon_c(\tau) \rightarrow 0$ with $p^*(f + \tau g_c) \leq 1 + \frac{\tau^2}{2}(H^{\text{sh}}(c) + \varepsilon_c(\tau))$.*

Proof. Blocks. $W_m(\tau) = (1 + \theta_m \tau^2)[w_m + \tau'(\omega_m - d_m w_m)]$ with $\tau' = \tau/(1 + \theta_m \tau^2)$, so Lemma 3.4(d) gives $N_m(W_m(\tau)) \leq 1 + \tau^2(\theta_m + H_m/2) + O(|\tau|^3)$. *Base.* Apply (3) with $t = \tau$, $B = b - \tau v_\theta$: $\tau B(\hat{z}) = -\tau^2 v_\theta(\hat{z})$; off F , $|\tau B_j| - z_j \tau B_j = \tau^2(|v_j| + z_j v_j)$; for $|\tau| \leq 1/(2\|b/a\|_\infty)$ each flip term is at most $2(\tau^2|v_j| - |a_j|/2)_+$, so $\text{Fl} \leq 2\tau^2 \sum_{j \in F, |v_j| > |a_j|/(2\tau^2)} |v_j| = o(\tau^2)$; and $\nu \Psi(\tau U^* B / \nu) = \frac{\tau^2}{2}(h(b) + O(\tau))$. Hence $q^*(A(\tau)) \leq 1 + \frac{\tau^2}{2}(h(b) - 2v_\theta(\hat{z}) + 2\kappa_q(v_\theta)) + o(\tau^2)$. Conclude with Lemma 1.6. $\square$

Theorem 3.13 (Shifted transport). *Let $f_n \rightarrow f$ in S_{p^*} be arbitrary, $c = (b, \omega, \theta)$ a shifted certificate at f , and $c_n := (b_n, \omega, \theta)$ with b_n as in Theorem 3.8. For every $\eta > 0$ there are $\tau_\eta > 0$ and n_η with $p^*(f_n + \tau g_{c_n}) \leq 1 + \frac{\tau^2}{2}(H^{\text{sh}}(c) + \eta)$ for $n \geq n_\eta$, $|\tau| \leq \tau_\eta$; moreover $\limsup_n H_n^{\text{sh}}(c_n) \leq H^{\text{sh}}(c)$ (computed at f_n). Consequently, if $g_c \in \mathcal{C}(f)$ and $H^{\text{sh}}(c) \leq 1$, then $\rho g_{c_n} \in \text{Cert}^{\text{sh}}(f_n)$ for every $\rho < 1$ and n large, and*

$$\text{cl } \text{Cert}^{\text{sh}}(f) \subseteq \text{Li } \text{cl } \text{Cert}^{\text{sh}}(f_n) \subseteq \text{Li } \mathcal{C}(f_n) \quad \text{for every sequence } f_n \rightarrow f.$$

Proof. The blocks are treated as in Proposition 3.12, with the uniform bounds of Theorem 3.8. For the base let $v_n := \sum_m \theta_m R_m^* w_{n,m} \rightarrow v_\theta$ in ℓ_1 and apply (3) at f_n with $B = b_n - \tau v_n$. For $|\tau| \leq 1/(2\|b_n/a_n\|_\infty)$ and $|\tau \tau_n| \leq 1$ one checks coordinatewise: for $j \in G_n$ the flip term is at most $2\tau^2|v_{n,j}|1[|v_{n,j}| > |a_j|/(4\tau^2)]$; for $j \in \text{supp } a_n \setminus G_n$ one has $b_n(j) = -\tau_n a_n(j)$ and the flip term is at most $\tau^2(|v_{n,j}| + z_n(j)v_{n,j})$; off $\text{supp } a_n$ the kink term equals $\tau^2(|v_{n,j}| + z_n(j)v_{n,j})$. Hence the flip and kink terms are at most τ^2 times

$$2 \sum_{j \in F} |v_{n,j}| 1[|v_{n,j}| > \frac{|a_j|}{4\tau^2}] + \sum_{j \notin F} (|v_{n,j}| + z_n(j)v_{n,j}) + 2 \sum_{j \in F \setminus G_n} |v_{n,j}|.$$

As $n \rightarrow \infty$ and $\tau \rightarrow 0$ the first and third sums tend to 0 (compare v_n with v_θ in ℓ_1 , use dominated convergence and that each $j \in F$ eventually lies in G_n), and the second tends to $\kappa_q(v_\theta)$ ($z_n \rightarrow z$ coordinatewise, $|z_n| \leq 1$). Also $v_n(\hat{z}_n) \rightarrow$

$v_\theta(\hat{z})$, $\sigma_{n,m} \rightarrow \sigma_m$, $q_0^{(n)} \rightarrow q_0$, and the Hilbert term is $\frac{\tau^2}{2}(h_n(b_n) + O(\tau))$ with $h_n(b_n) \rightarrow h(b)$. This gives the uniform estimate, and the same computation without τ gives the lim sup of $H_n^{\text{sh}}(c_n)$. For the mates, fix $\rho < 1$ and $\eta \leq (1 - \rho)/\rho$; for $|\tau| \leq t_1 := \min(\tau_\eta/\rho, 2\sqrt{1 - \rho})$ we get $p^*(f_n + \tau \rho g_{c_n}) \leq 1 + \rho \tau^2/2 \leq s(\tau)$, and for $|\tau| \geq t_1$ we use the slack (Lemma 3.6(a)) with $p^*(f_n - f) \rightarrow 0$ and $p^*(g_{c_n} - g_c) \rightarrow 0$. Finally $H_n^{\text{sh}}(\rho \cdot c_n) = \rho^2 H_n^{\text{sh}}(c_n) \leq 1$ for large n . $\square$

Remark 3.14. Shifts relax the condition $H \leq 1$ to $H^{\text{sh}} \leq 1$. Whether $\text{cl Cert}^{\text{sh}}(f)$ is strictly larger than $\text{cl Cert}(f)$ for some f in Martín's space is *open*: no mate with $H > 1 \geq H^{\text{sh}}$ outside $\text{cl Cert}(f)$ has been exhibited (finite-dimensional models suggest that shifts help, but they cannot distinguish $\text{Cert}(f)$ from its closure). Section 5 identifies the sharp second-order invariant at tame supports, Γ_w , and one checks that $\min_\theta H^{\text{sh}}(b, \omega, \theta) \geq \Gamma_w$ (Remark 5.10).

3.4. Several rows.

Theorem 3.15 (Joint fibres). *Let $c_1, \dots, c_d$ be finite certificates at f with $\mathbf{g} := (g_{c_1}, \dots, g_{c_d}) \in \mathcal{C}_d(f)$ and $\sup_{\theta \in S^{d-1}} H(\sum_i \theta_i c_i) \leq 1$. Then for every sequence $f_n \rightarrow f$ in S_{p^*} and every $\rho < 1$ the transported tuple $\rho \mathbf{g}_n := (\rho g_{c_{1,n}}, \dots, \rho g_{c_{d,n}})$ lies in $\mathcal{C}_d(f_n)$ for n large, and $\mathbf{g}_n \rightarrow \mathbf{g}$. In particular, if the f_n attain their norms, then $(f, \rho \mathbf{g})$ is a limit of norm-attaining operators into ℓ_2^{d+1} .*

Proof. The construction of Theorem 3.8 is linear in c (the index sets G_n of Theorem 3.8 do not depend on b), so $c(\theta) := \sum_i \theta_i c_i$ is transported to $c_n(\theta) = \sum_i \theta_i c_{i,n}$. On S^{d-1} the coefficients $H(c_n(\theta))$ converge to $H(c(\theta))$ uniformly (they are maxima of quadratic forms in θ with convergent coefficients), $\kappa(c_n(\theta))$ is bounded, and $r(c_n(\theta)) \geq r_0 > 0$ uniformly for n large, because only the finitely many coordinates of $\bigcup_i \text{supp } \omega_i$ and the bounded quantities $\|b_{i,n}/a_n\|_\infty$, $\|U^* b_{i,n}\|$, $|d_{i,n,m}|$ enter. Now run the proof of Proposition 3.7 along each ray $t = |t|\theta$: for $|t| \leq r_0$ use Lemma 3.6(b) for $\rho c_n(\theta)$; for $|t| \geq r_0$ use $p^*(f + \rho \sum_i t_i g_{c_i}) \leq s(\rho|t|)$ and the slack in $|t|$. If f_n attains its norm at x_n , then each row of $\rho \mathbf{g}_n$ vanishes at x_n and $(f_n, \rho \mathbf{g}_n)$ attains its norm there. $\square$

4. AVERAGING OVER SCALES

4.1. The averaging theorem.

Theorem 4.1 (Weighted averaging). *Let $g \in \mathcal{C}(f)$ and $\rho \in (0, 1)$. Let $c_1, \dots, c_n$ be finite certificates at f and $\mu_1, \dots, \mu_n \geq 0$ weights with $\sum_i \mu_i = 1$; put $r_i := r(c_i)$ and $\epsilon_i := p^*(g - g_{c_i})$. Let $s_0 \in (0, \min(1, \sqrt{1 - \rho^2})]$ and $\eta_0, \kappa_0 \geq 0$ satisfy $\rho^2(1 + \eta_0)(1 + \kappa_0) \leq (1 + \rho^2)/2$, and assume*

- (i) $H(c_i) \leq 1 + \eta_0$ and $\kappa(c_i) \min(r_i, s_0) \leq \kappa_0$ for all i ;
- (ii) $\sum_{i: r_i < \rho s} \mu_i \epsilon_i \leq (1 - \rho^2)s/(12\rho)$ for every $s \in (0, s_0]$;
- (iii) $\sum_i \mu_i \epsilon_i \leq (1 - \rho^2)s_0/(3\rho)$.

Then $c := \sum_i \mu_i c_i$ satisfies $\rho g_c \in \text{Cert}(f)$, and $p^(\rho g_c - \rho g) \leq \rho \sum_i \mu_i \epsilon_i$. Consequently, if for every $\rho < 1$ and $\varepsilon > 0$ such families exist with $\sum_i \mu_i \epsilon_i \leq \varepsilon$, then $g \in \text{cl Cert}(f)$.*

Proof. By linearity $g_c = \sum_i \mu_i g_{c_i}$, and by convexity of p^* we have $p^*(f + s \rho g_c) \leq \sum_i \mu_i p^*(f + s \rho g_{c_i})$ for real s . We use two bounds for each i . (A) If $\rho|s| \leq r_i$ and

$|s| \leq s_0$, Proposition 3.5 and (i) give $p^*(f + s\rho g_{c_i}) \leq 1 + \frac{\rho^2 s^2}{2}(1 + \eta_0)(1 + \kappa_0) \leq 1 + \frac{s^2}{4}(1 + \rho^2)$, since $\kappa(c_i)\rho|s| \leq \kappa(c_i)\min(r_i, s_0)$. (B) Always, since $g \in \mathcal{C}(f)$,

$$p^*(f + s\rho g_{c_i}) \leq p^*(f + s\rho g) + \rho|s|\epsilon_i \leq s(\rho s) + \rho|s|\epsilon_i.$$

Case $|s| \leq s_0$. Use (A) for $r_i \geq \rho|s|$ and (B) otherwise. By (ii),

$$p^*(f + s\rho g_c) \leq \max\left(1 + \frac{s^2}{4}(1 + \rho^2), s(\rho s)\right) + \frac{(1 - \rho^2)s^2}{12}.$$

Now $1 + \frac{s^2}{4}(1 + \rho^2) + \frac{(1 - \rho^2)s^2}{12} = 1 + \frac{s^2}{6}(2 + \rho^2) = 1 + \frac{s^2}{2} - \frac{(1 - \rho^2)s^2}{6} \leq 1 + \frac{s^2}{2} - \frac{s^4}{8} \leq s(s)$ because $s^2 \leq 1 - \rho^2$; and $s(\rho s) + (1 - \rho^2)s^2/12 \leq s(s)$ by Lemma 3.6(a).

Case $|s| \geq s_0$. Use (B) for every i and (iii): $p^*(f + s\rho g_c) \leq s(\rho s) + (1 - \rho^2)s_0|s|/3 \leq s(s)$, because $\min(s^2, |s|) \geq s_0|s|$.

Thus $\rho g_c \in \mathcal{C}(f)$. By convexity of H , $H(\rho c) \leq \rho^2 \max_i H(c_i) \leq \rho^2(1 + \eta_0) \leq 1$, so $\rho g_c \in \text{Cert}(f)$. The distance estimate is the triangle inequality. $\square$

The mechanism is that at each scale s only the certificates that are too fine for s contribute an error, of order s each, and these errors are summable against the weights.

Corollary 4.2 (Ladder form). *Let $g \in \mathcal{C}(f)$. Suppose there are $c_0 > 0$, $\kappa_1, K < \infty$, $t_* > 0$ and a function $\eta(t) \rightarrow 0$ ($t \rightarrow 0^+$) such that for every $t \in (0, t_*]$ there is a finite certificate c_t at f with*

$$H(c_t) \leq 1 + \eta(t), \quad r(c_t) \geq c_0 t, \quad \kappa(c_t) \leq \kappa_1, \quad p^*(g - g_{c_t}) \leq Kt.$$

Then $g \in \text{cl Cert}(f)$; hence g is recovered along every sequence $f_n \rightarrow f$ (Corollary 3.9). It suffices that the hypotheses hold for the scales $t = t_ 2^{1-i}$, $i \geq 1$.*

Proof. Fix $\rho < 1$; choose $\eta_0, \kappa_0 > 0$ with $\rho^2(1 + \eta_0)(1 + \kappa_0) \leq (1 + \rho^2)/2$, then $s_0 \in (0, \min(1, \sqrt{1 - \rho^2})]$ with $\kappa_1 s_0 \leq \kappa_0$, an integer $n \geq 24\rho^2 K/(c_0(1 - \rho^2))$, and $t_1 \in (0, t_*]$ with $\eta(t) \leq \eta_0$ for $t \leq t_1$ and $2\rho K t_1/n \leq (1 - \rho^2)s_0/3$; in the case of the last sentence of the statement we take t_1 of the form $t_* 2^{1-i_0}$, so that all the scales used below are among the given ones. Put $t_i := t_1 2^{1-i}$, $c_i := c_{t_i}$ and $\mu_i := 1/n$ ($i \leq n$). Condition (i) holds. For (ii), $r_i < \rho s$ implies $c_0 t_i < \rho s$, and the t_i with this property sum to less than $2\rho s/c_0$, so the left side of (ii) is at most $(K/n)(2\rho s/c_0) \leq (1 - \rho^2)s/(12\rho)$. For (iii), $\sum_i \mu_i K t_i \leq 2K t_1/n$. Theorem 4.1 gives $\rho g_c \in \text{Cert}(f)$ within $2\rho K t_1/n$ of ρg ; let $t_1 \rightarrow 0$ and then $\rho \rightarrow 1$. $\square$

Corollary 4.3 (Decomposition form). *Let $g \in \mathcal{C}(f)$. Suppose that for every small $t > 0$ there are a finite certificate $c_t = (b_t, \omega_t)$ with $r(c_t) \geq c_0 t$ and $\kappa(c_t) \leq \kappa_1$, and $e_t := g - g_{c_t}$ with $p^*(e_t) \leq Kt$, such that for $\tau = t$ or for $\tau = -t$ the decomposition*

$$f + \tau g = (a + \tau b_t + \tau e_t) + \sum_m R_m^*((1 - d_{t,m}\tau)w_m + \tau \omega_{t,m})$$

has all component norms at most $s(\tau) + o(\tau^2)$. Assume moreover that $t^2/C_m \rightarrow 0$ uniformly over the active blocks of c_t (this is automatic if I is finite). Then $H(c_t) \leq 1 + o(1)$, and Corollary 4.2 gives $g \in \text{cl Cert}(f)$.

Proof. *Blocks.* Since $\kappa(c_t) \leq \kappa_1$, $|d_{t,m}| \leq \kappa_1 C_m/(2M_m)$, so $Y = C_m + d_{t,m}\tau M_m = C_m(1 + O(t))$ and $1 - d_{t,m}\tau > 0$ for small t . Put $X := |\tau||h_\perp|$. Lemma 3.4(b) gives $\sqrt{Y^2 + X^2} - Y \leq \delta := s(\tau) - 1 + o(\tau^2)$, hence $X^2 \leq \delta(2Y + \delta)$ and

$$H_m(\omega_{t,m}) = \frac{\|h_\perp\|^2}{C_m} \leq \frac{\delta}{\tau^2} \left(\frac{2Y}{C_m} + \frac{\delta}{C_m} \right) = 1 + o(1) + O(\tau^2/C_m).$$

Base. $B := b_t + e_t$ satisfies $B(\hat{z}) = 0$, because $g(\xi) = 0$ and $g_{e_t}(\xi) = 0$ (Lemma 3.2). Also $\|U^*b_t\| \leq \kappa_1\nu/2$ and $\|e_t\|_1 \leq q^*(e_t) \leq (1 + \|L\|)p^*(e_t) \leq (1 + \|L\|)Kt$. Lemma 3.3(a) gives $h(B) \leq 1 + o(1)$, and since $\sqrt{h}$ is a seminorm and $h(e_t) \leq \|U\|^2\|e_t\|_1^2/\nu = O(t^2)$, also $h(b_t) \leq 1 + o(1)$. $\square$

Remark 4.4. For $I = \mathbb{N}$ the extra hypothesis in Corollary 4.3 cannot simply be dropped: $C_m \leq \|\Phi_m\|_2 \leq 2^{-m}$, and a block with $C_m \ll \tau^2$ can remain admissible at scale τ with $\|h_\perp\| \approx \tau/2 + C_m/\tau$, i.e. with $H_m \approx \tau^2/(4C_m) \gg 1$. Corollary 4.2 itself has no such restriction.

4.2. Weighted directions. For a block m , a function $\omega : \mathbb{N} \rightarrow \mathbb{R}$ vanishing on P_m and $\sigma > 0$, put

$$E_m(\sigma; \omega) := \{k \in Q_m : \sigma|\omega(k)| > \text{gap}_m(k)/2\},$$

$$T_m(\sigma; \omega) := \sum_{k \in E_m(\sigma; \omega)} \lambda_{k,m}|\omega(k)|.$$

Lemma 4.5 (Box tails). $T_m(\sigma; \omega) = o(\sigma)$ as $\sigma \rightarrow 0$ in each of the following cases: (a) $\sum_k \lambda_{k,m}\omega(k)^2/\text{gap}_m(k) < \infty$; (b) ω is bounded and $|\omega(k)| \leq K\sqrt{\text{gap}_m(k)}$ whenever $\text{gap}_m(k) \leq \gamma_0$, for some $K, \gamma_0 > 0$; (c) $\text{gap}_m \geq M_m/2$ on $\text{supp } \omega$ and $\sum_k \lambda_{k,m}\omega(k)^2 < \infty$ (the weighted directions of the weighted-directions theorem of [7]).

Proof. (a) On $E_m(\sigma)$, $\lambda_k|\omega(k)| < 2\sigma\lambda_k\omega(k)^2/\text{gap}(k)$, and the sets $E_m(\sigma)$ decrease to $\emptyset$ as $\sigma \downarrow 0$; use dominated convergence. (b) If $\sigma\|\omega\|_\infty < \gamma_0/2$, then $k \in E_m(\sigma)$ forces $\text{gap}(k) \leq \gamma_0$, hence $\text{gap}(k) < 2\sigma K\sqrt{\text{gap}(k)}$, i.e. $\text{gap}(k) < 4K^2\sigma^2$ and $|\omega(k)| < 2K^2\sigma$; so $T_m(\sigma) \leq 2K^2\sigma \sum_{\text{gap}(k) < 4K^2\sigma^2} \lambda_k = o(\sigma)$. (c) is a case of (a). $\square$

Theorem 4.6 (Weighted directions). Let $b \in \ell_1$ with $\text{supp } b \subseteq F$, $b(\hat{z}) = 0$ and $\sum_{j \in F, |b_j| > |a_j|/(2t)} |b_j| = O(t)$ (e.g. $\sum_{j \in F} b_j^2/|a_j| < \infty$). For finitely many blocks m let $\omega_m : \mathbb{N} \rightarrow \mathbb{R}$ vanish on P_m , with $\sum_k \lambda_{k,m}|\omega_m(k)| < \infty$ and $T_m(\sigma; \omega_m) = O(\sigma)$. Put $d_m := \langle D_m w_m, D_m \omega_m \rangle / C_m$, $g := b + \sum_m R_m^*(\omega_m - d_m w_m)$ and $H := \max(h(b), \max_m H_m(\omega_m))$. If $g \in \mathcal{C}(f)$ and $H \leq 1$, then $g \in \text{cl Cert}(f)$; in particular g is recovered along every sequence $f_n \rightarrow f$.

Proof. Since $\Phi_m \leq \lambda_{\cdot,m}$, $D_m \omega_m \in \ell_1 \subseteq \ell_2$. For small $t > 0$ choose $J_t \rightarrow \infty$ so large that the tails of b and of $\lambda_{\cdot,m}\omega_m$ beyond J_t have ℓ_1 -norm at most t , and put

$$G_t := F \cap [1, J_t] \cap \{|b_j| \leq |a_j|/(2t)\}, \quad b_t := b1_{G_t} - \varkappa_t a, \quad \varkappa_t := (b1_{G_t})(\hat{z}),$$

$$\omega_{m,t} := \omega_m 1_{\{k \leq J_t\} \setminus E_m(t; \omega_m)}.$$

Then $c_t := (b_t, (\omega_{m,t}))$ is a finite certificate. Since $|\varkappa_t| = |(b1_{G_t})(\hat{z})| \leq q^*(b1_{F \setminus G_t}) \leq (1 + \|U\|)\|b1_{F \setminus G_t}\|_1 = O(t)$, we get $\|b_t/a\|_\infty \leq 1/(2t) + O(t)$, so the base part of $r(c_t)$ is at least t for small t ; the block coordinates kept satisfy $t|\omega_{m,t}(k)| \leq \text{gap}_m(k)/2$, and the remaining terms of the radius are bounded below (the $d_{m,t}$ and $\|U^*b_t\|$ are bounded). Hence $r(c_t) \geq t$ and $\kappa(c_t)$ is bounded. The remainder is

$$g - g_{c_t} = (b - b_t) + \sum_m R_m^*((\omega_m - \omega_{m,t}) - (d_m - d_{m,t})w_m),$$

with $\|b - b_t\|_1 \leq \|b1_{F \setminus G_t}\|_1 + |\varkappa_t| = O(t)$, $\sum_k \lambda_{k,m}|\omega_m - \omega_{m,t}|(k) \leq T_m(t; \omega_m) + t = O(t)$ and $\|d_m - d_{m,t}\| \leq \|D_m(\omega_m - \omega_{m,t})\|_2 = O(t)$. So $p^*(g - g_{c_t}) \leq Kt$. Finally

$b_t \rightarrow b$ in ℓ_1 and $D_m \omega_{m,t} \rightarrow D_m \omega_m$ in ℓ_2 , so $H(c_t) \rightarrow H \leq 1$. Corollary 4.2 applies. $\square$

By Lemma 4.5(c), Theorem 4.6 contains the weighted-directions theorem of [7] (and removes its dependence on [9]); near-peak supports are allowed as long as the box tails are $O(\sigma)$. In particular the non-attaining examples of [7, Section 3], which admit no bounded first-order lift, are recovered along every sequence.

4.3. Locally split mates.

Definition 4.7. A mate $g \in \mathcal{C}(f)$ is *locally split* if there are $\tau > 0$, $b \in X^*$ and $v \in V^*$ with $g = b + L^*v$ and, for $|t| \leq \tau$,

$$q^*(a + tb) \leq s(t) \quad \text{and} \quad \|w + tv\|_{V^*} \leq s(t).$$

Equivalently, the linear path $f + tg = (a + tb) + L^*(w + tv)$ is admissible at level $s(t)$ near $t = 0$. Every finite certificate direction g_c with $H(c) < 1$ is locally split: by Lemma 3.3(b), Lemma 3.4(d) and the argument of Lemma 3.6(b), each component is at most $s(t)$ for $|t| \leq r_*(c, 1 - H(c))$. (For $H(c) = 1$ this can fail: the expansion only gives $1 + \frac{t^2}{2}(1 + \kappa(c)|t|)$, which exceeds $s(t)$ when $\kappa(c) > 0$.) The converse fails, since b and v may have infinite supports.

Lemma 4.8 (Base truncation with exact first-order correction). *Let $b \in \ell_1$ satisfy $q^*(a + tb) \leq s(t)$ for $|t| \leq \tau$. Then $\text{supp } b \subseteq F$ and $b(\hat{z}) = 0$. Put $\tau_a := \frac{1}{4} \min(\tau, 1, \nu/(2\|U^*b\| + 1))$. For every $\delta > 0$ there is N such that $b'' := P_N b + \beta_N a$, with P_N the coordinate projection onto $[1, N]$ and $\beta_N := ((I - P_N)b)(\hat{z})$, satisfies $b''(\hat{z}) = 0$, $\|b'' - b\|_1 < \delta$, and $q^*(a + tb'') \leq s(t) + \delta t^2$ for $|t| \leq \tau_a$.*

Proof. First order. The function $\varphi(t) = q^*(a + tb)$ is convex with $\varphi(0) = 1$ and $\varphi(t) \leq 1 + t^2/2$ on $[-\tau, \tau]$, so both one-sided derivatives at 0 vanish. The right derivative is $\sum_{j \in F} \text{sign}(a_j)b_j + \sum_{j \notin F} |b_j| + \langle U^*b, e \rangle$ and the left derivative is $\sum_{j \in F} \text{sign}(a_j)b_j - \sum_{j \notin F} |b_j| + \langle U^*b, e \rangle$; hence $\sum_{j \notin F} |b_j| = 0$ and $b(\hat{z}) = 0$.

Truncation. Let $r := (I - P_N)b$ and $\varepsilon_N := \|U^*r\|$; note that $\beta_N = r(\hat{z}) = \sum_{j > N} z_j r_j + \langle U^*r, e \rangle$ tends to 0. For $|s| \leq 2\tau_a$ and $j > N$ in F , $|a_j| \leq |a_j + sr_j| - sz_j r_j$; hence $\|a + sP_N b\|_1 \leq \|a + sb\|_1 - s \sum_{j > N} z_j r_j$. For the Hilbert part we use the elementary inequality

$$(5) \quad \|y - x\| \leq \|y\| - \langle y, x \rangle / \|y\| + \|x\|^2 / (2\|y\|) \quad (y \neq 0),$$

which follows from $(\langle y, x \rangle / \|y\|^2 - \|x\|^2 / (2\|y\|^2))^2 \geq 0$. With $y_s := U^*(a + sb)$ (so $\|y_s\| \geq \frac{3}{4}\nu$ and $\|y_s\|/y_s - e\| \leq 2|s|\|U^*b\|/\nu$) and $x = sU^*r$, $\|U^*(a + sP_N b)\| \leq \|y_s\| - s\langle e, U^*r \rangle + C_2 s^2 \varepsilon_N$, with C_2 depending only on $\nu, \|U^*b\|$. Adding,

$$q^*(a + sP_N b) \leq q^*(a + sb) - s\beta_N + C_2 s^2 \varepsilon_N.$$

Now $a + tb'' = (1 + t\beta_N)(a + sP_N b)$ with $s = t/(1 + t\beta_N)$; for N large ($|\beta_N| \leq 1$) and $|t| \leq \tau_a$ we have $|s| \leq 2\tau_a$, and

$$q^*(a + tb'') \leq (1 + t\beta_N)s(s) - t\beta_N + C_3 t^2 \varepsilon_N = 1 + F_t(\beta_N) + C_3 t^2 \varepsilon_N,$$

where $F_t(\beta) := \sqrt{(1 + t\beta)^2 + t^2} - (1 + t\beta)$. Since $F_t(0) = s(t) - 1$ and $|\partial_\beta F_t| = |t|(1 + t\beta)/\sqrt{(1 + t\beta)^2 + t^2} - 1| \leq 2|t|^3$ for $|t\beta| \leq \frac{1}{2}$, we get $q^*(a + tb'') \leq s(t) + 2|\beta_N||t|^3 + C_3 t^2 \varepsilon_N \leq s(t) + \delta t^2$ for N large. Finally $\|b'' - b\|_1 \leq \|r\|_1 + |\beta_N|\|a\|_1 \rightarrow 0$ and $b''(\hat{z}) = b(\hat{z}) - r(\hat{z}) + \beta_N = 0$. $\square$

Lemma 4.9 (Block truncation). *Fix a block (drop m) and let $v \in \ell_\infty$ satisfy $N(w + tv) \leq s(t)$ for $|t| \leq \tau$. Put $c := \langle Dw, Dv \rangle / C$, $d := c/M$ and $\omega := v + dw$. Then $\omega = 0$ on P , $|\text{sign}(w(k))v(k) + c| \leq \sqrt{2\text{gap}(k)}$ whenever $w(k) \neq 0$ and $\text{gap}(k) \leq \tau^2/2$, and $d = \langle Dw, D\omega \rangle / C$. Moreover there is $\tau_3 > 0$ such that for every $\delta > 0$ there is a finitely supported ω'' , vanishing on P , with $\min_{\text{supp } \omega''} \text{gap} > 0$ and $\langle Dw, D\omega'' \rangle = \langle Dw, D\omega \rangle$, such that $v'' := \omega'' - dw$ satisfies $\|R^*(v'' - v)\|_1 < \delta$ and $N(w + tv'') \leq N(w + tv) + \delta t^2$ for $|t| \leq \tau_3$.*

Proof. Structure. $\chi(t) := \|D(w + tv)\|$ is convex and differentiable at 0 with $\chi'(0) = c$, so $\chi(t) \geq C + tc$, and $\|w + tv\|_\infty \leq s(t) - \chi(t) \leq M - tc + t^2/2$ for $|t| \leq \tau$. For k with $w(k) \neq 0$ and $\varsigma_k := \text{sign } w(k)$ this gives $|w(k)| + t\varsigma_k v(k) \leq M - tc + t^2/2$, i.e. $t(\varsigma_k v(k) + c) \leq \text{gap}(k) + t^2/2$ for $|t| \leq \tau$. For a peak ($\text{gap} = 0$) both signs of t give $\varsigma_k v(k) = -c$; for $\text{gap}(k) \leq \tau^2/2$ take $t = \pm\sqrt{2\text{gap}(k)}$. Then $\omega(k) = \varsigma_k(\varsigma_k v(k) + c) = 0$ on P , and off P with $w(k) \neq 0$, $\omega(k) = \varsigma_k((\varsigma_k v(k) + c) - c\text{gap}(k)/M)$, which tends to 0 along every sequence with $\text{gap}(k) \rightarrow 0$; ω is bounded. Finally $\langle Dw, D\omega \rangle = cC + dC^2 = cC(M + C)/M = dC$.

Truncation. If $w(k) = 0$ for every $k \in Q$, then $\langle Dw, D\omega \rangle = 0$, $d = c = 0$, and we put $k_0 := \text{none}$, $\varkappa := 0$. Otherwise fix $k_0 \in Q$ with $w(k_0) \neq 0$, $\gamma_0 := \text{gap}(k_0)$. For $\gamma \in (0, \gamma_0)$ and $K \geq k_0$ let $S := \{k \leq K : \text{gap}(k) \geq \gamma\}$ (finite, contains k_0) and

$$\omega'' := \omega 1_S + \varkappa e_{k_0}, \quad \varkappa := \langle Dw, D(\omega - \omega 1_S) \rangle / (\Phi(k_0)^2 w(k_0)),$$

so that $\langle Dw, D\omega'' \rangle = \langle Dw, D\omega \rangle$. As $\gamma \rightarrow 0$ and $K \rightarrow \infty$, $\omega - \omega 1_S \rightarrow 0$ pointwise and boundedly ($\omega = 0$ on P), hence $\Delta := D(\omega'' - \omega) \rightarrow 0$ in ℓ_2 , $\varkappa \rightarrow 0$ and $\|R^*(\omega'' - \omega)\|_1 \leq \sum_k \lambda_k |\omega''(k) - \omega(k)| \rightarrow 0$.

Sup part. Let $W := w + tv = (1 - dt)w + t\omega$ and $W'' := w + tv'' = (1 - dt)w + t\omega''$. They agree on $S \setminus \{k_0\}$. For $k \notin S \cup \{k_0\}$, $|W''(k)| = |1 - dt| |w(k)| \leq |1 - dt| M = |W(k_*)|$ for any peak k_* ($\omega(k_*) = 0$), so $|W''(k)| \leq \|W\|_\infty$. At k_0 , $|W''(k_0)| \leq |1 - dt|(M - \gamma_0) + |t|(\|\omega\|_\infty + 1) \leq |1 - dt|M$ for $|t| \leq t_0 := \gamma_0/(4(|d|M + \|\omega\|_\infty + 2))$ (once $|\varkappa| \leq 1$). Hence $\|W''\|_\infty \leq \|W\|_\infty$ for $|t| \leq t_0$.

Hilbert part. Put $A_t := DW$. Since $\langle Dw, \Delta \rangle = 0$, $\langle A_t, \Delta \rangle = t \langle D\omega, \Delta \rangle$ and

$$\|A_t + t\Delta\|^2 = \|A_t\|^2 + 2t^2 \langle D\omega, \Delta \rangle + t^2 \|\Delta\|^2, \quad \text{so}$$

$$\|A_t + t\Delta\| \leq \|A_t\| + \frac{2t^2(\|D\omega\| \|\Delta\| + \|\Delta\|^2)}{C}$$

for $|t| \leq t_1 := C/(2\|D(\omega - dw)\| + 2)$ (then $\|A_t\| \geq C/2$). Adding, $N(W'') \leq N(W) + o(1)t^2$ on $|t| \leq \tau_3 := \min(\tau, t_0, t_1)$, where $o(1) \rightarrow 0$ as $\gamma \rightarrow 0$, $K \rightarrow \infty$. $\square$

Theorem 4.10 (Locally split mates). *Let the block set I be finite. Every locally split mate $g \in \mathcal{C}(f)$ lies in $\text{cl Cert}(f)$; in particular $(f, g) \in \text{cl NA}((c_0, p), \ell_2^2)$ and g is recovered along every sequence $f_n \rightarrow f$ in S_p^* .*

Proof. Let $g = b + L^*v$ be as in Definition 4.7. Fix $\rho \in (0, 1)$. Let τ_a be the range of Lemma 4.8 for b and $\tau_{3,m}$ the range of Lemma 4.9 for v_m , and put $\tau_5 := \min(\tau_a, \min_{m \in I} \tau_{3,m}) > 0$; since I is finite and these ranges do not depend on the truncation accuracy, τ_5 does not depend on δ below. Fix $\delta > 0$ and apply Lemma 4.8 to b and Lemma 4.9 to each v_m , $m \in I$, with accuracy δ : we obtain a finite certificate $c'' = (b'', (\omega''_m)_{m \in I})$ (its d -coefficients are the d_m of Lemma 4.9, because $\langle D_m w_m, D_m \omega''_m \rangle = \langle D_m w_m, D_m \omega_m \rangle$ with $\omega_m := v_m + d_m w_m$ as in that lemma) with

$$p^*(g_{c''} - g) \leq q^*(g_{c''} - g) \leq (1 + \|U\|) \|g_{c''} - g\|_1 \leq \delta' := (1 + \|U\|)(|I| + 1)\delta,$$

and, for $|t| \leq \tau_5$,

$$q^*(a + tb'') \leq s(t) + \delta t^2, \quad N_m(w_m + t(\omega_m'' - d_m w_m)) \leq s(t) + \delta t^2.$$

Coefficient. Lemma 3.3(a) gives $\frac{t^2}{2} h(b'') / (1 + |t| \|U^* b''\| / \nu) \leq s(t) - 1 + \delta t^2$; dividing by t^2 and letting $t \rightarrow 0$, $h(b'') \leq 1 + 2\delta$. Similarly Lemma 3.4(b) gives $H_m(\omega_m'') \leq 1 + 2\delta$. So $H(c'') \leq 1 + 2\delta$.

Mate. For $|t| \leq \tau_5 / \rho$ we have

$$p^*(f + t\rho g_{c''}) \leq s(\rho t) + \delta \rho^2 t^2 \leq s(t) \quad \text{as soon as} \quad \delta \rho^2 \max(1, \tau_5 / \rho) \leq (1 - \rho^2) / 3$$

(Lemma 3.6(a)). For $|t| \geq \tau_5 / \rho$ we have

$$p^*(f + t\rho g_{c''}) \leq s(\rho t) + \rho |t| \delta' \leq s(t) \quad \text{as soon as} \quad \rho \delta' \leq (1 - \rho^2) \min(1, \tau_5 / \rho) / 3.$$

Since τ_5 does not depend on δ and $\delta' \rightarrow 0$ with δ , both conditions hold for δ small; then $\rho g_{c''} \in \mathcal{C}(f)$ and $H(\rho c'') \leq \rho^2(1 + 2\delta) \leq 1$, i.e. $\rho g_{c''} \in \text{Cert}(f)$, at distance at most $\rho \delta'$ from ρg . Let $\delta \rightarrow 0$ and $\rho \rightarrow 1$. The last assertions follow from Corollary 3.9. $\square$

Corollary 4.11 (Split-contractive operators). *Let the block set I be finite. Let $S \in \mathcal{L}((c_0, q), \ell_2^2)$ and $\Phi \in \mathcal{L}(V, \ell_2^2)$ with $\|S\|_q \leq 1$, $\|\Phi\| \leq 1$, and suppose $A_0 := S + \Phi \circ L$ has $\|A_0\|_p = 1$. Then $A_0 \in \text{cl NA}((c_0, p), \ell_2^2)$.*

Proof. After a rotation $A_0 = (f, g)$ with $p^*(f) = 1$, $f = s_1 + L^* \varphi_1$, $g = s_2 + L^* \varphi_2$, where (s_1, s_2) is q -contractive and (φ_1, φ_2) is V -contractive. Then $q^*(s_1), \|\varphi_1\| \leq 1$, so by Proposition 1.9, $s_1 = a$ and $\varphi_1 = w$; the contractivity of the components gives $q^*(a + ts_2) \leq s(t)$ and $\|w + t\varphi_2\| \leq s(t)$ for all t . So g is locally split, and Theorem 4.10 applies. $\square$

Since the norm-attaining operators of the split form $S + \Phi \circ L$ are not dense [7, Section 3], the corollary shows that split operators are recovered by non-split norm-attaining ones.

4.4. Carriers with non-summable gaps.

Theorem 4.12 (Carrier mates). *Let $g \in \mathcal{C}(f)$, let $c^0 = (b^0, \omega^0)$ be a finite certificate, fix a block m and $c \in \mathbb{R} \setminus \{0\}$, and let $k_1, k_2, \dots$ be distinct strict non-peaks of w_m outside $\text{supp } \omega_m^0$, with $\lambda_i := \lambda_{k_i, m}$ and $\gamma_i := \text{gap}_m(k_i)$, such that*

- (a) $\gamma_{i+1} \lambda_{i+1} \leq \gamma_i \lambda_i / 2$ for all i ;
- (b) $p^*(g - g_{c^0} - c u_{k_i, m}) \leq K \lambda_i$ for all i ;
- (c) $H_* := \max(h(b^0), \max_{m' \neq m} H_{m'}(\omega_{m'}^0), H_m(\omega_m^0) + c^2 / (m^2 C_m)) \leq 1$;
- (d) $\sum_i \gamma_i = \infty$.

Then $g \in \text{cl Cert}(f)$. In particular this holds when $\inf_i \gamma_i > 0$, $\|u_{k_i, m} - v\| \leq K' \lambda_i$ for some v with $g = g_{c^0} + cv$, and (c) holds (after passing to a subsequence for which (a) holds): cross mates with rates through carriers whose gaps are bounded below are recovered along every sequence, provided (c) holds.

Proof. Let $c_i := c^0 + (0, (c/\lambda_i) e_{k_i})$ in block m . Its d -coefficient in block m is $d_i = d_m^0 + c \Phi_m(k_i) w_m(k_i) / (m C_m) \rightarrow d_m^0$, and $\|D_m \omega_i\|^2 = \|D_m \omega_m^0\|^2 + c^2 / m^2$ (because $\Phi_m(k_i) / \lambda_i = 1/m$), so $\limsup_i H(c_i) \leq H_* \leq 1$. The coordinatewise radius is $r_i = \min(r_*, \gamma_i \lambda_i / (2|c|))$ with $r_* > 0$ bounded below for large i , $\kappa(c_i)$ is bounded, and

$$\epsilon_i := p^*(g - g_{c_i}) \leq p^*(g - g_{c^0} - c u_{k_i, m}) + |d_i - d_m^0| p^*(R_m^* w_m) \leq K' \lambda_i.$$

Fix $\rho < 1$, choose η_0, κ_0, s_0 as in Theorem 4.1 with moreover $s_0 \sup_i \kappa(c_i) \leq \kappa_0$. For a stretch $i_0 \leq i \leq i_1$ put $\Gamma := \sum_{i_0}^{i_1} \gamma_i$ and $\mu_i := \gamma_i/\Gamma$. If $s \leq r_*/\rho$, then by (a)

$$\sum_{r_i < \rho s} \mu_i \epsilon_i \leq \frac{K'}{\Gamma} \sum_{\gamma_i \lambda_i < 2|c|\rho s} \gamma_i \lambda_i \leq \frac{4K'|c|\rho s}{\Gamma} \leq \frac{(1-\rho^2)s}{12\rho}$$

once $\Gamma \geq 48K'|c|\rho^2/(1-\rho^2)$, which is possible with i_0 arbitrarily large by (d). If $r_*/\rho < s \leq s_0$, then $\sum_i \mu_i \epsilon_i \leq K' \sup_{i \geq i_0} \lambda_i \leq (1-\rho^2)r_*/(12\rho^2)$ for i_0 large (note $\lambda_i \rightarrow 0$), which gives (ii) as well; (iii) holds similarly, and (i) holds for i_0 large because $\limsup H(c_i) \leq 1$ and we may take $\eta_0 > 0$. Theorem 4.1 gives $\rho g_c \in \text{Cert}(f)$ within $\rho K' \sup_{i \geq i_0} \lambda_i$ of ρg . For the last assertion, $\epsilon_i \leq |c|K'\lambda_i + O(\lambda_i)$, and a subsequence with $\lambda_{i+1} \leq \lambda_i \inf \gamma/2$ satisfies (a). $\square$

Remark 4.13. If $\sum_i \gamma_i < \infty$ the argument breaks down: condition (ii) at $s \approx \gamma_i \lambda_i$ forces $\mu_i \lesssim \gamma_i$. Whether switching mates carried by such “super-near-threshold” coordinates, or by weak peaks (small margins), lie in $\text{cl Cert}(f)$ is open; their limit directions lie in the closed span of the non-peak vectors, so the linear obstruction of Section 6 does not apply to them.

5. SECOND-ORDER REBALANCING AND TAME SUPPORTS

In this section the block set I is finite. Let $f \in S_{p^*}$ have a normer $\eta \in S_{p^{**}}$; either $\eta = \xi$ is the bidual normer of an arbitrary f , or $\eta = x' \in S_p$ when $f = f'$ attains its norm. We use the forced data of Section 1 at η , writing $c_\eta := q^{**}(\eta)$ (so $c_\eta = q_0$ when $\eta = \xi$), $\hat{z} := \eta/c_\eta = z + Ue$, $\zeta_m := R_m^{**}\eta$ and $\sigma_m := |\zeta_m|_m$, so that $c_\eta + \sum_m \sigma_m = 1$.

5.1. Slices, excess and tame points.

Lemma 5.1 (Bidual slice). *Let $g \in \mathcal{C}(f)$. Then $g(\eta) = 0$, and for every $\chi \in X^{**}$ with $\chi(f) = 0$,*

$$\chi(g)^2 \leq \varphi(\chi)(2 + \varphi(\chi)), \quad \varphi(\chi) := p^{**}(\eta + \chi) - 1.$$

In particular, if $\varphi(s\chi) = o(s^2)$ as $s \rightarrow 0^+$, then $\chi(g) = 0$.

Proof. For $\theta \in X^{**}$ and $t \in \mathbb{R}$, $\theta(f) + t\theta(g) = \theta(f + tg) \leq p^{**}(\theta)s(t)$; the supremum of the left side divided by $s(t)$ over t is $\sqrt{\theta(f)^2 + \theta(g)^2}$, so $\theta(f)^2 + \theta(g)^2 \leq p^{**}(\theta)^2$. With $\theta = \eta$ we get $g(\eta) = 0$, and with $\theta = \eta + \chi$ we get $1 + \chi(g)^2 \leq (1 + \varphi(\chi))^2$. For the last claim apply this to $s\chi$ and divide by s^2 . $\square$

Lemma 5.2 (Excess splitting). *For $\chi \in X^{**}$ put $\Delta(\eta + \chi) := p^{**}(\eta + \chi) - f(\eta + \chi) \geq 0$. Then*

$$\Delta(\eta + \chi) = E_q(\chi) + \sum_m E_m(R_m^{**}\chi),$$

$$E_q(\chi) := q^{**}(\eta + \chi) - c_\eta - a(\chi), \quad E_m(\delta) := |\zeta_m + \delta|_m - \sigma_m - w_m(\delta),$$

and all these terms are nonnegative. If $f(\chi) = 0$, then $\Delta(\eta + \chi) = \varphi(\chi)$.

Proof. $p^{**}(\eta) = 1 = f(\eta) = c_\eta + \sum_m \sigma_m$, $f = a + \sum_m R_m^* w_m$ and $p^{**} = q^{**} + \sum_m |R_m^{**} \cdot|_m$; nonnegativity holds because a and w_m have norm one. $\square$

Lemma 5.3 (Flatness on free coordinates). *If $\chi \in \ell_\infty$ is supported in J and $|\chi_j| \leq c_\eta(1 - |z_j|)$ for all j , then $E_q(\chi) = 0$ and $a(\chi) = 0$.*

Proof. $\eta + \chi = c_\eta((z + \chi/c_\eta) + Ue)$ with $\|z + \chi/c_\eta\|_\infty \leq 1$, so $q^{**}(\eta + \chi) \leq c_\eta$; and $q^{**}(\eta + \chi) \geq a(\eta + \chi) = c_\eta$ because $J \cap F = \emptyset$. $\square$

Lemma 5.4 (Overshoot bound). *Fix a block and drop m . Let $\delta = \mu\zeta + \delta'$ with $\mu > -1$, $\delta' \in \ell_1$ supported in P and $\sum_{k \in P} s_k \delta'_k = 0$, where $s_k := \text{sign } w(k)$. Then*

$$E(\delta) \leq 2 \sum_{k \in P} (-s_k \delta'_k - (1 + \mu)|\zeta| |\alpha(k)|)_+ \leq 2 \sum_{k \in P} (|\delta'_k| - (1 + \mu)\lambda_k \mu_k)_+.$$

Proof. By Lemma 1.7, $\zeta + \delta = [(1 + \mu)|\zeta| \alpha + \delta'] + (1 + \mu)|\zeta| D(Dw/C)$. Using $|x + y| = |x| + sy + 2(-sy - |x|)_+$ for $s \in \{\pm 1\}$ with $sx = |x|$, the ℓ_1 -norm of the bracket is $(1 + \mu)|\zeta| + \sum_P s_k \delta'_k + 2\text{ov} = (1 + \mu)|\zeta| + 2\text{ov}$, where ov is the sum in the statement; the second term lies in $(1 + \mu)|\zeta| D(B_{\ell_2})$. Hence $|\zeta + \delta| \leq (1 + \mu)|\zeta| + 2\text{ov}$, while $w(\delta) = \mu|\zeta| + M \sum_P s_k \delta'_k = \mu|\zeta|$. The second inequality is (2). $\square$

Note that $|(R_m \chi)(k)| = \lambda_{k,m} |u_{k,m}(\chi)| \leq \lambda_{k,m} q(\chi)$ for $\chi \in X$.

Definition 5.5 (Tame points, certificate span). Let $P_m^\circ := \{k \in P_m : \mu_{k,m} > 0\}$ and $\text{Dg}_m := P_m \setminus P_m^\circ$ (degenerate peaks), $\bar{Q}_m := Q_m \cup \text{Dg}_m$. The normer η satisfies *margin sparsity* (MS) if $\sum_{k \in P_m^\circ, \mu_{k,m} \leq s} \Phi_m(k) = o(s)$ as $s \rightarrow 0^+$ for every m . It is *C-tame* if $a \in c_{00}$, K is finite, every $\bar{Q}_m$ is finite, and (MS) holds. (At $\eta = x' \in c_0$ the first two conditions are automatic.) The *certificate span* at f is

$$S(f) := \left\{ b + \sum_m R_m^*(\omega_m - d_m(\omega_m)w_m) : b \in \ell_1(F), b(\eta) = 0, \omega_m \in c_{00}(Q_m) \right\},$$

$$d_m(\omega) := \frac{\langle D_m w_m, D_m \omega \rangle}{C_m},$$

and its elements are called *balanced finite certificates* (the base part need not satisfy $\|b/a\|_\infty < \infty$ when F is infinite). Equivalently $S(f) = (\ell_1(F) \cap \eta^\perp) + \text{span}\{y_{k,m} : k \in Q_m, m \in I\}$ with $y_{k,m} := \lambda_{k,m} u_{k,m} - (\Phi_m(k)^2 w_m(k)/C_m) R_m^* w_m$. For $g \in S(f)$ with data (b, ω) put $H_b := h(b)$, $H_m := H_m(\omega_m)$ and

$$\Gamma_w(g) := c_\eta H_b + \sum_m \sigma_m H_m, \quad \Gamma_{\max}(g) := \max(H_b, \max_m H_m).$$

When F and all Q_m are finite the data (b, ω) of $g \in S(f)$ are unique (Lemma 1.8), so Γ_w is well defined; in general we regard Γ_w as a function of the data. Since $c_\eta + \sum_m \sigma_m = 1$, $\Gamma_w \leq \Gamma_{\max}$.

Theorem 5.6 (Fibres at C-tame points). *If the normer η of f is C-tame, then $\mathcal{C}(f) \subseteq S(f)$; that is, every mate is a balanced finite certificate $g = b + \sum_m R_m^*(\omega_m - d_m w_m)$ with $\text{supp } b \subseteq F$, $b(\eta) = 0$ and ω_m supported in the finite set Q_m . In particular $\mathcal{C}(f)$ is finite dimensional.*

Proof. J is cofinite. By Lemma 1.8(b) with $P'_m := P_m^\circ$ (cofinite, hence infinite) and $\pi_m := \sum_{k \in P'_m} s_k \lambda_{k,m} u_{k,m}$ (so that $\pi_m(\chi) = \sum_{P'_m} s_k (R_m \chi)_k$; the degenerate peaks are among the finitely many $u_{k,m}$, $k \in \bar{Q}_m$), the restrictions to $c_{00}(J)$ of the finite family

$$\Psi := \{u_{k,m} : k \in \bar{Q}_m, m \in I\} \cup \{\pi_m : m \in I\}$$

are linearly independent. Note $R_m^* w_m = M_m \pi_m + \sum_{k \in Q_m} w_m(k) \lambda_{k,m} u_{k,m}$. We use repeatedly: if $\chi \in X$ and $R_m \chi = \mu \zeta_m + \delta'$ where δ' vanishes on $\bar{Q}_m$ and has zero peak sum, then by Lemma 5.4 and (MS), $E_m(t\mu \zeta_m + t\delta') \leq 2m \sum_{k \in P'_m} \Phi_m(k) (|t|B - \frac{1}{2}\mu_{k,m})_+ = o(t^2)$ for $|t\mu| \leq \frac{1}{2}$, where B bounds $|\delta'_k|/\lambda_{k,m}$.

Step 1: $\mathcal{C}(f) \subseteq W := \text{span}\{e_j^ : j \in F \cup K\} + \text{span}\Psi$.* Let $\nu \in c_{00}(J)$ with $\psi(\nu) = 0$ for all $\psi \in \Psi$. Then $a(\nu) = 0$, $R_m\nu$ vanishes on $\bar{Q}_m$, and its peak sum is $\pi_m(\nu) = 0$; also $f(\nu) = a(\nu) + \sum_m w_m(R_m\nu) = 0$. For small $|t|$, $E_q(t\nu) = 0$ (Lemma 5.3: ν is finitely supported in J) and $E_m(tR_m\nu) = o(t^2)$, so $\varphi(t\nu) = o(t^2)$ and $g(\nu) = 0$ for every $g \in \mathcal{C}(f)$ (Lemma 5.1). By linear algebra, the restriction of g to $c_{00}(J)$ is a combination of the restrictions of Ψ ; hence g minus that combination is supported in $F \cup K$.

Step 2: unique form. Since π_m and the $u_{k,m}$ are images under R_m^* of $s1_{P_m}$ and $\lambda_{k,m}^{-1}e_k$, every $g \in W$ can be written $g = b + \sum_m R_m^*(\omega_m + c_m w_m)$ with $\text{supp } b \subseteq F \cup K$ and $\omega_m \in c_{00}(\bar{Q}_m)$; the representation is unique by Lemma 1.8(a) and because $w_m \neq 0$ on the infinite set P_m° .

Step 3: no contact components. Let $j \in K$. Choose $\nu_J \in c_{00}(J)$ with $\psi(\nu_J) = z_j\psi(e_j)$ for $\psi \in \Psi$ and put $\nu := -z_j e_j + \nu_J$, so $\psi(\nu) = 0$ for $\psi \in \Psi$. For small $s > 0$, $\eta + s\nu = c_\eta((z - sz_j e_j/c_\eta + s\nu_J/c_\eta) + Ue)$ has cube part of norm ≤ 1 (the contact moves inward), so $q^{**}(\eta + s\nu) \leq c_\eta = a(\eta + s\nu)$ and $E_q(s\nu) = 0$. As in Step 1, $E_m(sR_m\nu) = o(s^2)$ and $f(\nu) = 0$. Hence $g(\nu) = 0$. But $g(\nu) = -z_j b_j$, because $b(\nu_J) = 0$ and $(\omega_m + c_m w_m)(R_m\nu) = 0$. So $b_j = 0$.

Step 4: no degenerate components. Let $k_0 \in \text{Dg}_{m_0}$ and $s_0 := \text{sign } w_{m_0}(k_0)$. Choose $\nu \in c_{00}(J)$ with $u_{k_0,m_0}(\nu) = s_0$ and $\psi(\nu) = 0$ for all other $\psi \in \Psi$. Then $R_{m_0}\nu$ vanishes on $\bar{Q}_{m_0} \setminus \{k_0\}$, equals $s_0 \lambda_{k_0,m_0}$ at k_0 , and has zero peak sum. For $s > 0$ the first bound of Lemma 5.4 contributes nothing at k_0 (there $-s_0 \cdot s s_0 \lambda_{k_0,m_0} < 0$) and nothing at the other degenerate peaks; the non-degenerate peaks give $o(s^2)$ by (MS). Other blocks and the base are as in Step 1. So $\varphi(s\nu) = o(s^2)$ for $s > 0$ and $g(\nu) = 0$; but $g(\nu) = \omega_{m_0}(k_0) \lambda_{k_0,m_0} s_0$. So $\omega_{m_0}(k_0) = 0$.

Step 5: balance. Fix m_0 . Choose $\nu_J \in c_{00}(J)$ with $\psi(\nu_J) = -\psi(\eta)/\sigma_{m_0}$ for $\psi \in \{u_{k,m_0} : k \in \bar{Q}_{m_0}\} \cup \{\pi_{m_0}\}$ and $\psi(\nu_J) = 0$ for the other elements of Ψ , and put $\nu := \eta + \nu_J \in X^{**}$. Then $f(\nu) = 1 + \sum_m w_m(R_m\nu_J) = 1 - w_{m_0}(\zeta_{m_0})/\sigma_{m_0} = 0$. The base: $\eta + t\nu = (1+t)(\eta + t\nu_J/(1+t))$, so $E_q(t\nu) = 0$ for small $|t|$ by Lemma 5.3 and homogeneity. Blocks $m \neq m_0$: $R_m^{**}(t\nu) = t\zeta_m + tR_m\nu_J$ with $R_m\nu_J$ vanishing on $\bar{Q}_m$ and of zero peak sum. Block m_0 : $R_{m_0}^{**}(t\nu) = t(1 - 1/\sigma_{m_0})\zeta_{m_0} + t\delta''$ with $\delta'' := R_{m_0}\nu_J + \zeta_{m_0}/\sigma_{m_0}$ vanishing on $\bar{Q}_{m_0}$ and of zero peak sum. Hence $\varphi(t\nu) = o(t^2)$, and $g(\nu) = 0$ by Lemma 5.1. Writing $v_m := \omega_m + c_m w_m$ we compute $g(\nu) = g(\eta) + b(\nu_J) + \sum_m v_m(R_m\nu_J) = -v_{m_0}(\zeta_{m_0})/\sigma_{m_0}$, since $v_m(\delta) = 0$ whenever δ vanishes on $\bar{Q}_m$ and has zero peak sum. Thus $v_m(\zeta_m) = 0$ for all m . As ω_m lives on Q_m , where $\zeta_m(k) = \sigma_m \Phi_m(k)^2 w_m(k)/C_m$, we have $\omega_m(\zeta_m) = \sigma_m d_m(\omega_m)$, and $v_m(\zeta_m) = \sigma_m(d_m + c_m)$; hence $c_m = -d_m$. Finally $b(\eta) = g(\eta) - \sum_m v_m(\zeta_m) = 0$. $\square$

Remark 5.7. At a norm-attaining f' with C-tame normer x' (i.e. finitely many strict non-peaks and degenerate peaks, and (MS)), Theorem 5.6 says $\mathcal{C}(f') \subseteq S(f')$. This is a *linear obstruction at the approximants*: if $f_n \rightarrow f$ are norm attaining with C-tame normers, then $\text{Li}_n \mathcal{C}(f_n) \subseteq \text{Li}_n S(f_n)$. It will be used in Section 6.

5.2. Transfer peaks and the mass-weighted coefficient. In this subsection $\eta = \xi$. The *canonical truncations* of f are defined when $a \in c_{00}$: let $z_j^N := z_j$ for $j \in F \cup [1, N]$ and $z_j^N := 0$ otherwise, $\tilde{x}_N := z^N + Ue \in c_0$, $x'_N := \tilde{x}_N/p(\tilde{x}_N)$ and $f'_N := \nabla p(x'_N)$. By Proposition 1.10(c), $q(\tilde{x}_N) = 1$, $\nabla q(\tilde{x}_N) = a$ and $f'_N = a + \sum_m R_m^* w'_{m,N}$ with $w'_{m,N} = J_m(R_m x'_N)$; by Proposition 1.12, $f'_N \rightarrow f$. We write

$c_N := q(x'_N) \rightarrow q_0$, and $M'_{m,N}, C'_{m,N}, \sigma'_{m,N}, \dots$ for the data at x'_N ; they converge to those at ξ (Proposition 1.11).

Theorem 5.8 (Transfer peaks). *Let $f \in S_{p^*}$ with $a \in c_{00}$, and let $g = b + \sum_m R_m^*(\omega_m - d_m w_m) \in S(f)$ be a balanced finite certificate such that (f, g) is contractive and $\Gamma_w(g) \leq 1$. Then for every $\rho \in (0, 1)$ there are $g'_N \in X^*$ with $g'_N \rightarrow g$ and $\rho g'_N \in \mathcal{C}(f'_N)$ for N large. In particular $(f, \rho g) \in \text{clNA}((c_0, p), \ell_2^2)$, and all such mates are recovered simultaneously along the canonical truncations, which do not depend on g .*

Proof. Fix $\rho \in (0, 1)$ and $\varepsilon_0 \in (0, \frac{1}{2}]$ with $\rho^2(\Gamma_w(g) + 4\varepsilon_0) \leq \rho$, and put $\ell_* := \Gamma_w(g)/2 + \varepsilon_0$. Let $\gamma_0 := \frac{1}{2} \min_m \min_{k \in \text{supp } \omega_m} \text{gap}_m(k) > 0$.

Step 1: the transported natural certificate. Put $b'_N := b - (b(x'_N)/c_N)a$, $d'_{m,N} := \langle D_m w'_{m,N}, D_m \omega_m \rangle / C'_{m,N}$, $v'_{m,N} := \omega_m - d'_{m,N} w'_{m,N}$ and $g'_N := b'_N + \sum_m R_m^* v'_{m,N}$. Since $b(x'_N) \rightarrow b(\xi) = 0$, $d'_{m,N} \rightarrow d_m$ and $R_m^* w'_{m,N} \rightarrow R_m^* w_m$, we have $g'_N \rightarrow g$. For N large every $k \in \text{supp } \omega_m$ has $M'_{m,N} - |w'_{m,N}(k)| \geq \gamma_0$. *Base:* b'_N is supported in the finite set F , where $z^N = \text{sign } a$. The computation in the proof of Lemma 3.3, with $\hat{z}$ replaced by $\tilde{x}_N = z^N + Ue$, gives $q^*(a + tb'_N) = 1 + t b'_N(\tilde{x}_N) + \nu \Psi(tU^* b'_N / \nu)$ as long as no coordinate of $a + tb'_N$ changes sign on F ; and $b'_N(\tilde{x}_N) = b'_N(x'_N)/c_N = 0$. Since $P^\perp U^* b'_N = P^\perp U^* b$ and $b'_N \rightarrow b$, the bounds for Ψ give $q^*(a + tb'_N) \leq 1 + \frac{t^2}{2} H_b \theta(t)$, where here and below $\theta(t) \rightarrow 1$ as $t \rightarrow 0$, uniformly in large N . *Blocks:* Lemma 3.4(c),(d) at x'_N give $\|w'_{m,N} + tv'_{m,N}\|_\infty \leq (1 - td'_{m,N})M'_{m,N}$ and $\|D_m(w'_{m,N} + tv'_{m,N})\| \leq C'_{m,N} + td'_{m,N}M'_{m,N} + \frac{t^2}{2} H'_{m,N} \theta(t)$, with $H'_{m,N} := (\|D_m \omega_m\|^2 - d_{m,N}^2) / C'_{m,N} \rightarrow H_m$.

Step 2: transfer peaks at ξ . Fix m and put $\varsigma_m := \text{sign}(H_m/2 - \ell_*) \in \{-1, 0, 1\}$; if $\varsigma_m = 0$ set $\tau_m := 0$ and skip this step. Choose $\gamma_m \in (\max_{k \in \text{supp } \omega_m} |w_m(k)|, M_m)$ with $|w_m(k)| \neq \gamma_m$ for all k , let $\mathfrak{L}_m := \{k \notin \text{supp } \omega_m : |w_m(k)| \geq \gamma_m\} \supseteq P_m$, $y_m^0 := \text{sign}(w_m)1_{\mathfrak{L}_m}$, $\hat{\pi}_m := R_m^* y_m^0$ and $\tau_{*,m} := (\hat{\pi}_m(\xi)/q_0)a - \hat{\pi}_m$, so $\tau_{*,m}(\xi) = 0$. For $\delta_0 \in (0, 1]$ let $\mathcal{T}_m := \varsigma_m \tau_{*,m} + \delta_0(q^*(\tau_{*,m}) + 1)a$ and $\ell_m := q^*(\mathcal{T}_m)$; then $\mathcal{T}_m(\xi) = \delta_0(q^*(\tau_{*,m}) + 1)q_0 > 0$. For $\delta_1 \in (0, \mathcal{T}_m(\xi)/(2\ell_m q_0))$ choose, by density of the tails of $(u_{k,m})_k$, an index $k_* = k_{*,m} \notin \text{supp } \omega_m$ with $q^*(u_{k_*,m} - \mathcal{T}_m/\ell_m) \leq \delta_1$, so large that $\ell_m \Phi_m(k_*)M_m/(mC_m) \leq \frac{1}{2}$ and $\vartheta_m \Phi_m(k_*) < \mathcal{T}_m(\xi)/(2\ell_m)$. Then $u_{k_*,m}(\xi) \geq \mathcal{T}_m(\xi)/\ell_m - \delta_1 q_0 \geq \mathcal{T}_m(\xi)/(2\ell_m) > \vartheta_m \Phi_m(k_*)$, so k_* is a non-degenerate peak with $w_m(k_*) = M_m$ and margin $\mu_* > 0$. Put

$$y_m := y_m^0 + \varsigma_m \frac{\ell_m}{\lambda_{k_*,m}} e_{k_*}, \quad R_m^* y_m = \hat{\pi}_m + \varsigma_m \ell_m u_{k_*,m},$$

$$e_{\infty,m} := R_m^* y_m - \frac{(R_m^* y_m)(\xi)}{q_0} a.$$

Since $R_m^* y_m = (\text{multiple of } a) + \varsigma_m(\ell_m u_{k_*,m} - \mathcal{T}_m)$, we get $e_{\infty,m} = \varsigma_m[(\ell_m u_{k_*,m} - \mathcal{T}_m) - \frac{(\ell_m u_{k_*,m} - \mathcal{T}_m)(\xi)}{q_0} a]$ and $q^*(e_{\infty,m}) \leq 2\ell_m \delta_1$. Also $\ell_m \mu_* \leq \ell_m u_{k_*,m}(\xi) \leq \mathcal{T}_m(\xi) + \ell_m \delta_1 q_0$. Hence the *inefficiency* $\epsilon_{1,m} := \ell_m \mu_*/q_0 + q^*(e_{\infty,m}) \leq \delta_0(q^*(\tau_{*,m}) + 1) + 3\ell_m \delta_1$ can be made as small as we wish by choosing first δ_0 and then δ_1 (note $\ell_m \leq (1 + \delta_0)(q^*(\tau_{*,m}) + 1)$). Put $e_{y,m} := 1 + \langle D_m y_m, D_m w_m \rangle / C_m$. As $y_m^0 w_m \geq 0$ and $\ell_m \Phi_m(k_*)M_m/(mC_m) \leq \frac{1}{2}$, we have $e_{y,m} \geq \frac{1}{2}$. Finally let $\tau_m := (H_m/2 - \ell_*)/e_{y,m}$, whose sign is ς_m , and note $|\tau_m| \leq H_m + 2$ independently of δ_0, δ_1, k_* . We choose δ_0, δ_1 so that $\sum_m (H_m + 2)\epsilon_{1,m} \leq \varepsilon_0/2$.

Step 3: the limiting levels. Put $\Lambda_b := H_b/2 + \sum_m (\tau_m \sigma_m e_{y,m}/q_0 + |\tau_m| \epsilon_{1,m})$ and $\Lambda_m := H_m/2 - \tau_m e_{y,m} = \ell_*$. Using $\sum_m \sigma_m = 1 - q_0$,

$$q_0 \Lambda_b = \frac{\Gamma_w}{2} - \ell_*(1 - q_0) + q_0 \sum_m |\tau_m| \epsilon_{1,m}, \quad \text{so} \quad \Lambda_b = \ell_* - \frac{\varepsilon_0}{q_0} + \sum_m |\tau_m| \epsilon_{1,m} \leq \ell_*.$$

Step 4: transport of the transfer data. For N large let $\mathfrak{L}_{m,N} := \{k \notin \text{supp } \omega_m : |w'_{m,N}(k)| \geq \gamma_m\}$ and $y_{m,N} := \text{sign}(w'_{m,N})1_{\mathfrak{L}_{m,N}} + \varsigma_m(\ell_m/\lambda_{k_*,m})e_{k_*}$. For large N , k_* is a peak of $w'_{m,N}$ of sign $+1$ with margin $\mu'_{*,N} \rightarrow \mu_*$, and the peak set of $w'_{m,N}$ is contained in $\mathfrak{L}_{m,N}$. Since $w'_{m,N} \rightarrow w_m$ coordinatewise and $|w_m(k)| \neq \gamma_m$, $y_{m,N} \rightarrow y_m$ coordinatewise and boundedly; hence $R_m^* y_{m,N} \rightarrow R_m^* y_m$ in ℓ_1 and $D_m y_{m,N} \rightarrow D_m y_m$ in ℓ_2 . Put $\beta_{m,N} := ((R_m^* y_{m,N})(x'_N)/c_N)a$, $e_{m,N} := R_m^* y_{m,N} - \beta_{m,N} \rightarrow e_{\infty,m}$, and $e_{y,m,N} := 1 + \langle D_m y_{m,N}, D_m w'_{m,N} \rangle / C'_{m,N} \rightarrow e_{y,m}$. Writing $R_m x'_N = \sigma'_{m,N}(\alpha' + D_m^2 w'_{m,N}/C'_{m,N})$ (Lemma 1.7) and using (2) at k_* , one obtains the bookkeeping identity

$$(6) \quad y_{m,N}(R_m x'_N) = \sigma'_{m,N} e_{y,m,N} + \varsigma_m \ell_m \mu'_{*,N}.$$

Indeed $\langle \text{sign}(w')1_{\mathfrak{L}_{m,N}}, \alpha' \rangle = 1$, and

$$\frac{\ell_m}{\lambda_{k_*}}(R_m x'_N)(k_*) = \ell_m u_{k_*}(x'_N) = \ell_m \mu'_{*,N} + \frac{\ell_m}{\lambda_{k_*}} \frac{\sigma' \Phi(k_*)^2 M'}{C'},$$

the last term being the contribution of e_{k_*} to $\sigma' \langle D y_{m,N}, D w' \rangle / C'$.

Step 5: the decomposition. For $t \in \mathbb{R}$ put $\varepsilon_m := \tau_m \rho^2 t^2$ and

$$A(t) := a + t \rho b'_N + \sum_m \varepsilon_m R_m^* y_{m,N}, \quad W_m(t) := w'_{m,N} + t \rho v'_{m,N} - \varepsilon_m y_{m,N},$$

so that $A(t) + \sum_m R_m^* W_m(t) = f'_N + t \rho g'_N$. *Base.* By (6), $A(t) = (1 + E)a + t \rho b'_N + \sum_m \varepsilon_m e_{m,N}$ with $E := \rho^2 t^2 \sum_m (\tau_m \sigma'_{m,N} e_{y,m,N} + |\tau_m| \ell_m \mu'_{*,N})/c_N$. By homogeneity and Step 1,

$$q^*(A(t)) \leq 1 + E + \frac{\rho^2 t^2}{2(1 + E)} H_b \theta(t) + \rho^2 t^2 \sum_m |\tau_m| q^*(e_{m,N}) = 1 + \rho^2 t^2 (\Lambda_b + o(1)),$$

where $o(1) \rightarrow 0$ as $t \rightarrow 0$ and $N \rightarrow \infty$. *Blocks, sup norm.* For $|t|$ small and N large, $\|W_m(t)\|_\infty \leq (1 - t \rho d'_{m,N})M'_{m,N} - \varepsilon_m$: on $\mathfrak{L}_{m,N} \setminus \{k_*\}$ the coordinates are $(1 - t \rho d')w'(k) - \varepsilon_m \text{sign } w'(k)$, whose moduli are at most this bound; at k_* the value is $(1 - t \rho d')M' - \varepsilon_m(1 + \varsigma_m \ell_m/\lambda_{k_*})$, which lies between $\pm((1 - t \rho d')M' - \varepsilon_m)$ once $|\varepsilon_m| \ell_m/\lambda_{k_*} \leq M'$ (here $\varsigma_m \varepsilon_m \geq 0$ is used); on $\text{supp } \omega_m$ the moduli are at most $(1 - t \rho d')(M' - \gamma_0) + \rho|t|\|\omega_m\|_\infty$; elsewhere at most $(1 - t \rho d')\gamma_m$; both are below the bound for small $|t|$ and large N because $M'_{m,N} \rightarrow M_m > \gamma_m$. *Blocks, Hilbert part.* With $h_0 := D_m(w' + t \rho v')$, inequality (5) gives $\|h_0 - \varepsilon_m D_m y_{m,N}\| \leq \|h_0\| - \varepsilon_m \langle h_0/\|h_0\|, D_m y_{m,N} \rangle + O(t^4)$, and $\langle h_0/\|h_0\|, D_m y_{m,N} \rangle = \langle D_m w', D_m y_{m,N} \rangle / C' + O(|t|)$ uniformly in N . Together with Step 1,

$$N_m(W_m(t)) \leq 1 + \rho^2 t^2 \left(\frac{H'_{m,N} \theta(t)}{2} - \tau_m e_{y,m,N} \right) + O(|t|^3) = 1 + \rho^2 t^2 (\ell_* + o(1)).$$

By Lemma 1.6 and Step 3, there are $t_4 > 0$ and N_4 with $p^*(f'_N + t \rho g'_N) \leq 1 + \rho^2 t^2 (\ell_* + \varepsilon_0) \leq 1 + \frac{t^2}{2} \rho^2 (\Gamma_w + 4\varepsilon_0) \leq 1 + \rho t^2/2 \leq s(t)$ for $|t| \leq t_5 := \min(t_4, 2\sqrt{1-\rho}, 1)$ and $N \geq N_4$.

Step 6: large t . Since (f, g) is contractive, $p^*(f'_N + t \rho g'_N) \leq s(\rho t) + \varepsilon_N + \rho|t|\varepsilon'_N$ with $\varepsilon_N := p^*(f'_N - f) \rightarrow 0$ and $\varepsilon'_N := p^*(g'_N - g) \rightarrow 0$; by Lemma 3.6(a) this

is at most $s(t)$ for $|t| \geq t_5$ and N large. Hence $\rho g'_N \in \mathcal{C}(f'_N)$ for N large, and $g'_N \rightarrow g$. $\square$

Remark 5.9. (a) If $\Gamma_{\max}(g) \leq 1$, no transfer peaks are needed ($\tau_m = 0$); this is the natural-certificate case, and together with Corollary 3.10 it re-proves the local-certificate theorem of [9]. (b) The transfer peak is the genuinely infinite-dimensional ingredient: lowering a block's sup norm requires lowering every near-peak coordinate, and the image $\hat{\pi}_m$ of such a uniform lowering is a fixed vector of Y which the base cannot absorb; density of the tails of $(u_{k,m})_k$ provides a robust deep peak whose functional converts it into a multiple of a with arbitrarily small inefficiency. In finite-dimensional models of the construction this mechanism is absent, and numerical computations in such models (numerical evidence only, not a theorem) give second-order coefficients strictly larger than Γ_w . (c) For every balanced finite certificate g at f with $a \in c_{00}$, $\limsup_{t \rightarrow 0} 2(p^*(f + tg) - 1)/t^2 \leq \Gamma_w(g)$; this is proved in Proposition 7.9, by running the rebalancing at ξ instead of x'_N , and Theorem 7.10 identifies the exact one-sided coefficients at (BT) points.

Remark 5.10 (Γ_w and shifts). For every shifted certificate $c = (b, \omega, \theta)$ at f one has $H^{\text{sh}}(c) \geq \Gamma_w(g_c)$. Indeed, if $\Lambda := H^{\text{sh}}(c)$, then $2\theta_m \leq \Lambda - H_m$ for every m , and $\Lambda \geq h(b) - 2 \sum_m \theta_m \sigma_m / q_0 \geq h(b) - \sum_m (\Lambda - H_m) \sigma_m / q_0$; multiplying by q_0 and using $q_0 + \sum_m \sigma_m = 1$ gives $\Lambda \geq \Gamma_w$. Thus, when $a \in c_{00}$, Theorem 5.8 covers every $g_c \in \mathcal{C}(f)$ with c a shifted certificate and $H^{\text{sh}}(c) \leq 1$ (the direction of a finite certificate always lies in $S(f)$).

5.3. The sharp second-order coefficient at tame supports.

Lemma 5.11 (Exact local block formula). *Fix a block and drop m . Recall that $\sum_k \Phi(k)^2 < 1$. Let $y \in \ell_1$, let $P \neq \emptyset$ be a set of indices with complement Q , and $s \in \{\pm 1\}^P$. Put $S := \sum_{k \in P} s_k y_k$, $Y := (\sum_{k \in Q} y_k^2 / \Phi(k)^2)^{1/2}$ and $F_2 := \sum_{k \in P} \Phi(k)^2 \in (0, 1)$, and assume $0 < S$, $Y < \infty$ and $Y \leq S$. Define*

$$\Lambda(S, Y) := \frac{S - \sqrt{F_2} \sqrt{S^2 - (1 - F_2)Y^2}}{1 - F_2}, \quad \lambda := \Lambda(S, Y), \quad \rho_0 := (S/\lambda - 1)/F_2.$$

Then $\lambda > 0$ and

$$(7) \quad |y| \leq \lambda + 2 \sum_{k \in P} (\lambda \rho_0 \Phi(k)^2 - s_k y_k)_+.$$

Moreover, if $\zeta \neq 0$ with norming functional w , and (P, s) are its peak set and signs, then with $c_Q := \|D(w1_Q)\|_2^2$: $S_P(\zeta) = |\zeta|(1 + F_2 M/C)$, $Y_Q(\zeta) = |\zeta|\sqrt{c_Q}/C < S_P(\zeta)$, $\Lambda(S_P(\zeta), Y_Q(\zeta)) = |\zeta|$, $\lambda \rho_0 = |\zeta|M/C$, and at this point

$$\partial_S \Lambda = M, \quad \partial_Y \Lambda = CY/|\zeta|, \quad \partial_S^2 \Lambda = \frac{\sqrt{F_2} Y^2}{(S^2 - (1 - F_2)Y^2)^{3/2}} = \frac{C c_Q}{|\zeta| F_2}.$$

Proof. $\Delta := S^2 - (1 - F_2)Y^2 \geq F_2 S^2 > 0$; λ is the smaller root of $(1 - F_2)l^2 - 2Sl + (S^2 + F_2 Y^2) = 0$, i.e. of $(S - l)^2 + F_2 Y^2 = F_2 l^2$, and $\lambda > 0$ because $\sqrt{F_2} \Delta < S$. The root equation reads $\rho_0^2 F_2 + Y^2 / \lambda^2 = 1$. Define $h \in \ell_2$ by $h_k := \rho_0 \Phi(k) s_k$ ($k \in P$), $h_k := y_k / (\lambda \Phi(k))$ ($k \in Q$); then $\|h\|_2 = 1$, and $\alpha := y/\lambda - Dh$ vanishes on Q , with $\|\alpha\|_1 = \sum_P s_k \alpha_k + 2 \sum_P (-s_k \alpha_k)_+ = S/\lambda - \rho_0 F_2 + \frac{2}{\lambda} \sum_P (\lambda \rho_0 \Phi(k)^2 - s_k y_k)_+$, and $S/\lambda - \rho_0 F_2 = 1$. Hence $y = \lambda \alpha + \lambda Dh \in \lambda \|\alpha\|_1 (B_{\ell_1} + D(B_{\ell_2}))$, which is (7).

At ζ , Lemma 1.7 gives $s_k \zeta_k = |\zeta|(|\alpha_k| + \Phi(k)^2 M/C)$ on P and $\zeta_k = |\zeta|\Phi(k)^2 w_k/C$ on Q , whence the formulas for S and Y ; $Y < S$ because $\sqrt{c_Q} \leq C$ and $F_2 M >$

0. Using $C^2 = M^2 F_2 + c_Q$ one checks that $l = |\zeta|$ satisfies $(S - l)^2 + F_2 Y^2 = F_2 l^2$; since $|\zeta|(1 - F_2) < S$, it is the smaller root. Then $\rho_0 = M/C$, and from the root equation $\sqrt{F_2 \Delta} = S - (1 - F_2)\lambda = \lambda F_2(1 + \rho_0)$, i.e. $\sqrt{\Delta} = |\zeta| \sqrt{F_2}/C$. Differentiating, $\partial_S \Lambda = (1 - \sqrt{F_2} S / \sqrt{\Delta}) / (1 - F_2) = (1 - C - F_2 M) / (1 - F_2) = M$, $\partial_Y \Lambda = \sqrt{F_2} Y / \sqrt{\Delta} = C Y / |\zeta|$, and $\partial_S^2 \Lambda = \sqrt{F_2} (S^2 - \Delta) / ((1 - F_2) \Delta^{3/2}) = \sqrt{F_2} Y^2 / \Delta^{3/2} = C c_Q / (|\zeta| F_2)$. $\square$

Proposition 5.12 (Lower bound). *Let I be finite and let $f \in S_{p^*}$ satisfy: $a \in c_{00}$, K is finite, every Q_m is finite, there are no degenerate peaks, and (MS) holds at ξ . Then for every balanced finite certificate $g \in S(f)$,*

$$\liminf_{t \rightarrow 0} \frac{p^*(f + tg) - 1}{t^2} \geq \frac{\Gamma_w(g)}{2}.$$

In particular $\Gamma_w(g) \leq 1$ for every $g \in \mathcal{C}(f) \cap S(f)$.

Proof. Write $g = b + \sum_m R_m^* v_m$, $v_m = \omega_m - d_m w_m$. For $\chi, \chi_2 \in X$ put $\eta_t := \xi + t\chi + t^2\chi_2$. Since $g(\xi) = 0$ and $p^{**}(\eta_t) = f(\eta_t) + \Delta(\eta_t)$ with $f(\eta_t) = 1 + O(t)$,
(8)

$$p^*(f + tg) \geq \frac{(f + tg)(\eta_t)}{p^{**}(\eta_t)} = 1 + t^2 g(\chi) - \Delta(\eta_t) + O(t^3) \quad \text{whenever } \Delta(\eta_t) = O(t^2).$$

By Lemma 5.2, $\Delta(\eta_t) = E_q(t\chi + t^2\chi_2) + \sum_m E_m(t\delta_{1,m} + t^2\delta_{2,m})$ with $\delta_{i,m} := R_m \chi_i$.

Base test. Let $E_F := \text{span}\{U^* e_j^* : j \in F\} \ni e$ and fix $k_\perp \in E_F$ with $\langle k_\perp, e \rangle = 0$. Put $\kappa_2 := \nu \|k_\perp\|^2 / (2q_0)$, $\kappa'_2 := -\kappa_2 \|a\|_1 / \nu$, and let $k_{2\perp} \in E_F$ solve $\langle k_{2\perp}, U^* e_j^* \rangle = -\kappa_2 z_j - \kappa'_2 (Ue)_j$ for $j \in F$ (the Gram matrix of the independent vectors $U^* e_j^*$ is invertible); then $\langle k_{2\perp}, e \rangle = (-\kappa_2 \|a\|_1 - \kappa'_2 \nu) / \nu = 0$. Put $k_2 := \kappa'_2 e + k_{2\perp}$, so that $(Uk_2)_j = -\kappa_2 z_j$ on F . Let $\chi := Uk_\perp + \chi_c$ with $\chi_c \in c_{00}(J)$ to be chosen, and $\chi_2 := (Uk_2)1_{\mathbb{N} \setminus F}$. With $\mathbf{h} := q_0 e + tk_\perp + t^2 k_2$ and $Z := \eta_t - U\mathbf{h} = q_0 z + t\chi_c - t^2 (Uk_2)1_F$ we have $|Z_j| = q_0 + t^2 \kappa_2$ on F and $|Z_j| \leq q_0$ off F for small t (χ_c moves finitely many free coordinates), while $\|\mathbf{h}\| = q_0 + t^2 (\kappa'_2 + \|k_\perp\|^2 / (2q_0)) + O(t^3) = q_0 + t^2 \kappa_2 (1 - \|a\|_1) / \nu + O(t^3) = q_0 + t^2 \kappa_2 + O(t^3)$, using $\|a\|_1 + \nu = 1$. Since $B_{q^{**}} = B_{\ell_\infty} + U(B_H)$, $q^{**}(\eta_t) \leq q_0 + t^2 \kappa_2 + O(t^3)$; and $a(\chi) = \langle U^* a, k_\perp \rangle = 0 = a(\chi_2)$. Hence $E_q(t\chi + t^2\chi_2) \leq t^2 \nu \|k_\perp\|^2 / (2q_0) + O(t^3)$, and $b(\chi) = \langle U^* b, k_\perp \rangle$.

Block test. Fix m and drop it. Apply (7) to $y := \zeta + t\delta_1 + t^2\delta_2$ with the peak set and signs of ζ . As Q is finite, $G(y) := \Lambda(S_P(y), Y_Q(y))$ is a smooth function of the finitely many numbers $(S_P(y), (y_k)_{k \in Q})$ near those of ζ (Λ depends smoothly on (S, Y^2)). By Lemma 5.11 its differential at ζ is $\delta \mapsto MS_P(\delta) + \sum_{k \in Q} w_k \delta_k = w(\delta)$, so $G(y) = |\zeta| + w(t\delta_1 + t^2\delta_2) + \frac{t^2}{2} \mathcal{H}(\delta_1) + O(t^3)$, where $\mathcal{H}$ is the Hessian of G at ζ . The overshoot: $\lambda(y)\rho_0(y) = |\zeta| M/C + O(t)$ and, by (2), $s_k y_k = \lambda_k \mu_k + |\zeta| \Phi(k)^2 M/C + t s_k \delta_{1,k} + t^2 s_k \delta_{2,k}$ on P , with $|\delta_{i,k}| \leq \lambda_k q(\chi_i)$; so each overshoot term is at most $\lambda_k (|t|B - \mu_k)_+$ for a constant B , and their sum is $o(t^2)$ by (MS), all peaks being non-degenerate. Hence $E_m(t\delta_1 + t^2\delta_2) \leq \frac{t^2}{2} \mathcal{H}_m(\delta_{1,m}) + o(t^2)$.

Decoupling. $\mathcal{H}_m(\delta)$ and $v_m(\delta)$ depend only on $\sigma_1 := S_{P_m}(\delta)$ and $\delta|_{Q_m}$. Since $F \cup K$ is finite, Lemma 1.8(b) (with $P'_m = P_m$, which is cofinite) allows us to choose $\chi_c \in c_{00}(J)$ so that, for the fixed $k_\perp$, the finitely many numbers $u_{k,m}(\chi)$ ($k \in Q_m$)

and $\pi_m(\chi) = S_{P_m}(R_m\chi)$ take arbitrary prescribed values. By (8),

$$\begin{aligned} \liminf_{t \rightarrow 0} \frac{p^*(f + tg) - 1}{t^2} \\ \geq \frac{1}{2} \left[\sup_{k_\perp} \left(2\langle U^*b, k_\perp \rangle - \frac{\nu}{q_0} \|k_\perp\|^2 \right) + \sum_m \sup_\delta (2v_m(\delta) - \mathcal{H}_m(\delta)) \right]. \end{aligned}$$

Base supremum. $U^*b \in E_F$; the maximizer is $k_\perp = (q_0/\nu)P^\perp U^*b \in E_F \cap e^\perp$, with value $(q_0/\nu)\|P^\perp U^*b\|^2 = q_0 H_b$.

Block supremum. Drop m . Suppose first $c_Q > 0$, so $Y > 0$. Write $\zeta' := D^{-1}\delta_Q$, $\hat{h} := D^{-1}\zeta_Q/Y = Dw_Q/\sqrt{c_Q}$, $\sigma_2 := \langle \hat{h}, \zeta' \rangle$, $\zeta'_\perp := \zeta' - \sigma_2 \hat{h}$ and $r := Y\sigma_1 - S\sigma_2$. Since Λ is 1-homogeneous, its Hessian is $(\partial_S^2 \Lambda/Y^2)(Y, -S)^{\otimes 2}$, and $Y_Q(\zeta + \delta) = Y + \sigma_2 + \|\zeta'_\perp\|^2/(2Y) + O(\|\delta\|^3)$; with Lemma 5.11, $\mathcal{H}(\delta) = (C/|\zeta|)\|\zeta'_\perp\|^2 + (\partial_S^2 \Lambda/Y^2)r^2$. Next, with $\tilde{\omega} := (\omega - dw)1_Q$, $v(\delta) = \langle D\tilde{\omega}, \zeta'_\perp \rangle + (\langle D\tilde{\omega}, \hat{h} \rangle \sigma_2 - dM\sigma_1)$. The linear form in brackets vanishes at $(\sigma_1, \sigma_2) = (S, Y)$, since there it equals $v(\zeta) = \omega(\zeta) - d|\zeta| = 0$ ($\omega(\zeta) = |\zeta|d$ because $\zeta = |\zeta|D^2w/C$ on Q); hence it equals $-(dM/Y)r$. As $\zeta'_\perp \in \hat{h}^\perp$ and r are independent variables,

$$\begin{aligned} \sup_\delta (2v - \mathcal{H}) &= \frac{|\zeta|}{C} \|P_{\hat{h}^\perp} D\tilde{\omega}\|^2 + \frac{d^2 M^2}{\partial_S^2 \Lambda} = \frac{|\zeta|}{C} \left(\|D\omega\|^2 - \frac{d^2 C^2}{c_Q} \right) + \frac{d^2 M^2 |\zeta| F_2}{C c_Q} \\ &= \frac{|\zeta|}{C} (\|D\omega\|^2 - d^2), \end{aligned}$$

where we used $\langle D\omega, Dw \rangle = dC$, $\|P_{\hat{h}^\perp} D\tilde{\omega}\|^2 = \|D\omega\|^2 - d^2 C^2/c_Q$ (a direct computation) and $C^2 = c_Q + M^2 F_2$. This is $\sigma_m H_m$. If $c_Q = 0$, then $w_Q = 0$, $d = 0$, $\Lambda(S, \cdot)$ depends on Y^2 with $\partial \Lambda / \partial (Y^2) = C/(2|\zeta|)$ at ζ , $\mathcal{H}(\delta) = (C/|\zeta|)\|D^{-1}\delta_Q\|^2$, $v(\delta) = \langle D\omega, D^{-1}\delta_Q \rangle$, and the supremum is $(|\zeta|/C)\|D\omega\|^2 = \sigma_m H_m$ directly.

Summing, $\liminf_{t \rightarrow 0} (p^*(f + tg) - 1)/t^2 \geq \frac{1}{2}(q_0 H_b + \sum_m \sigma_m H_m) = \Gamma_w(g)/2$. If $g \in \mathcal{C}(f)$, then $(p^*(f + tg) - 1)/t^2 \leq (s(t) - 1)/t^2 \rightarrow \frac{1}{2}$. $\square$

Theorem 5.13 (Tame supports are recoverable). *Let I be finite and let $f \in S_{p^*}$ have a C -tame normer without degenerate peaks, i.e. $a \in c_{00}$, K finite, every Q_m finite, $\text{Dg}_m = \emptyset$ and (MS). Then $\mathcal{C}(f) \subseteq \text{Li}_N \mathcal{C}(f'_N)$ for the canonical truncations f'_N of f . In particular $f \in \mathcal{R}$.*

Proof. By Theorem 5.6, every $g \in \mathcal{C}(f)$ is a balanced finite certificate; by Proposition 5.12, $\Gamma_w(g) \leq 1$; by Theorem 5.8, $\rho g \in \text{Li}_N \mathcal{C}(f'_N)$ for every $\rho < 1$, and the lower limit is closed. Corollary 2.3 gives $f \in \mathcal{R}$. $\square$

Remark 5.14 (On the hypotheses). (a) The finiteness of K enters Theorem 5.13 twice: in Step 1 of Theorem 5.6 and in the decoupling step of Proposition 5.12, both of which need the set J of free coordinates to be cofinite so that Lemma 1.8(b) applies. Lemma B does not exclude elements of $\text{span}\{u_{k,m}, \pi_m\} \subseteq Y$ supported in $F \cup K$ when K is infinite, and then the lower bound may fail. In a finite-dimensional analogue in which every off-support coordinate is a contact, an exact numerical computation (in that finite model only; this is not a theorem about Martín's space) gives a second-order coefficient of a pure block certificate of at most 0.296 on both sides, while $\Gamma_w = 0.485$; the mechanism is that re-splitting through the contacts is free at first order for one sign of t . Whether a finite certificate with $\Gamma_2 \leq 1 < \Gamma_w$ exists at a support with infinitely many contacts in Martín's space is *open*. Theorem 5.8 would not recover it, but Corollary 7.22 does whenever

f satisfies (BT): by Theorem 7.10 the exact one-sided coefficients are the minima $\gamma^\pm$ of Γ_w over side-admissible decompositions, which re-split through contacts with the cheap sign, and the proof of the lower bound there needs no decoupling. (b) Earlier formulations assumed $K = \emptyset$; Step 3 of Theorem 5.6 removes this. (c) The existence of tame non-attaining supports for a *given* admissible T is not known in general. In the open part of the bidual face where $K = \emptyset$, where the residuality argument of [7, Section 3] for half-peaks applies, they form a meagre set. For the operator T of Section 6 the canonical truncations of the special first row are C-tame norm-attaining points (Proposition 6.8).

6. THE RESONANCE OBSTRUCTION

Throughout this section I is finite.

6.1. The linear obstruction. For $f \in S_{p^*}$ let $\mathfrak{B}(f) := \{g \in \mathcal{C}(f) \cap S(f) : \Gamma_w(g) \leq 1\}$ if $a \in c_{00}$ (then the data of $g \in S(f)$ are unique by Lemma 1.8(a)), and $\mathfrak{B}(f) := \emptyset$ otherwise. The *intrinsic class* $\mathcal{K}(f)$ is the closed convex hull of $\text{Cert}^{\text{sh}}(f) \cup \mathfrak{B}(f)$, and the *defect* is $\text{Def}(f) := \mathcal{C}(f) \setminus \mathcal{K}(f)$.

Proposition 6.1 (Intrinsic classes). (a) $\mathcal{K}(f)$ contains every mate shown to be recoverable in Sections 3–5: $\text{cl Cert}(f)$ (hence the mates of Corollaries 3.10 and 4.2 and Theorems 4.6, 4.10, 4.12), $\text{cl Cert}^{\text{sh}}(f)$ and $\mathfrak{B}(f)$. (b) $\mathcal{K}(f) \subseteq \text{cl } S(f)$, a closed linear subspace of $\{g : g(\xi) = 0\}$. (c) If $a \in c_{00}$, then $\mathcal{K}(f) \subseteq \text{Li}_N \mathcal{C}(f'_N)$ for the canonical truncations; in any case $\text{cl Cert}^{\text{sh}}(f) \subseteq \text{Li}_n \mathcal{C}(f_n)$ for every sequence $f_n \rightarrow f$.

Proof. (a) is a summary of the quoted results ($\text{Cert}(f) \subseteq \text{Cert}^{\text{sh}}(f)$). (b) Conditions (C1), (C2) imply that $g_c \in S(f)$, for shifted certificates as well, and $\mathfrak{B}(f) \subseteq S(f)$ by definition. (c) Theorem 3.13 for Cert^{sh} , Theorem 5.8 for $\mathfrak{B}(f)$, and closedness and convexity of lower limits of convex sets. $\square$

Corollary 6.2 (Linear obstruction). $\mathcal{C}(f) \setminus \text{cl } S(f) \subseteq \text{Def}(f)$. If F and every Q_m are finite, then $S(f)$ is finite dimensional, and a mate g lies outside $\mathcal{K}(f)$ as soon as $g \notin \text{span}\{e_j^* : j \in F\} + \text{span}\{u_{k,m} : k \in Q_m\} + \text{span}\{R_m^* w_m\}$.

When some block has infinitely many strict non-peaks whose vectors are spread in $\ker \xi$, the closure of $S(f)$ may be all of $\ker \xi$ and the obstruction is void. It bites at first rows with few non-peaks. We now construct such a first row with an infinite-dimensional fibre.

6.2. An admissible operator with an exact resonance. Let $\nu_j := \|U^* e_j^*\| \rightarrow 0$ and $J_0 := \{j \geq 2 : \nu_j \leq \frac{1}{2}\}$ (cofinite). Put $\bar{\alpha} := 1/(1 + \nu_1)$, $a := \bar{\alpha} e_1^*$ (so $q^*(a) = 1$), $e := U^* e_1^*/\nu_1$, and note $(Ue)_1 = \nu_1$. Fix a bijection $\ell : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ with $\ell(1, 1) = 1$ and $\ell(n, m) < \ell(n', m)$ for $n < n'$, pairwise disjoint infinite sets $S_l \subseteq J_0$ ($l \in \mathbb{N}$), and put $h^{(l)} := \sum_{s \in S_l} 2^{-s} e_s^*$, $c_l := 2^{-(l-1)^2}$ (so $c_1 = 1$ and $\sum_{l' \geq l} c_{l'} \leq 2c_l$). Let $l_0 := \ell(2, 1)$, $K_0 := S_{l_0}$,

$$z := e_1 + 1_{K_0} \in \ell_\infty, \quad \hat{z} := z + Ue, \quad z^{(n)} := e_1 + 1_{K_0 \cap [1, n]}, \quad x^{(n)} := z^{(n)} + Ue \in c_0.$$

Note $\hat{z}_1 = x_1^{(n)} = 1 + \nu_1 = 1/\bar{\alpha}$. For $k \geq 2$, $(k, m) \neq (2, 1)$, put $\varpi_{k,m} := 2^{1-m-k} c_{\ell(k,m)} / c_{\ell(1,m)} \leq \frac{1}{4}$ and $r_{\ell(k,m)} := \sqrt{\varpi_{k,m}}$, and call (k, m) *exceptional* if $\varpi_{k,m} \geq \epsilon_*^2$, $\epsilon_* := 1/(72(1 + \|U\|))$ (finitely many). Let $\delta_l := \min(2^{-l}, 1/(8(1 + \|U\|)))$ for $l = \ell(1, m)$, $\delta_{l_0} := 1$, $\delta_l := \min(2^{-l}, 1/(16(1 + \|U\|)))$ for exceptional l , and

$\delta_l := \min(2^{-l}, r_l/(1 + \|U\|))$ otherwise. Fix a dense sequence $(y^{(i)})_{i \geq 1}$ in $c_{00} \cap S_{q^*}$ with $y^{(1)} := e_1^*/q^*(e_1^*)$, let $\Lambda_i := \{n : \text{supp } y^{(i)} \cap K_0 \cap (n, \infty) \neq \emptyset\}$ (finite), and call i *allowed* at a non-exceptional $l = \ell(k, m)$ ($k \geq 2, l \neq l_0$) if

$$\begin{aligned} \text{supp } y^{(i)} \cap S_l &= \emptyset, & 3r_l(2|\Lambda_i| + 3) &\leq \frac{1}{8}, \\ 2c_l &\leq 2^{-2s} c_{l'} \delta_{l'} & \text{for } s \in \text{supp } y^{(i)} \cap S_{l'}, l' \neq l. \end{aligned}$$

Each i is allowed at all large l , and $i = 1$ is allowed everywhere ($\Lambda_1 = \emptyset$ and $9r_l \leq \frac{1}{8}$). Fix a sequence (i_n) in which every positive integer occurs infinitely often, and let $i(n, m) := i_n$ if i_n is allowed at $\ell(n, m)$ and $i(n, m) := 1$ otherwise. Define, with n_l the q^* -norm of the bracket,

- $u_{1,m} := (e_1^*/q^*(e_1^*) + \delta_l h^{(l)})/n_l$ ($l = \ell(1, m)$) and, for exceptional (k, m) , $u_{k,m} := (e_1^*/q^*(e_1^*) + \delta_l h^{(l)})/n_l$ ($l = \ell(k, m)$);
- $u_{2,1} := (-\kappa_u e_1^* + h^{(l_0)})/n_{l_0}$, with $\kappa_u := (\|h^{(l_0)}\|_1 + \langle U^* h^{(l_0)}, e \rangle)/(1 + \nu_1)$;
- for the remaining (k, m) , with $l = \ell(k, m)$ and $y := y^{(i(k, m))}$: $u_{k,m} := (y + s_l \bar{\alpha} e_1^* + \delta_l h^{(l)})/n_l$, where $|s_l| \leq 3r_l(2|\Lambda_i| + 3)$ is chosen so that $|y(x) + s_l| \geq 3r_l$ for every x in the finite set $\{\hat{z}\} \cup \{x^{(n)} : n \in \Lambda_i\}$;

and $Te_{n,m} := c_{\ell(n,m)} u_{n,m}$. The choice of s_l is possible: the excluded values form $|\Lambda_i| + 1$ intervals of length $6r_l$ inside an interval of length $6r_l(2|\Lambda_i| + 3)$; and $(\bar{\alpha} e_1^*)(x) = \bar{\alpha} x_1 = 1$ for all these x .

Lemma 6.3 (The operator T). *T is admissible, $\Phi_m(k) = 2^{-m-k} c_{\ell(k,m)}$, and:*

- (P1) $\text{supp } u_{2,1} = \{1\} \cup K_0$, $u_{2,1}(1) < 0 < u_{2,1}(j)$ for $j \in K_0$, and $u_{2,1}(\hat{z}) = 0$;
- (P2) $|u_{1,m}(x)| \geq 7/9$ for $x \in X_0 := \{\hat{z}\} \cup \{x^{(n)} : n \in \mathbb{N}\}$ and every m ;
- (P3) $|u_{k,m}(x)| \geq \frac{3}{2} r_{\ell(k,m)}$ for $x \in X_0$, every non-exceptional (k, m) with $k \geq 2$, $(k, m) \neq (2, 1)$, and $|u_{k,m}(x)| \geq \frac{1}{2}$ for exceptional (k, m) ; in both cases $|u_{k,m}(x)| > \varpi_{k,m}$.

Proof. Norms. $q^*(h^{(l)}) \leq (1 + \|U\|)\|h^{(l)}\|_1 \leq 1 + \|U\|$, so $q^*(\delta_l h^{(l)}) \leq \frac{1}{8}$ for $l = \ell(1, m)$, $\leq \frac{1}{16}$ for exceptional l , and $\leq r_l \leq \frac{1}{72}$ otherwise; also $q^*(s_l \bar{\alpha} e_1^*) = |s_l| \leq \frac{1}{8}$. Hence $n_l \in [\frac{7}{8}, \frac{9}{8}]$ in the first two cases and $n_l \in [\frac{3}{4}, \frac{5}{4}]$ in the last. (T-a): $q^*(Te_{n,m}) = c_{\ell(n,m)} \leq 1 = c_1$.

(P1): $|\langle U^* h^{(l_0)}, e \rangle| \leq \sum_{s \in K_0} 2^{-s} \nu_s \leq \|h^{(l_0)}\|_1/2$, so $\kappa_u > 0$, and $(-\kappa_u e_1^* + h^{(l_0)})(\hat{z}) = -\kappa_u(1 + \nu_1) + \|h^{(l_0)}\|_1 + \langle U^* h^{(l_0)}, e \rangle = 0$ because $z = 1$ on K_0 .

(P2), (P3): every $x \in X_0$ satisfies $x_1 = 1/\bar{\alpha}$, $x = Ue$ on $\bigcup_{l \neq l_0} S_l$ and $|x_s| \leq \|U\|$ there. Hence $(e_1^*/q^*(e_1^*))(x) = 1$ and $|\delta_l h^{(l)}(x)| \leq \delta_l \|U\|$; this gives $|u_{1,m}(x)| \geq (1 - \frac{1}{8})/\frac{9}{8} = \frac{7}{9}$ and, for exceptional (k, m) , $\geq (1 - \frac{1}{16})/\frac{17}{16} > \frac{1}{2} \geq 2\varpi_{k,m}$. For the remaining (k, m) and $n \notin \Lambda_i$, $x^{(n)} - \hat{z} = -1_{K_0 \cap (n, \infty)}$ is annihilated by y , by e_1^* and by $h^{(l)}$, so the choice of s_l gives $|(y + s_l \bar{\alpha} e_1^*)(x)| \geq 3r_l$ for all $x \in X_0$, and $|u_{k,m}(x)| \geq (3r_l - r_l)/\frac{5}{4} = \frac{8}{5} r_l > r_l \geq \varpi_{k,m}$.

(T-d): fix m and i . Infinitely many n have $i(n, m) = i$, and along them $\|u_{n,m} - y^{(i)}\|_1 \rightarrow 0$, because $r_l, \delta_l \rightarrow 0$, $|s_l| \leq 3r_l(2|\Lambda_i| + 3) \rightarrow 0$ and $n_l \rightarrow 1$ ($l = \ell(n, m)$). Hence every $y^{(i)}$ is a limit point of $(u_{n,m})_n$.

(T-b), (T-c): let $x \in \ell_1(\mathbb{N} \times \mathbb{N})$, indexed by l , with $\mathcal{Z} := \sum_l x_l c_l u_l \in c_{00}$, and suppose $x_{l'} \neq 0$. Write $u_l = (y_l + \delta_l h^{(l)})/n_l$ with $y_l \in c_{00}$; the y_l of the first two items and the multiples of e_1^* live on $\{1\}$, and $1 \notin \bigcup S_l$. For $s \in S_{l'}$ only $h^{(l')}$ among the $h^{(l)}$ is nonzero at s , and $y_{l'}(s) = 0$ (allowedness). If $s \in \text{supp } y_l$ for some l , let $L(s)$ be the least such l ; allowedness at $L(s)$ gives $2c_{L(s)} \leq 2^{-2s} c_{l'} \delta_{l'}$, and with

$$|y_l(s)| \leq 1, n_l \geq \frac{3}{4},$$

$$|\mathcal{Z}(s)| \geq \frac{|x_{l'}|c_{l'}\delta_{l'}2^{-s}}{n_{l'}} - \frac{4}{3}\|x\|_\infty \sum_{l \geq L(s)} c_l \geq c_{l'}\delta_{l'}2^{-s} \left(\frac{|x_{l'}|}{n_{l'}} - \frac{4}{3}\|x\|_\infty 2^{-s} \right) > 0$$

for all large $s \in S_{l'}$. As $S_{l'}$ is infinite, $\mathcal{Z} \notin c_{00}$, a contradiction. Hence $x = 0$: T is injective and $Y \cap c_{00} = \{0\}$. $\square$

The construction is flexible (any finite set of prescribed vectors can be inserted, and the targets may be the same in all blocks, as in Martín's proof); only (T-a)–(T-d) and (P1)–(P3) are used below.

Lemma 6.4 (The first row). *With T as in Lemma 6.3 and any finite $I \ni 1$, put $\xi := \hat{z}/p^{**}(\hat{z})$ and $f := a + L^*J_V(L^{**}\xi)$. Then $f \in S_{p^*}$ is not norm attaining, ξ is its normer, $q_0 = 1/p^{**}(\hat{z})$, its forced data are $a = \bar{\alpha}e_1^*$, $w = J_V(L^{**}\xi)$ and $z = e_1 + 1_{K_0}$, $F = \{1\}$, and $K = K_0$ is infinite. Moreover $(2, 1)$ is a strict non-peak with $w_1(2) = 0$, every other coordinate (k, m) , $m \in I$, is a non-degenerate peak, and $S(f) = \mathbb{R}u_{2,1}$.*

Proof. $\hat{z} \in B_{\ell_\infty} + U(B_H) = B_{q^{**}}$ and $a(\hat{z}) = \bar{\alpha}\hat{z}_1 = 1 = q^*(a)$, so $q^{**}(\hat{z}) = 1$ and $p^{**}(\hat{z}) = 1 + \|L^{**}\hat{z}\|$. Then $f(\xi) = (1 + \|L^{**}\hat{z}\|)/p^{**}(\hat{z}) = 1$ and $p^*(f) \leq \max(q^*(a), \|J_V\|) = 1$ by Lemma 1.6. Proposition 1.9 gives the forced data, and $\xi \notin c_0$ shows that f does not attain its norm (a norming point in X would be the unique normer).

Blocks: $\zeta_m(k) = \lambda_{k,m}q_0u_{k,m}(\hat{z})$, and $\sigma_m \leq \|\zeta_m\|_1 \leq q_0m2^{-m}$. By Lemma 1.7, off P_m we have $|\zeta_m(k)| = \sigma_m\Phi_m(k)^2|w_m(k)|/C_m \leq \sigma_m\Phi_m(k)$, i.e. $|u_{k,m}(\xi)| \leq \sigma_m/m$, and $|u_{k,m}(\xi)| > \vartheta_m\Phi_m(k)$ implies $k \in P_m$ with positive margin. By (P2), $|u_{1,m}(\xi)| \geq \frac{7}{9}q_0 > q_02^{-m} \geq \sigma_m/m$, so $1 \in P_m$; hence $C_m \geq \Phi_m(1)M_m$ and $\vartheta_m\Phi_m(k) \leq \sigma_m\Phi_m(k)/(m\Phi_m(1)) \leq q_0\varpi_{k,m}$. By (P3), every $(k, m) \neq (2, 1)$ with $k \geq 2$ is a peak with positive margin; and $u_{1,m}(\xi)$ exceeds $q_0/2 \geq \vartheta_m\Phi_m(1)$. Finally $u_{2,1}(\xi) = 0$ by (P1), so $\zeta_1(2) = 0$, $2 \notin P_1$ and $w_1(2) = 0$. Thus $Q_1 = \{2\}$, $Q_m = \emptyset$ for $m \neq 1$, and $S(f) = (\ell_1(\{1\}) \cap \xi^\perp) + \mathbb{R}u_{2,1} = \mathbb{R}\lambda_{2,1}u_{2,1}$ because $\xi_1 \neq 0$ and $w_1(2) = 0$. $\square$

Proposition 6.5 (Two-piece mates). *Let $u := u_{2,1}$, $\lambda_0 := \lambda_{2,1} = \Phi_1(2)$, $c > 0$, $v := cu$, and for $K_1 \subseteq K_0$ put $K_2 := K_0 \setminus K_1$, $\beta := (v1_{K_1})(\hat{z})$ and $g_{K_1} := v1_{K_1} - \beta a$. There is $c_* > 0$, depending only on $\|U\|, \nu_1, M_1, C_1, \lambda_0$, such that for $0 < c \leq c_*$ every g_{K_1} lies in $\mathcal{C}(f)$, with the admissible decompositions*

$$f + tg = (a + tg) + L^*w \quad (t \geq 0), \quad f + tg = (a + t(g - v)) + L^*(w + tD) \quad (t \leq 0),$$

where $D := (c/\lambda_0)e_2$ in block 1 (so $L^*D = v$).

Proof. We use $\|xe + y\| \leq x + \langle e, y \rangle + \|y\|^2/(2x)$ for $x > 0$, $y \in H$, which follows from $(\langle e, y \rangle + \|y\|^2/(2x))^2 \geq 0$. Let $|t| \leq 1$ and $c \leq c_1 := \min(\bar{\alpha}/4, 1/(4(1 + \|U\|)))$; then $|\beta| \leq \|v\|_1\|\hat{z}\|_\infty \leq c(1 + \|U\|) \leq \frac{1}{4}$. Put $\nu := \|U^*a\| = \bar{\alpha}\nu_1$.

Side $t \geq 0$. $a + tg = (1 - t\beta)\bar{\alpha}e_1^* + tv1_{K_1}$, with $1 - t\beta \geq \frac{3}{4}$ and $v > 0$ on K_1 . So $q^*(a + tg) \leq (1 - t\beta)(\bar{\alpha} + \nu) + t(\|v1_{K_1}\|_1 + \langle e, h_1 \rangle) + \frac{2}{3}t^2\|h_1\|^2/\nu$ with $h_1 := U^*(v1_{K_1})$. Here $\bar{\alpha} + \nu = 1$ and $\|v1_{K_1}\|_1 + \langle e, h_1 \rangle = (v1_{K_1})(z + Ue) = \beta$ since $z = 1$ on K_1 . Hence $q^*(a + tg) \leq 1 + \frac{2}{3}t^2\|h_1\|^2/\nu$, and the block part is w .

Side $t = -s \leq 0$. $g - v = -\beta a - v1_{K_1}^* - v1_{K_2}$ with $v_1 := v(1) < 0$, $|v_1| \leq c$, so $a + t(g - v) = (\bar{\alpha}(1 + s\beta) + sv_1)e_1^* + sv1_{K_2}$ with first coefficient $\geq \bar{\alpha}/2$. As before, $q^*(a + t(g - v)) \leq (1 + s\beta) + sv_1/\bar{\alpha} + s(v1_{K_2})(\hat{z}) + s^2\|h_2\|^2/\nu$ with $h_2 := U^*(v1_{K_2})$, and

the first-order bracket equals $sv(\hat{z}) = scu(\hat{z}) = 0$ (P1). In block 1, $W := w_1 + tD$ differs from w_1 only at 2, where $|W(2)| = sc/\lambda_0 \leq M_1$ if $c \leq M_1\lambda_0$; the peaks keep the value M_1 , so $\|W\|_\infty = M_1$, and $\|D_1W\|^2 = C_1^2 + s^2c^2$ because $w_1(2) = 0$ and $\Phi_1(2)/\lambda_0 = 1$. Thus $N_1(W) \leq 1 + s^2c^2/(2C_1)$.

Conclusion. For $|t| \leq 1$, $p^*(f+tg) \leq 1 + t^2 \max(\|U\|^2c^2/\nu, c^2/(2C_1)) \leq 1 + \frac{3}{8}t^2 \leq s(t)$ if moreover $c^2\|U\|^2 \leq 3\nu/8$ and $c^2 \leq \frac{3}{4}C_1$. For $|t| \geq 1$, $p^*(f+tg) \leq 1 + |t|q^*(g) \leq 1 + |t| \cdot 2c(1 + \|U\|) \leq 1 + |t|(\sqrt{2} - 1) \leq s(t)$ if $c \leq 0.2/(1 + \|U\|)$, since $(s(t) - 1)/|t|$ increases. Take c_* the minimum of these bounds. $\square$

Theorem 6.6 (Nonempty defect). *For the admissible operator T of Lemma 6.3 and every finite $I \ni 1$, the first row f of Lemma 6.4 satisfies $\mathcal{K}(f) \subseteq \mathbb{R}u_{2,1}$. Let c_* be the constant of Proposition 6.5 and $0 < c \leq c_*$. Then the mates $g_{K_1} = g_{K_1}(c)$ of Proposition 6.5 satisfy $g_{K_1} \in \mathcal{C}(f)$ for every $K_1 \subseteq K_0$, and if $\emptyset \neq K_1 \neq K_0$, then $g_{K_1} \notin \mathbb{R}u_{2,1}$, so $g_{K_1} \in \text{Def}(f)$. The mates $g_{\{j\}} = cu_{2,1}(j)(e_j^* - \hat{z}_j \bar{a} e_1^*)$, $j \in K_0$, are finitely supported, and the defect spans an infinite-dimensional space.*

Proof. By Lemma 6.4 and Proposition 6.1(b), $\mathcal{K}(f) \subseteq \mathbb{R}u_{2,1}$, and $g_{K_1} \in \mathcal{C}(f)$ for $0 < c \leq c_*$ by Proposition 6.5. If $g_{K_1} = \mu u_{2,1}$, comparing coordinates $j \in K_0$ (where a vanishes and $u_{2,1}(j) > 0$) gives $c1_{K_1}(j) = \mu$ for all $j \in K_0$, which forces $K_1 \in \{\emptyset, K_0\}$. For $j \in K_0$ we have $\beta = cu_{2,1}(j)\hat{z}_j$, which gives the formula for $g_{\{j\}}$; since K_0 is infinite, $\{j\} \neq K_0$, so every $g_{\{j\}}$ lies in $\text{Def}(f)$, and these vectors are linearly independent because their e_j^* -coordinates $cu_{2,1}(j) \neq 0$ sit at distinct $j \notin \{1\}$. $\square$

Remark 6.7. (a) The mechanism is an *exact resonance*: the block vector $u_{2,1}$ of a strict non-peak with $w = 0$ is, as a base functional, supported on $F \cup K_0$ with the contact signs on K_0 , and $u_{2,1}(\hat{z}) = 0$; hence the block carrier and the contact vector represent the same functional, and splitting the contacts between the two signs of t produces switching mates. For $K_1 \in \{\emptyset, K_0\}$, g_{K_1} is a two-sided certificate. (b) If g is supported in $\{1\} \cup K_0$, $g(\xi) = 0$ and $\sup_{j \in K_0} |g_j/u_{2,1}(j)| \leq c \leq c_*$, then the decompositions $f + tg = (a + t(g \pm cu_{2,1})) + L^*(w \mp t(c/\lambda_0)e_2)$ for $\pm t \geq 0$ (block part in block 1) are estimated exactly as in the proof of Proposition 6.5, the base parts using the contacts with their signs; so, possibly after decreasing c_* by a fixed factor, $\mathcal{C}(f)$ contains an infinite-dimensional slab. Conversely, every mate is supported in $\{1\} \cup K_0$ with $g_j/u_{2,1}(j)$ bounded on K_0 (Corollary 7.23(b)). (c) Since K_0 is infinite, f is not C-tame, and Theorem 5.6 does not apply at f ; the infinite contact set is unavoidable (Proposition 6.10). But f satisfies the block-tameness condition (BT) of Section 7, which allows infinite contact sets (Corollary 7.23).

6.3. Non-recovery along canonical truncations. The defect of Theorem 6.6 is defined relative to the known intrinsic classes; a priori its elements could still be recovered along every sequence. For the canonical truncations this is not the case.

Proposition 6.8 (Non-recovery). *Let T , I and f be as in Theorem 6.6, and let $f_n := \nabla p(x^{(n)})$, $x^{(n)} = z^{(n)} + Ue$, be the canonical truncations of f . Then:*

- (a) $f_n \in \text{NA} \cap S_{p^*}$ and $f_n \rightarrow f$;
- (b) for large n the normer $x'_n := x^{(n)}/p(x^{(n)})$ of f_n is C-tame, with $Q_{n,1} = \{2\}$, $Q_{n,m} = \emptyset$ for $m \neq 1$, and no degenerate peaks;
- (c) $\mathcal{C}(f_n) \subseteq \mathbb{R}y^{(n)}$, where $y^{(n)} := \lambda_0 u_{2,1} - (\Phi_1(2)^2 w_1^{(n)}(2)/C_1^{(n)}) R_1^* w_1^{(n)} \rightarrow \lambda_0 u_{2,1}$;

- (d) $\text{Li}_n \mathcal{C}(f_n) \subseteq \mathbb{R}u_{2,1}$. In particular, for $\emptyset \neq K_1 \neq K_0$ and $\rho \in (0, 1]$, $\rho g_{K_1} \notin \text{Li}_n \mathcal{C}(f_n)$: the mates g_{K_1} are not recovered along the canonical truncations.

Proof. (a) $z^{(n)} \in c_{00}$, $\|z^{(n)}\|_\infty \leq 1$ and $z_1^{(n)} = 1 = \text{sign } a_1$, so $\nabla q(x^{(n)}) = a$ and $f_n \in \text{NA} \cap S_{p^*}$ by Proposition 1.10(c); $z^{(n)} \rightarrow z$ coordinatewise, and Proposition 1.12 gives $f_n \rightarrow f$.

(b) Let $c'_n := q(x'_n) = 1/p(x^{(n)}) \rightarrow q_0 > 0$. Since $|u_{k,m}(x^{(n)})| \leq q(x^{(n)}) = 1$, $\sigma'_{n,m} \leq c'_n m 2^{-m}$. Exactly as in the proof of Lemma 6.4, using (P2) and (P3) at $x^{(n)} \in X_0$: 1 is a peak of every block, $\vartheta'_{n,m} \Phi_m(k) \leq c'_n \varpi_{k,m}$, and every $(k, m) \neq (2, 1)$ is a peak with margin $\mu'_{k,m} \geq c'_n r_{\ell(k,m)}$ if (k, m) is non-exceptional with $k \geq 2$, and $\mu'_{k,m} \geq c'_n/4$ otherwise. As $r_{\ell(k,m)} = (2\Phi_m(k)/c_{\ell(1,m)})^{1/2} \geq (2\Phi_m(k))^{1/2}$ and $\Phi_m(k+1) \leq \Phi_m(k)/2$, for $s < c'_n/4$ we get $\sum_{\mu'_{k,m} < s} \Phi_m(k) \leq \sum_{\Phi_m(k) < s^2/(2c_n'^2)} \Phi_m(k) \leq s^2/c_n'^2 = o(s)$: (MS) holds. Next, $u_{2,1}(x^{(n)}) = u_{2,1}(\hat{z}) - \sum_{j \in K_0, j > n} u_{2,1}(j) \rightarrow 0$, while $\vartheta'_{n,1} \Phi_1(2) \rightarrow \vartheta_1 \Phi_1(2) > 0$ (Proposition 1.11); so for large n , $|u_{2,1}(x'_n)| < \vartheta'_{n,1} \Phi_1(2)$ and $(2, 1)$ is a strict non-peak. Finally $a \in c_{00}$ and the contact set of x'_n is $K_0 \cap [1, n]$, which is finite.

(c) By Theorem 5.6 at $\eta = x'_n$, $\mathcal{C}(f_n) \subseteq S(f_n) = (\ell_1(\{1\}) \cap (x'_n)^\perp) + \mathbb{R}y^{(n)} = \mathbb{R}y^{(n)}$, since $x'_n(1) \neq 0$. By Proposition 1.11, $w_1^{(n)}(2) \rightarrow w_1(2) = 0$ and $C_1^{(n)} \rightarrow C_1 > 0$, so $y^{(n)} \rightarrow \lambda_0 u_{2,1}$.

(d) If $\gamma_n y^{(n)} \rightarrow g$, then (γ_n) is bounded because $\|y^{(n)}\| \rightarrow \lambda_0 \|u_{2,1}\| > 0$; along a subsequence $\gamma_n \rightarrow \gamma$ and $g = \gamma \lambda_0 u_{2,1}$. The last claim follows from Theorem 6.6. $\square$

Thus, for this T , the mates recovered along *every* sequence are contained in the certificate line $\mathbb{R}u_{2,1}$: universal recovery fails, and a recovery of the switching mates must use norm-attaining approximants whose certificate spans grow, for instance by putting base mass on the contact windows. This is done in Section 7 (Remark 6.9). By Remark 5.7, along any sequence of norm-attaining approximants with C-tame normers, recovery is a linear-algebra design problem: $\text{Li}_n \mathcal{C}(f_n) \subseteq \text{Li}_n S(f_n)$.

Remark 6.9 (Engineered approximants). Proposition 1.12 shows that, outside a finite window containing $\text{supp } a'_n$, the base coordinates z'_n of a norm-attaining approximant are free. Section 7 uses this freedom: engineered approximants put masses on finitely many contacts and truncate z far out, and transfer peaks rebalance levels between the base and the blocks. It is proved there that every mate of the first row f of Theorem 6.6 is recovered (Corollary 7.23); more generally every (BT) point lies in $\mathcal{R}$ (Corollary 7.22), and two-piece mates are recovered at every f with finite base support under the conditions of Theorem 7.18. Thus the defect of Theorem 6.6 is a defect of the intrinsic mechanisms only, and Proposition 6.8 is a statement about the canonical truncations, not about recoverability.

6.4. Exact resonances.

Proposition 6.10 (Exact resonances need infinite one-sided sets). *Let $f \in S_{p^*}$, $\Omega \in V^*$ with $D := L^* \Omega \neq 0$, and suppose that for a sequence $s_n \rightarrow 0^+$,*

$$q^*(a + s_n D) \leq s(2s_n) \quad \text{and} \quad N_m(w_m - s_n \Omega_m) \leq s(2s_n) \quad (m \in I).$$

Then $\sum_{j \notin F} (|D_j| - z_j D_j) = 0$: off F , D is supported in the contact set K with $z_j D_j = |D_j|$. Since $D \in Y \setminus \{0\}$ is not finitely supported, $F \cup K$ is infinite. In particular there are no such transfers when $a \in c_{00}$ and K is finite (e.g. at every norm-attaining f).

Proof. By Lemma 3.3 (dropping nonnegative terms), $q^*(a + sD) \geq 1 + sD(\hat{z}) + s \sum_{j \notin F} (|D_j| - z_j D_j)$ for $s > 0$, and $N_m(w_m - s\Omega_m) \geq \langle w_m - s\Omega_m, \zeta_m \rangle / \sigma_m = 1 - s\langle \Omega_m, \zeta_m \rangle / \sigma_m$. With $s = s_n$, multiply the first inequality by q_0 and the m -th by σ_m and add: since $q_0 D(\hat{z}) = D(\xi) = \langle \Omega, L^{**}\xi \rangle = \sum_m \langle \Omega_m, \zeta_m \rangle$ and $q_0 + \sum_m \sigma_m = 1$,

$$q_0 s_n \sum_{j \notin F} (|D_j| - z_j D_j) \leq s(2s_n) - 1 \leq 2s_n^2.$$

Divide by s_n and let $n \rightarrow \infty$. Each term $|D_j| - z_j D_j$ is nonnegative and vanishes only if $D_j = 0$ or $|z_j| = 1$ with $z_j D_j = |D_j|$. $\square$

Corollary 6.11 (Two-sided linear decompositions). *Let $g \in \mathcal{C}(f)$ and suppose there are $\tau > 0$ and $(B_\pm, \Omega_\pm) \in \ell_1 \times V^*$ with $g = B_\pm + L^*\Omega_\pm$, $\max(q^*(a + tB_+), \|w + t\Omega_+\|) \leq s(t)$ for $0 \leq t \leq \tau$ and $\max(q^*(a + tB_-), \|w + t\Omega_-\|) \leq s(t)$ for $-\tau \leq t \leq 0$. If $F \cup K$ is finite, then $B_+ = B_-$, $\Omega_+ = \Omega_-$, and $g \in \text{cl Cert}(f)$.*

Proof. For $0 < t \leq \tau$ average the decompositions of $f + tg$ and $f - tg$: $f = (a + \frac{t}{2}D) + L^*(w - \frac{t}{2}\Omega_D)$ with $D := B_+ - B_-$, $\Omega_D := \Omega_- - \Omega_+$, $L^*\Omega_D = D$, and both component norms are at most $s(t)$ by convexity. If $D \neq 0$, Proposition 6.10 with $s_n = t_n/2$ contradicts the finiteness of $F \cup K$. Hence $D = 0$, and $\Omega_+ = \Omega_-$ by injectivity of L^* . The common decomposition makes g locally split, and Theorem 4.10 applies. $\square$

Remark 6.12 (Resonance types; the sign rule). The transfer in the proof of Corollary 6.11 exists for every mate and every pair of admissible decompositions at scales $\pm t$; on the contacts, for instance, the cheaply used parts of B_+ and of B_- have opposite signs, so they add up in $D = B_+ - B_-$. (A general “matching at scale” statement for all one-sided resources is only available as a sketch and is not used.) A mate outside $\mathcal{K}(f)$ therefore needs *resonances*: cheap transfers with large one-sided mass at arbitrarily small scales. Proposition 6.10 shows that exact (scale-free) resonances require infinitely many contacts (or near-flip coordinates of F when $a \notin c_{00}$); Theorem 6.6 realizes one. Approximate resonances run through one-sided block carriers: weak peaks and near-threshold strict non-peaks at depth comparable to the scale. For these we only record an elementary fact: if $u_{k,m}(\xi) \neq 0$, then $\text{sign } w_m(k) = \text{sign } u_{k,m}(\xi)$ (on P_m by Lemma 1.7, off P_m because $w_m(k) = C_m \zeta_m(k) / (\Phi_m(k)^2 \sigma_m)$). Consequently a one-sided usage of (k, m) pushes in the direction $-\text{sign } u_{k,m}(\xi)$, and two near-threshold carriers whose vectors differ by $o(\Phi)$ in the ξ -direction (for instance Martín’s cross-block near duplicates with small perturbations) serve the same side of $t = 0$ and cannot switch with each other. This is a heuristic guide; no classification of approximate resonances is proved here. When the gaps of the carriers are not summable, Theorem 4.12 puts the corresponding mates in $\text{cl Cert}(f)$; the summable case and the weak-peak case are open (Remark 4.13).

7. ENGINEERED APPROXIMANTS

Throughout this section the block set I is finite and T is an arbitrary admissible operator. We prove that the switching mates of Section 6 are recovered, and much more: every first row satisfying a block-tameness condition that allows arbitrary contact sets lies in $\mathcal{R}$ (Corollary 7.22). The approximants are norm attaining and are *engineered*: unlike the canonical truncations of Section 5, they put base mass on finitely many contacts. The mechanism that makes this work is *first-order rebalancing* through the transfer peaks of Theorem 5.8: a linear mismatch between

the base part and the block parts of a decomposition is a pure shift of levels, and transfer peaks move levels between the base and the blocks at a relative cost that can be fixed in advance.

7.1. Levels, excesses and rebalancing. Let $f \in S_{p^*}$ have forced data as in Section 1; the results of this subsection apply both to non-attaining f (normer $\xi \in S_{p^{**}}$) and to norm-attaining $f' = \nabla p(x')$ (normer $x' \in S_p$, forced data at x' , with $\hat{z}' = x'/q(x')$). For $A \in \ell_1$ and $W_m \in \ell_\infty$ put

$$\mathfrak{l}_b(A) := (A - a)(\hat{z}), \quad G_b(A) := q^*(A) - A(\hat{z}) = q^*(A) - 1 - \mathfrak{l}_b(A),$$

$$\mathfrak{l}_m(W_m) := \frac{\langle W_m - w_m, \zeta_m \rangle}{\sigma_m},$$

$$G_m(W_m) := N_m(W_m) - \frac{\langle W_m, \zeta_m \rangle}{\sigma_m} = N_m(W_m) - 1 - \mathfrak{l}_m(W_m).$$

The *first-order terms* $\mathfrak{l}$ are linear and the *excesses* G are nonnegative: $q^{**}(\hat{z}) = 1$ and $|\langle W, \zeta_m \rangle| \leq N_m(W)|\zeta_m|$.

Lemma 7.1 (Bookkeeping and exact base excess). (a) If $f + h = A + \sum_m R_m^* W_m$ with $h \in X^*$, then $q_0 \mathfrak{l}_b(A) + \sum_m \sigma_m \mathfrak{l}_m(W_m) = h(\xi)$.

(b) For $B \in \ell_1$, $G_b(a + B) = \text{Exc}(B) + \nu \Psi(U^* B / \nu)$, where $\text{Exc}(B) := \sum_j (|a_j + B_j| - |a_j| - z_j B_j) \geq 0$ and Ψ is the function of Lemma 3.3. Moreover $\text{Exc}(B + B') \leq \text{Exc}(B) + 2\|B'\|_1$, and if $|B_j| \leq |a_j|$ for $j \in F$, then $\text{Exc}(B) = \sum_{j \notin F} (|B_j| - z_j B_j)$.

(c) For every block, $G_m(W) = \sum_{k \in P_m} |\alpha_m(k)| (\|W\|_\infty - \text{sign}(w_m(k))W(k)) + (\|D_m W\|_2 - \langle D_m W, D_m w_m \rangle / C_m)$, and both terms are nonnegative.

Proof. (a) $q_0 \mathfrak{l}_b(A) = (A - a)(\xi)$ and $\sigma_m \mathfrak{l}_m(W_m) = \langle W_m - w_m, R_m^{**} \xi \rangle$; their sum is $(A + L^* W - a - L^* w)(\xi) = h(\xi)$. (b) This is (3) with $t = 1$: on F , $z_j = \text{sign } a_j$ and $|a_j + B_j| - |a_j| - z_j B_j = 2(-\text{sign}(a_j)B_j - |a_j|)_+$; off F the summand is $|B_j| - z_j B_j$. Each summand is ≥ 0 because $|z_j| \leq 1$, and it changes by at most $2|B'_j|$ when B_j is replaced by $B_j + B'_j$. If $|B_j| \leq |a_j|$ the summand at $j \in F$ vanishes. (c) By Lemma 1.7, $\zeta_m / \sigma_m = \alpha_m + D_m^2 w_m / C_m$ with $\|\alpha_m\|_1 = 1$, α_m supported in P_m with the signs of w_m ; hence $\langle W, \zeta_m \rangle / \sigma_m = \sum_{P_m} |\alpha_m(k)| \text{sign}(w_m(k))W(k) + \langle D_m W, D_m w_m \rangle / C_m$, and $\|W\|_\infty = \sum_{P_m} |\alpha_m(k)| \|W\|_\infty$. The second term is ≥ 0 by Cauchy–Schwarz. $\square$

Lemma 7.2 (Transfer vectors). Fix a block and drop its index. Let $\gamma \in (0, M)$, $L := \{k : |w(k)| \geq \gamma\}$, let $k_* \in L$ with $w(k_*) = M$, let $\Lambda \geq 0$, and put

$$y^\downarrow := \text{sign}(w)1_L + \Lambda e_{k_*}, \quad y^\uparrow := \text{sign}(w)1_L - \Lambda e_{k_*}.$$

Let $W \in \ell_\infty$ with $\|W - w\|_\infty \leq \eta \leq (M - \gamma)/4$.

- (a) If $\varepsilon \geq 0$, $\varepsilon(2 + \Lambda) \leq \gamma - \eta$ and $\varepsilon \leq (M - \gamma)/4$, then $\|W - \varepsilon y^\downarrow\|_\infty \leq \|W\|_\infty - \varepsilon$.
- (b) If $\varepsilon \leq 0$ and $|\varepsilon|(2 + \Lambda) \leq \gamma - \eta$, then $\|W - \varepsilon y^\uparrow\|_\infty \leq \|W\|_\infty - \varepsilon$.
- (c) If $\|D(W - w)\|_2 \leq C/2$, then for all $y \in \ell_\infty$ and $\varepsilon \in \mathbb{R}$,

$$\|D(W - \varepsilon y)\|_2 \leq \|DW\|_2 - \varepsilon \frac{\langle Dw, Dy \rangle}{C} + \frac{4|\varepsilon| \|D(W - w)\|_2 \|Dy\|_2 + \varepsilon^2 \|Dy\|_2^2}{C}.$$

Proof. We may assume $\varepsilon \neq 0$; then $\eta < \gamma$. For $k \in L$, $|W(k) - w(k)| \leq \eta < \gamma \leq |w(k)|$, so $\text{sign } W(k) = \text{sign } w(k)$ and $|W(k)| \geq \gamma - \eta$; also $\|W\|_\infty \geq W(k_*) \geq M - \eta$.

(a) For $k \in L \setminus \{k_*\}$, $|W(k)| \geq \gamma - \eta \geq \varepsilon$, so $|W(k) - \varepsilon \text{sign } w(k)| = |W(k)| - \varepsilon$. At k_* , $W(k_*) - \varepsilon(1 + \Lambda) \leq \|W\|_\infty - \varepsilon$, and $W(k_*) - \varepsilon(1 + \Lambda) \geq -(\|W\|_\infty - \varepsilon)$

because $W(k_*) + \|W\|_\infty \geq 2(\gamma - \eta) \geq \varepsilon(2 + \Lambda)$. For $k \notin L$, $|W(k)| < \gamma + \eta \leq M - \eta - \varepsilon \leq \|W\|_\infty - \varepsilon$, since $2\eta + \varepsilon \leq \frac{3}{4}(M - \gamma)$.

(b) Put $e := -\varepsilon > 0$. For $k \in L \setminus \{k_*\}$, $|W(k) + e \operatorname{sign} w(k)| = |W(k)| + e$. At k_* , $W(k_*) + e(1 - \Lambda) \leq \|W\|_\infty + e$, and $W(k_*) + e(1 - \Lambda) + \|W\|_\infty + e \geq 2(\gamma - \eta) - e(\Lambda - 2) \geq 0$. For $k \notin L$, $|W(k)| \leq \|W\|_\infty + e$.

(c) Put $\mathbf{a} := DW$, $\mathbf{b} := Dw$ (so $\|\mathbf{b}\| = C$ and $\|\mathbf{a}\| \geq C/2$). From $\sqrt{A^2 + x} \leq A + x/(2A)$, $\|\mathbf{a} - \varepsilon Dy\| \leq \|\mathbf{a}\| - \varepsilon \langle \mathbf{a}, Dy \rangle / \|\mathbf{a}\| + \varepsilon^2 \|Dy\|^2 / (2\|\mathbf{a}\|)$, and

$$\left| \frac{\langle \mathbf{a}, Dy \rangle}{\|\mathbf{a}\|} - \frac{\langle \mathbf{b}, Dy \rangle}{\|\mathbf{b}\|} \right| \leq \frac{\|\mathbf{a} - \mathbf{b}\| \|Dy\|}{\|\mathbf{a}\|} + \frac{\|Dy\| \|\mathbf{b}\| - \|\mathbf{a}\|}{\|\mathbf{a}\|} \leq \frac{4\|\mathbf{a} - \mathbf{b}\| \|Dy\|}{C}. \quad \square$$

Proposition 7.3 (Rebalancing). *Let $f \in S_{p^*}$, $h \in X^*$ with $h(\xi) = 0$, and $f + h = A + \sum_{m \in I} R_m^* W_m$ with $A \in \ell_1$, $W_m \in \ell_\infty$. For each $m \in I$ let $y_m \in \ell_\infty$ and $\varepsilon_m \in \mathbb{R}$ satisfy*

$$(9) \quad \|W_m - \varepsilon_m y_m\|_\infty \leq \|W_m\|_\infty - \varepsilon_m, \quad \|D_m(W_m - w_m)\|_2 \leq C_m/2.$$

Put $Y_m := \langle y_m, \zeta_m \rangle$, $e_{y,m} := 1 + \langle D_m w_m, D_m y_m \rangle / C_m$, the inefficiency

$$\iota_m := \frac{|Y_m - \sigma_m e_{y,m}|}{q_0} + q^* \left(R_m^* y_m - \frac{Y_m}{q_0} a \right),$$

and $\lambda := \sum_m \varepsilon_m Y_m / q_0$; assume $\lambda \geq -1$. Let $\hat{G}_b \geq G_b(A)$ and $\hat{G}_m \geq G_m(W_m)$, put $\hat{\Gamma} := q_0 \hat{G}_b + \sum_m \sigma_m \hat{G}_m$, and assume that $e_{y,m} > 0$ and

$$\varepsilon_m = (\iota_m(W_m) + \hat{G}_m - \hat{\Gamma}) / e_{y,m} \quad (m \in I).$$

Then $p^*(f + h) \leq 1 + \hat{\Gamma} + E$, where

$$E := \sum_m |\varepsilon_m| \iota_m + |\lambda| (|q^*(A) - 1| + q^*(A - a)) + \max_m \frac{4|\varepsilon_m| \|D_m(W_m - w_m)\|_2 \|D_m y_m\|_2 + \varepsilon_m^2 \|D_m y_m\|_2^2}{C_m}.$$

Proof. Put $A' := A + \sum_m \varepsilon_m R_m^* y_m$ and $W'_m := W_m - \varepsilon_m y_m$; then $A' + \sum_m R_m^* W'_m = f + h$, and Lemma 1.6 gives $p^*(f + h) \leq \max(q^*(A'), \max_m N_m(W'_m))$.

Blocks. By (9) and Lemma 7.2(c), $N_m(W'_m) \leq N_m(W_m) - \varepsilon_m e_{y,m} + E$, and $N_m(W_m) = 1 + \iota_m(W_m) + G_m(W_m) \leq 1 + \iota_m(W_m) + \hat{G}_m$; by the choice of ε_m the right side is at most $1 + \hat{\Gamma} + E$.

Base. Since $(R_m^* y_m)(\xi) = Y_m$, we have $R_m^* y_m = (Y_m/q_0)a + e_m$ with $e_m := R_m^* y_m - (Y_m/q_0)a$, so $A' = (1 + \lambda)a + B + \sum_m \varepsilon_m e_m$ with $B := A - a$. For $\lambda \geq 0$, $q^*((1 + \lambda)a + B) \leq q^*(a + B) + \lambda$; for $-1 \leq \lambda < 0$, $(1 + \lambda)a + B = (1 + \lambda)(a + B) - \lambda B$ gives $q^*((1 + \lambda)a + B) \leq q^*(a + B) + \lambda + |\lambda|(|q^*(a + B) - 1| + q^*(B))$. Writing $\lambda = \sum_m \varepsilon_m \sigma_m e_{y,m} / q_0 + \sum_m \varepsilon_m (Y_m - \sigma_m e_{y,m}) / q_0$, we get

$$q^*(A') \leq q^*(A) + \sum_m \frac{\varepsilon_m \sigma_m e_{y,m}}{q_0} + E.$$

By the choice of ε_m and Lemma 7.1(a) (with $h(\xi) = 0$),

$$\sum_m \frac{\varepsilon_m \sigma_m e_{y,m}}{q_0} = \frac{-q_0 \iota_b(A) + (\hat{\Gamma} - q_0 \hat{G}_b) - (1 - q_0) \hat{\Gamma}}{q_0} = -\iota_b(A) - \hat{G}_b + \hat{\Gamma},$$

using $\sum_m \sigma_m = 1 - q_0$. Since $q^*(A) \leq 1 + \iota_b(A) + \hat{G}_b$, we obtain $q^*(A') \leq 1 + \hat{\Gamma} + E$. $\square$

Thus, up to the error E , every piece of the rebalanced decomposition sits at the *mass-weighted average* $\hat{\Gamma}$ of the estimated excesses, whatever the first-order terms $\mathfrak{l}_m$ are; the price of moving a level ε_m is $|\varepsilon_m|\iota_m$. The next lemma provides transfer vectors of arbitrarily small inefficiency; it is Step 2 of the proof of Theorem 5.8, for both signs.

Lemma 7.4 (Transfer data). *Let $f \in S_{p^*}$, $m \in I$ and $\gamma \in (0, M_m)$ with $|w_m(k)| \neq \gamma$ for all k ; put $L := \{k : |w_m(k)| \geq \gamma\}$. There is $\Lambda_0 < \infty$, depending only on f, m, γ , such that for every $\eta_1 \in (0, \frac{1}{2}]$ and $k_0 \in \mathbb{N}$ there are, for $\varsigma \in \{\downarrow, \uparrow\}$, indices $k_*^\varsigma \geq k_0$ and numbers $\Lambda^\varsigma \geq 0$ with the following properties, where y^ς is the vector of Lemma 7.2 built with $(\gamma, k_*^\varsigma, \Lambda^\varsigma)$ and $Y^\varsigma, e_y^\varsigma, \iota^\varsigma$ are as in Proposition 7.3:*

- (a) $k_*^\varsigma \in P_m$, $w_m(k_*^\varsigma) = M_m$, and its margin $\mu_*^\varsigma := \mu_{k_*^\varsigma, m}$ is positive;
- (b) $Y^\varsigma = \sigma_m e_y^\varsigma \pm \Lambda^\varsigma \lambda_{k_*^\varsigma, m} \mu_*^\varsigma$ (+ for $\downarrow$, - for $\uparrow$), and $\Lambda^\varsigma \lambda_{k_*^\varsigma, m} \leq \Lambda_0$;
- (c) $\iota^\varsigma \leq \eta_1$;
- (d) $|e_y^\varsigma - e_y^0| \leq \eta_1$, where $e_y^0 := 1 + \sum_{k \in L} \Phi_m(k)^2 |w_m(k)| / C_m \geq 1$; and $\|D_m y^\varsigma\|_2 \leq \|\Phi_m\|_2 + \Lambda_0 / m$.

Proof. Drop m . Put $\hat{\pi} := R^*(\text{sign}(w)1_L) \in \ell_1$ and $\tau_* := (\hat{\pi}(\xi)/q_0)a - \hat{\pi}$, so $\tau_*(\xi) = 0$, and $\Lambda_0 := 2(q^*(\tau_*) + 1)$. For $\delta_0 \in (0, 1]$ put $\mathcal{T}^\downarrow := \tau_* + \delta_0(q^*(\tau_*) + 1)a$ and $\mathcal{T}^\uparrow := -\tau_* + \delta_0(q^*(\tau_*) + 1)a$. Then $\mathcal{T}^\varsigma(\xi) = \delta_0(q^*(\tau_*) + 1)q_0 > 0$ and $\ell^\varsigma := q^*(\mathcal{T}^\varsigma) \in (0, \Lambda_0]$. Recall that $|v(\xi)| = q_0|v(\hat{z})| \leq q_0q^*(v)$ for $v \in \ell_1$, because $q^{**}(\hat{z}) = 1$.

Fix ς and drop it. Let $\delta_1 \in (0, \mathcal{T}(\xi)/(2\ell q_0))$. By (T-d) the tail $\{u_k : k \geq k_0\}$ is dense in S_{q^*} , and $\Phi(k) \rightarrow 0$; choose $k_* \geq k_0$ with $q^*(u_{k_*} - \mathcal{T}/\ell) \leq \delta_1$, $\vartheta\Phi(k_*) < \mathcal{T}(\xi)/(2\ell)$ and $\ell\Phi(k_*)M/(mC) \leq \eta_1$. Then $u_{k_*}(\xi) \geq \mathcal{T}(\xi)/\ell - \delta_1 q_0 \geq \mathcal{T}(\xi)/(2\ell) > \vartheta\Phi(k_*)$, so by Lemma 1.7 k_* is a peak with $w(k_*) = M$ and positive margin μ_* :

- (a). Put $\Lambda := \ell/\lambda_{k_*}$.

- (b) By Lemma 1.7, $\langle y, \zeta \rangle = \sigma(\langle y, \alpha \rangle + \langle Dy, Dw \rangle / C)$. As $P \subseteq L$ and α lives on P with the signs of w , $\langle \text{sign}(w)1_L, \alpha \rangle = \|\alpha\|_1 = 1$, and $\alpha(k_*) \geq 0$; so $Y = \sigma e_y \pm \sigma \Lambda \alpha(k_*) = \sigma e_y \pm \Lambda \lambda_{k_*} \mu_*$ by (2). Also $\Lambda \lambda_{k_*} = \ell \leq \Lambda_0$.

- (c) For $\downarrow$, $R^*y = \hat{\pi} + \ell u_{k_*} = \hat{\pi} + \mathcal{T} + \ell(u_{k_*} - \mathcal{T}/\ell) = (\hat{\pi}(\xi)/q_0 + \delta_0(q^*(\tau_*) + 1))a + \ell(u_{k_*} - \mathcal{T}/\ell)$; for $\uparrow$, $R^*y = \hat{\pi} - \ell u_{k_*}$ and the same computation applies with $-\tau_*$. In both cases $R^*y - (Y/q_0)a = \varrho - (\varrho(\xi)/q_0)a$ with $\varrho := \pm \ell(u_{k_*} - \mathcal{T}/\ell)$, whose q^* -norm is at most $2q^*(\varrho) \leq 2\ell\delta_1$. Moreover $\ell\mu_* \leq \ell u_{k_*}(\xi) \leq \mathcal{T}(\xi) + \ell\delta_1 q_0$, so $\iota \leq \delta_0(q^*(\tau_*) + 1) + 3\Lambda_0\delta_1$. Choosing first δ_0 and then δ_1 small gives $\iota \leq \eta_1$.

- (d) $e_y - e_y^0 = \pm \Lambda \Phi(k_*)^2 M / C = \pm \ell \Phi(k_*) M / (mC)$, since $\lambda_{k_*} = m\Phi(k_*)$; and $\|Dy\|_2 \leq \|\Phi\|_2 + \Lambda \Phi(k_*) = \|\Phi\|_2 + \ell/m$. $\square$

Lemma 7.5 (Persistence of transfer data). *Let $f_n \rightarrow f$ in S_{p^*} , and let $(\gamma, k_*^\varsigma, \Lambda^\varsigma)$ be transfer data of f in block m as in Lemma 7.4. Put $L_n := \{k : |w_{n,m}(k)| \geq \gamma\}$ and $y_n^\varsigma := \text{sign}(w_{n,m})1_{L_n} \pm \Lambda^\varsigma e_{k_*^\varsigma}$. Then for n large: k_*^ς is a peak of $w_{n,m}$ with $w_{n,m}(k_*^\varsigma) = M_{n,m}$ and positive margin, and $P_{n,m} \subseteq L_n$; and, computing the quantities of Proposition 7.3 at f_n , $Y_n^\varsigma/q_0^{(n)} \rightarrow Y^\varsigma/q_0$, $e_{y,n}^\varsigma \rightarrow e_y^\varsigma$, $\iota_n^\varsigma \rightarrow \iota^\varsigma$, and $D_m y_n^\varsigma \rightarrow D_m y^\varsigma$ in ℓ_2 .*

Proof. By Proposition 1.11, $\xi_n \rightarrow \xi$ weak*, $w_{n,m} \rightarrow w_m$ coordinatewise, $D_m w_{n,m} \rightarrow D_m w_m$, and the scalar data and a_n converge. Since $u_{k_*,m} \in \ell_1$, $u_{k_*,m}(\xi_n) \rightarrow u_{k_*,m}(\xi)$, and $\vartheta_{n,m} \rightarrow \vartheta_m$; so the strict threshold inequality of Lemma 1.7 at k_* persists, with the positive sign. Peaks of $w_{n,m}$ have modulus $M_{n,m} \rightarrow M_m > \gamma$. As $|w_m(k)| \neq \gamma$, $y_n^\varsigma(k) \rightarrow y^\varsigma(k)$ for every k , with $|y_n^\varsigma(k)| \leq 1$ off k_* ; dominated convergence with the summable majorants $\lambda_{k,m}\|u_{k,m}\|_1$ and $\Phi_m(k)$ gives $R_m^* y_n^\varsigma \rightarrow R_m^* y^\varsigma$

in ℓ_1 and $D_m y_n^\zeta \rightarrow D_m y^\zeta$ in ℓ_2 . Hence $Y_n = (R_m^* y_n)(\xi_n) \rightarrow Y$, and all the listed quantities converge. $\square$

Lemma 7.6 (Assembly). *Let $f, f' \in S_{p^*}$, $g \in \mathcal{C}(f)$, $\rho \in (0, 1)$, $g' \in X^*$, $0 < \delta \leq 1$ and $0 < T_0 \leq 1$ with $T_0^2 \leq 4\delta$. If $p^*(f' + \tau g') \leq 1 + \frac{\tau^2}{2}(1 - \delta)$ for $|\tau| \leq T_0$, $p^*(f' - f) \leq (1 - \rho^2)T_0^2/6$ and $p^*(g' - \rho g) \leq (1 - \rho^2)T_0/6$, then $g' \in \mathcal{C}(f')$. If moreover f' attains its norm, then (f', g') attains its norm and $\|(f', g') - (f, \rho g)\| \leq p^*(f' - f) + p^*(g' - \rho g)$.*

Proof. For $|\tau| \leq T_0$: $\frac{\tau^2}{2}\delta \geq \frac{\tau^4}{8}$, so the bound is at most $1 + \frac{\tau^2}{2} - \frac{\tau^4}{8} \leq s(\tau)$. For $|\tau| \geq T_0$, $p^*(f' + \tau g') \leq s(\rho\tau) + (1 - \rho^2)(T_0^2 + |\tau|T_0)/6$, and the last term is at most $(1 - \rho^2)\min(\tau^2, |\tau|)/3$; conclude with Lemma 3.6(a). The last assertions are Proposition 1.13(d) and $\|(f_1, g_1)\| \leq p^*(f_1) + p^*(g_1)$. $\square$

7.2. Two-piece data and the exact one-sided coefficient.

Definition 7.7 (Side-admissible data, two-piece data). Let $f \in S_{p^*}$ with F finite, and $g \in X^*$. A pair (b, ω) with $b \in \ell_1$ and $\omega = (\omega_m)_{m \in I}$, $\omega_m \in c_{00}$, $\text{supp } \omega_m \subseteq Q_m$, represents g if $g = b + \sum_m R_m^*(\omega_m - d_m(\omega_m)w_m)$, where $d_m(\omega) := \langle D_m w_m, D_m \omega \rangle / C_m$. It is *side-+ admissible* (resp. *side-− admissible*) if moreover $b_j = 0$ for $j \notin F \cup K$ and $z_j b_j \geq 0$ (resp. $z_j b_j \leq 0$) for $j \in K$. *Two-piece data* for g are a side-+ admissible pair (b^+, ω^+) and a side-− admissible pair (b^-, ω^-) ; put $\Delta d_m := d_m(\omega_m^-) - d_m(\omega_m^+)$,

$$\Gamma_w(b, \omega) := q_0 h(b) + \sum_m \sigma_m H_m(\omega_m), \quad \kappa_w := \max(\Gamma_w(b^+, \omega^+), \Gamma_w(b^-, \omega^-)).$$

The data are *d-neutral* if $\Delta d_m = 0$ for all m . For g with $g(\xi) = 0$ the *one-sided coefficients* are $\gamma^\pm(g) := \inf\{\Gamma_w(b, \omega) : (b, \omega) \text{ side-}\pm \text{ admissible for } g\} \in [0, \infty]$.

If (b, ω) represents g , then $b(\xi) = g(\xi)$, because $\langle \omega_m - d_m(\omega_m)w_m, \zeta_m \rangle = 0$ (Lemma 3.2). A pair with b supported in F is admissible on both sides; if moreover $g(\xi) = 0$ (equivalently $b(\xi) = 0$), it is then a balanced finite certificate in the sense of Definition 5.5, and $\Gamma_w(b, \omega) = \Gamma_w(g)$. The mates of Proposition 6.5 carry the two-piece data $(b^+, \omega^+) = (g_{K_1}, 0)$ and $(b^-, \omega^-) = (g_{K_1} - cu_{2,1}, (c/\lambda_0)e_2 \text{ in block } 1)$, which are *d-neutral* because $w_1(2) = 0$. Since h and H_m are positive semidefinite quadratic forms, Γ_w is convex in (b, ω) .

Definition 7.8 (Block-tame points). $f \in S_{p^*}$ satisfies (BT) if F is finite, every Q_m is finite, there are no degenerate peaks, and (MS) holds at ξ . Nothing is assumed about the contact set K , which may be infinite.

Thus the points of Theorem 5.13 are the (BT) points with K finite, and the first row f of Lemma 6.4 is a (BT) point with $K = K_0$ infinite (Corollary 7.23 below).

Proposition 7.9 (One-sided upper bound). *Let $f \in S_{p^*}$ with F finite, $g \in X^*$ with $g(\xi) = 0$, and let (b, ω) be side-+ admissible for g . Then*

$$\limsup_{t \rightarrow 0^+} \frac{2(p^*(f + tg) - 1)}{t^2} \leq \Gamma_w(b, \omega),$$

and the same holds for $t \rightarrow 0^-$ and side-− admissible pairs. In particular, for every balanced finite certificate $g \in S(f)$ with $a \in c_{00}$, $\limsup_{t \rightarrow 0} 2(p^(f + tg) - 1)/t^2 \leq \Gamma_w(g)$.*

Proof. Let $t > 0$; the case $t < 0$ is the same with $-g$, $-b$ and $-\omega$, which turns side $-$ into side $+$. We have the exact decomposition $f + tg = A + \sum_m R_m^* W_m$, $A := a + tb$, $W_m := w_m + t(\omega_m - d_m w_m)$, $d_m := d_m(\omega_m)$.

First-order terms and excesses. $\mathfrak{l}_b(A) = tb(\hat{z}) = 0$, since $b(\xi) = g(\xi) = 0$, and $\mathfrak{l}_m(W_m) = 0$ by Lemma 3.2. For $t\|b\|_\infty \leq \min_F |a_j|$ there are no flips on F ; off F , b vanishes on J and $z_j b_j = |b_j|$ on K ; so $\text{Exc}(tb) = 0$ by Lemma 7.1(b), and by Lemma 3.3, $G_b(A) = \nu \Psi(tU^*b/\nu) \leq \frac{t^2}{2} h(b)(1 + 2t\|U^*b\|/\nu)$ for $t\|U^*b\| \leq \nu/2$. By Lemma 3.4(d), for t below the coordinatewise radius of ω_m , $G_m(W_m) = N_m(W_m) - 1 \leq \frac{t^2}{2} H_m(\omega_m)(1 + 2t|d_m|M_m/C_m)$. Hence there is K_1 with $G_b(A) \leq \hat{G}_b := \frac{t^2}{2} h(b)(1 + K_1 t)$ and $G_m(W_m) \leq \hat{G}_m := \frac{t^2}{2} H_m(\omega_m)(1 + K_1 t)$ for small t , and $\hat{\Gamma} = \frac{t^2}{2} \Gamma_w(b, \omega)(1 + K_1 t)$.

Rebalancing. Fix $\eta_1 \in (0, \frac{1}{4}]$. In every block choose $\gamma_m \in (0, M_m)$ with $|w_m(k)| \neq \gamma_m$ for all k and transfer data as in Lemma 7.4; then $e_{y,m}^s \geq \frac{1}{2}$, $t_m^s \leq \eta_1$, and $\|D_m y_m^s\|_2 \leq K_2$, $|Y_m^s|/q_0 \leq K_2$ with K_2 independent of η_1 . Put $\varepsilon_m := (\hat{G}_m - \hat{\Gamma})/e_{y,m}$, where $y_m := y_m^\dagger$ if $\hat{G}_m \geq \hat{\Gamma}$ and $y_m := y_m^\dagger$ otherwise. Then $|\varepsilon_m| \leq K_3 t^2$ with K_3 independent of η_1 . Since $\|W_m - w_m\|_\infty + \|D_m(W_m - w_m)\|_2 \leq K_4 t$, the hypotheses of Lemma 7.2(a),(b),(c) hold for small t (the radii depend on Λ_m^s , which is fixed once η_1 is), so (9) holds and $|\lambda| \leq |I|K_2 K_3 t^2 \leq 1$. Proposition 7.3 gives

$$p^*(f + tg) \leq 1 + \frac{t^2}{2} \Gamma_w(b, \omega)(1 + K_1 t) + |I|K_3 \eta_1 t^2 + O(t^3),$$

because $|q^*(A) - 1| + q^*(A - a) = O(t)$. Hence the limsup is at most $\Gamma_w(b, \omega) + 2|I|K_3 \eta_1$, and η_1 is arbitrary. The last sentence follows because such a g has a pair (b, ω) with $\text{supp } b \subseteq F$. $\square$

Theorem 7.10 (Exact one-sided coefficient). *Let f satisfy (BT) and let $g \in X^*$ with $g(\xi) = 0$. Then:*

- (a) $\lim_{t \rightarrow 0^\pm} \frac{2(p^*(f + tg) - 1)}{t^2} = \gamma^\pm(g)$ in $[0, \infty]$;
- (b) if $\gamma^\pm(g) < \infty$, the infimum defining it is attained;
- (c) every $g \in \mathcal{C}(f)$ satisfies $\gamma^\pm(g) \leq 1$, and therefore has two-piece data with $\kappa_w = \max(\gamma^+(g), \gamma^-(g)) \leq 1$.

Proof. By Proposition 7.9, $\limsup_{t \rightarrow 0^+} \leq \gamma^+(g)$. We prove $\liminf_{t \rightarrow 0^+} \geq \gamma^+(g)$ and (b) for side $+$; side $-$ is symmetric (replace t by $-t$ and the cone C_+ below by $C_- := -C_+$).

Step 1: test points. Let $C_+ := \{\chi_c \in c_{00} : \chi_c = 0 \text{ on } F, z_j \chi_{c,j} \leq 0 \text{ for } j \in K\}$, a convex cone (inward moves of contacts, arbitrary moves of the free coordinates). Let $k \in H$ with $\langle k, e \rangle = 0$ and $\chi_c \in C_+$, and put $\chi := Uk + \chi_c \in c_0$. As in the base test of the proof of Proposition 5.12 (with k in place of $k_\perp$, which is no longer restricted to E_F), let $\kappa_2 := \nu\|k\|^2/(2q_0)$, choose $k_2 \in H$ with $(Uk_2)_j = -\kappa_2 z_j$ for $j \in F$ and $\langle e, k_2 \rangle = \kappa_2 - \|k\|^2/(2q_0)$ (possible by the Gram system there, which only involves F), put $\chi_2 := (Uk_2)1_{\mathbb{N} \setminus F}$, $\eta_t := \xi + t\chi + t^2\chi_2$, $\mathbf{h}_t := q_0 e + tk + t^2 k_2$ and $Z_t := \eta_t - U\mathbf{h}_t$. Then $Z_{t,j} = (q_0 + t^2 \kappa_2)z_j$ for $j \in F$ and $Z_{t,j} = q_0 z_j + t\chi_{c,j}$ for $j \notin F$: the vector k does not move any coordinate of Z_t . Since χ_c is finitely supported, moves contacts inward and $|z_j| < 1$ on its support off $F \cup K$, we have $\|Z_t\|_\infty \leq q_0 + t^2 \kappa_2$ for small t ; and $\|\mathbf{h}_t\| = q_0 + t^2 \kappa_2 + O(t^3)$ because $k \perp e$. As

$B_{q^{**}} = B_{\ell_\infty} + U(B_H)$, $q^{**}(\eta_t) \leq q_0 + t^2 \kappa_2 + O(t^3)$, while $a(\eta_t - \xi) = t \langle U^* a, k \rangle = 0$. Hence the base excess satisfies $E_q(t\chi + t^2\chi_2) \leq \frac{t^2}{2} \frac{\nu}{q_0} \|k\|^2 + O(t^3)$.

Step 2: the test inequality. For each block let $\mathfrak{d}_m(\delta) := (S_{P_m}(\delta), (\delta_k)_{k \in Q_m}) \in \mathbb{R} \times \mathbb{R}^{Q_m}$, where $S_{P_m}(\delta) = \sum_{k \in P_m} \text{sign}(w_m(k)) \delta_k$. The block test of the proof of Proposition 5.12 (which uses only that Q_m is finite, that there are no degenerate peaks, and (MS)) gives $E_m(tR_m\chi + t^2R_m\chi_2) \leq \frac{t^2}{2} \mathcal{Q}_m(\mathfrak{d}_m(R_m\chi)) + o(t^2)$, where $\mathcal{Q}_m$ is the Hessian of the function G there, a quadratic form on $\mathbb{R} \times \mathbb{R}^{Q_m}$. The formulas of the block suprema in that proof show that $\mathcal{Q}_m$ is positive semidefinite and vanishes at $\mathfrak{d}_m(\zeta_m)$, and that for every $\omega \in \mathbb{R}^{Q_m}$

$$(10) \quad \sup_{\mathfrak{d} \in \mathbb{R} \times \mathbb{R}^{Q_m}} (2v(\mathfrak{d}) - \mathcal{Q}_m(\mathfrak{d})) = \sigma_m H_m(\omega), \quad v := \omega - d_m(\omega)w_m,$$

where a vector $V \in \ell_\infty$ which is a constant multiple of $\text{sign}(w_m)$ on P_m is regarded as the linear form $\mathfrak{d}_m(\delta) \mapsto \langle V, \delta \rangle$ on the data (recall $P_m \cup Q_m = \mathbb{N}$, and every $\mathfrak{d} \in \mathbb{R} \times \mathbb{R}^{Q_m}$ is $\mathfrak{d}_m(\delta)$ for some $\delta \in \ell_1$). Every linear form on $\mathbb{R} \times \mathbb{R}^{Q_m}$ is $\mathfrak{d} \mapsto \langle V, \delta \rangle$ with $V = c \text{sign}(w_m) 1_{P_m} + \tilde{\omega}$, $\tilde{\omega} \in \mathbb{R}^{Q_m}$, and then $V = \omega + (c/M_m)w_m$ with $\omega := \tilde{\omega} - (c/M_m)w_m 1_{Q_m}$; since $\langle \omega, \zeta_m \rangle = \sigma_m d_m(\omega)$, V is of the form $2v$ in (10) iff $\langle V, \zeta_m \rangle = 0$. If $\langle V, \zeta_m \rangle \neq 0$, the supremum of $\langle V, \delta \rangle - \mathcal{Q}_m$ is $+\infty$ (take $\delta = s\zeta_m$). As in (8), $p^*(f + tg) \geq (f + tg)(\eta_t)/p^{**}(\eta_t) = 1 + t^2 g(\chi) - \Delta(\eta_t) + O(t^3)$, so

$$\liminf_{t \rightarrow 0^+} \frac{2(p^*(f + tg) - 1)}{t^2} \geq \mathcal{J}(k, \chi_c) := 2g(\chi) - \frac{\nu}{q_0} \|k\|^2 - \sum_m \mathcal{Q}_m(\mathfrak{d}_m(R_m\chi)).$$

Step 3: weak duality. Let (b, ω) be side-+ admissible for g . Then $b(\chi_c) \leq 0$, $2b(Uk) = 2\langle P^\perp U^* b, k \rangle \leq q_0 h(b) + \frac{\nu}{q_0} \|k\|^2$, and by (10) $2(\omega_m - d_m w_m)(R_m\chi) - \mathcal{Q}_m(\mathfrak{d}_m(R_m\chi)) \leq \sigma_m H_m(\omega_m)$. Adding, $\mathcal{J}(k, \chi_c) \leq \Gamma_w(b, \omega)$.

Step 4: strong duality. Let $n_Q := \sum_m (|Q_m| + 1)$, $\mathfrak{d}(\chi) := (\mathfrak{d}_m(R_m\chi))_m \in \mathbb{R}^{n_Q}$, $\mathcal{Q}(y) := \sum_m \mathcal{Q}_m(y_m)$ (convex and finite on $\mathbb{R}^{n_Q}$), and

$$\mathcal{P}(y) := \sup \left\{ 2g(Uk + \chi_c) - \frac{\nu}{q_0} \|k\|^2 : k \perp e, \chi_c \in C_+, \mathfrak{d}(Uk + \chi_c) = y \right\}$$

($\sup \emptyset := -\infty$). $\mathcal{P}$ is concave (the constraint set is convex and the objective jointly concave), $\mathcal{P}(0) \geq 0$, and $\sup \mathcal{J} = \sup_y (\mathcal{P}(y) - \mathcal{Q}(y))$. If $\mathcal{P}(y_0) = +\infty$ for some y_0 , then $\sup \mathcal{J} = +\infty$, so by Step 3 no side-+ admissible pair exists, $\gamma^+(g) = +\infty$, and Step 2 gives the claim. Otherwise $\mathcal{P}$ is proper, $\text{ri}(\text{dom } \mathcal{Q}) = \mathbb{R}^{n_Q}$ meets $\text{ri}(\text{dom } \mathcal{P})$, and Fenchel's duality theorem [5, Theorem 31.1] gives

$$\sup_y (\mathcal{P}(y) - \mathcal{Q}(y)) = \min_{\pi \in \mathbb{R}^{n_Q}} (\mathcal{Q}^*(\pi) - \mathcal{P}_*(\pi)),$$

the minimum being attained (both sides may be $+\infty$), where $\mathcal{Q}^*(\pi) = \sup_y (\langle \pi, y \rangle - \mathcal{Q}(y))$ and $\mathcal{P}_*(\pi) = \inf_y (\langle \pi, y \rangle - \mathcal{P}(y))$. Write $\langle \pi, \mathfrak{d}(\chi) \rangle = \sum_m \langle V_m, R_m\chi \rangle = \varphi_\pi(\chi)$ with V_m as in Step 2 and $\varphi_\pi := \sum_m R_m^* V_m \in \ell_1$. Then, with $b_\pi := g - \varphi_\pi/2$,

$$-\mathcal{P}_*(\pi) = \sup_{k \perp e} (2\langle P^\perp U^* b_\pi, k \rangle - \frac{\nu}{q_0} \|k\|^2) + \sup_{\chi_c \in C_+} 2b_\pi(\chi_c),$$

the first supremum is $(q_0/\nu) \|P^\perp U^* b_\pi\|^2 = q_0 h(b_\pi)$, and the second is 0 if $(b_\pi)_j = 0$ on J and $z_j(b_\pi)_j \geq 0$ on K , and $+\infty$ otherwise. By Step 2, $\mathcal{Q}^*(\pi) = \sum_m \sigma_m H_m(\omega_m)$ if $V_m = 2(\omega_m - d_m(\omega_m)w_m)$ with $\omega_m \in \mathbb{R}^{Q_m}$ for every m , and $+\infty$ otherwise. In the finite case $\varphi_\pi/2 = \sum_m R_m^*(\omega_m - d_m(\omega_m)w_m)$, so $\mathcal{Q}^*(\pi) - \mathcal{P}_*(\pi)$ is finite iff (b_π, ω) is a side-+ admissible pair for g , and then equals $\Gamma_w(b_\pi, \omega)$. Conversely

every side-+ admissible pair arises from some π . Hence $\sup \mathcal{J} = \gamma^+(g)$, and a minimizing π yields a pair attaining $\gamma^+(g)$ when it is finite. With Step 2 this proves (a) and (b).

(c) If $g \in \mathcal{C}(f)$, then $g(\xi) = 0$ and $p^*(f + tg) \leq s(t) \leq 1 + t^2/2$, so both limits in (a) are at most 1; the minimizers of (b) are two-piece data with $\kappa_w \leq 1$. $\square$

Remark 7.11. (a) If K is finite, two pairs representing g whose base parts are supported in $F \cup K$ coincide (Lemma 1.8(a)); a pair that is admissible on both sides has $b = 0$ on K . Hence $\max(\gamma^+(g), \gamma^-(g)) < \infty$ iff $g \in S(f)$, and then $\gamma^+(g) = \gamma^-(g) = \Gamma_w(g)$; so at (BT) points with K finite, Theorem 7.10 contains Proposition 5.12 and gives again $\mathcal{C}(f) \subseteq S(f)$. The proof above needs no decoupling, so the cofiniteness of J (Remark 5.14(a)) is not needed. (b) When K is infinite, $\gamma^\pm(g)$ can be smaller than Γ_w of a balanced finite certificate representing g : the optimal side decompositions re-split through contacts with the cheap sign. For instance, let f be the first row of Lemma 6.4 (a (BT) point, Corollary 7.23), $u := u_{2,1}$, $\lambda_0 := \lambda_{2,1} = \Phi_1(2)$, $c > 0$ and $g := cu$. Since $Q_1 = \{2\}$, $Q_m = \emptyset$ for $m \neq 1$ and $w_1(2) = 0$, the pairs representing g are exactly $(\theta cu, (1 - \theta)(c/\lambda_0)e_2)$ in block 1), $\theta \in \mathbb{R}$; the pair with $\theta = 0$ is a balanced finite certificate with $\Gamma_w(g) = \sigma_1 H_1((c/\lambda_0)e_2) = \sigma_1 c^2/C_1 =: B > 0$ ($\sigma_1 > 0$ by (P2)). By (P1), $\text{supp } u = \{1\} \cup K_0 = F \cup K$ and $u > 0$ on K_0 , where $z = 1$; so the pair is side-+ admissible iff $\theta \geq 0$ and side-− admissible iff $\theta \leq 0$, and its weighted coefficient is $\theta^2 A + (1 - \theta)^2 B$ with $A := q_0 h(cu)$. Here $A > 0$: otherwise $U^*u \in \mathbb{R}U^*e_1^*$, hence $u \in \mathbb{R}e_1^*$ by injectivity of U^* , which contradicts $\text{supp } u \cap K_0 \neq \emptyset$. Minimizing over θ ($\theta = B/(A + B) \in (0, 1)$ on side +, $\theta = 0$ on side −) gives

$$\gamma^+(g) = \frac{AB}{A + B} < B = \Gamma_w(g) = \gamma^-(g).$$

Whether $\max(\gamma^+(g), \gamma^-(g)) < \Gamma_w(g)$ can occur for a balanced finite certificate g is the open question of Remark 5.14(a).

7.3. The engineered approximants. Throughout this subsection $f \in S_{p^*}$ has F finite, $g \in \mathcal{C}(f)$ carries two-piece data $(b^\pm, \omega^\pm)$, and $\rho \in (0, 1)$. Put

$$b^\theta := \frac{1}{2}(b^+ + b^-), \quad \omega^\theta := \frac{1}{2}(\omega^+ + \omega^-), \quad v := b^+ - b^-, \quad \Omega_m := \text{supp } \omega_m^+ \cup \text{supp } \omega_m^-.$$

Then $(b^\theta, \omega^\theta)$ represents g , v is supported in $F \cup K$ with $z_j v_j = |v_j|$ on K , and $b^\pm = b^\theta \pm v/2$.

Definition 7.12 (Engineered approximants). For a *window* $N_w \geq \max F$, a *scale* $s_1 \in (0, 1]$ and a *cut-off* $N'' > N_w$ put $\mathfrak{m}_j := 4\rho s_1 |b_j^\theta|$ for $j \in K \cap [1, N_w]$ and

$$\begin{aligned} a'' &:= a + \sum_{j \in K \cap [1, N_w]} \mathfrak{m}_j z_j e_j^*, & a' &:= a''/q^*(a''), & e' &:= U^*a'/\|U^*a'\|, \\ z'_j &:= z_j \quad (j \leq N''), & z'_j &:= 0 \quad (j > N''), & \hat{x}' &:= z' + Ue', \\ x' &:= \hat{x}'/p(\hat{x}'), & f' &:= \nabla p(x'). \end{aligned}$$

A *construction sequence* is a sequence of parameters $(N_{w,i}, s_{1,i}, N''_i)$ with $N_{w,i} \rightarrow \infty$ and $s_{1,i} \rightarrow 0$; a statement holds *at late stages* if it holds for all large i . We write $t_{k,m} := \|u_{k,m} 1_{(N'', \infty)}\|_1$ and $\mathfrak{t}(N'') := \sum_{m \in I} \sum_k \lambda_{k,m} t_{k,m}$; by dominated convergence $\mathfrak{t}(N'') \rightarrow 0$ as $N'' \rightarrow \infty$. We shall always assume

$$(11) \quad \mathfrak{t}(N''_i) \leq s_{1,i}^2.$$

Lemma 7.13 (The approximants). (a) $f' \in \text{NA}((X, p), \mathbb{R}) \cap S_{p^*}$; its forced data are $a', e', z', \hat{z}' = \hat{x}'$ and $q'_0 = c' := q(x') = 1/p(\hat{x}')$; $\text{supp } a' = F \cup \{j \in K \cap [1, N_w] : b_j^\theta \neq 0\}$, with $\text{sign } a'_j = z_j$ there. If $4\rho s_1 \|b^\theta\|_1(1 + \|U\|) \leq 1$, then $q^*(a'') \leq 2$, so $|a'_j| \geq |a_j|/2$ on F and $|a'_j| \geq m_j/2$ on window contacts.

(b) Along every construction sequence $f' \rightarrow f$, and all forced data of f' converge to those of f as in Proposition 1.11.

(c) With $K_e := 8\rho\|U\|\|b^\theta\|_1/\nu$:

- (E1) $\|a'' - a\|_1 \leq 4\rho s_1 \|b^\theta\|_1$ and $\|e' - e\| \leq K_e s_1$;
- (E2) $|u(\hat{x}') - u(\hat{z})| \leq K_e s_1 q^*(u) + \|u\|_{(N'', \infty)} \|1\|_1$ for $u \in \ell_1$;
- (E3) $||R_m \hat{x}'|_m - |R_m^{**} \hat{z}|_m| \leq \sum_k \lambda_{k,m} |u_{k,m}(\hat{x}' - \hat{z})| \leq K_e s_1 + t(N'')$, and $|1/c' - 1/q_0| \leq |I|(K_e s_1 + t(N''))$;
- (E4) at late stages (where $|R_m \hat{x}'|_m \geq |R_m^{**} \hat{z}|_m/2$), for every $\omega \in c_{00}$ vanishing on $P_m \cup P'_m$ we have $|d'_m(\omega) - d_m(\omega)| \leq K(\omega)(s_1 + t(N''))$, where $d'_m(\omega) := \langle D_m w'_m, D_m \omega \rangle / C'_m$ and the constant $K(\omega)$ depends only on f, g, ρ, ω ;
- (E5) the Bregman terms $Bx_m := \langle w'_m - w_m, R_m \hat{x}' \rangle$ satisfy

$$0 \leq Bx_m \leq K_e s_1 \sum_k \lambda_{k,m} |w'_m(k) - w_m(k)| + 2t(N'');$$

hence $Bx_m/s_1 \rightarrow 0$ along every construction sequence satisfying (11).

Proof. (a) $a'' \in c_{00}$, $z' \in c_{00}$, $\|z'\|_\infty \leq 1$, and $z' = \text{sign } a'$ on $\text{supp } a'$ (on F because $z = \text{sign } a$ there, on window contacts because $a'_j = m_j z_j$). By Proposition 1.10(c), $q(\hat{x}') = 1$, $\nabla q(\hat{x}') = a'$, and $f' = \nabla p(\hat{x}') = \nabla p(x')$ attains its norm at x' , which is its unique normer; Proposition 1.9 at x' gives the forced data. Finally $q^*(a'') \leq 1 + (1 + \|U\|)\|a'' - a\|_1$. (b) $a' \rightarrow a$ in ℓ_1 and $z' \rightarrow z$ coordinatewise; apply Proposition 1.12 and Proposition 1.11. (E1) $\|e' - e\| = \|U^* a'' / \|U^* a''\| - U^* a / \|U^* a\|\| \leq 2\|U^*(a'' - a)\|/\nu$. (E2) $\hat{x}' - \hat{z} = -z1_{(N'', \infty)} + U(e' - e)$ and $\|U^* u\| \leq q^*(u)$. (E3) w_m norms $R_m^{**} \hat{z}$ and w'_m norms $R_m \hat{x}'$, so $\langle w_m, R_m(\hat{x}' - \hat{z}) \rangle \leq |R_m \hat{x}'|_m - |R_m^{**} \hat{z}|_m \leq \langle w'_m, R_m(\hat{x}' - \hat{z}) \rangle$; both sides are bounded by $\sum_k \lambda_{k,m} |u_{k,m}(\hat{x}' - \hat{z})|$, and $\sum_k \lambda_{k,m} \leq 1$. Moreover $1/c' = p(\hat{x}') = 1 + \sum_m |R_m \hat{x}'|_m$ and $1/q_0 = p^{**}(\hat{z}) = 1 + \sum_m |R_m^{**} \hat{z}|_m$. (E4) Off P'_m , Lemma 1.7 at x' gives $\Phi_m(k)^2 w'_m(k)/C'_m = (R_m \hat{x}')(k)/|R_m \hat{x}'|_m$, so $d'_m(\omega) = \omega(R_m \hat{x}')/|R_m \hat{x}'|_m$, and likewise $d_m(\omega) = \omega(R_m^{**} \hat{z})/|R_m^{**} \hat{z}|_m$; use (E2), (E3). (E5) $\langle w'_m, R_m \hat{x}' \rangle = |R_m \hat{x}'|_m \geq \langle w_m, R_m \hat{x}' \rangle$ and $\langle w'_m - w_m, R_m^{**} \hat{z} \rangle \leq 0$; hence $0 \leq Bx_m \leq \langle w'_m - w_m, R_m(\hat{x}' - \hat{z}) \rangle$, which is bounded by (E2). Since $w'_m \rightarrow w_m$ coordinatewise and $|w'_m - w_m| \leq 2$, $\sum_k \lambda_{k,m} |w'_m(k) - w_m(k)| \rightarrow 0$. $\square$

Lemma 7.14 (Lipschitz control of the block data). Along a construction sequence satisfying (11) there is K_F , depending only on f, g, ρ , such that at late stages, for all m and k ,

$$|C'_m - C_m| = |M'_m - M_m| \leq K_F s_1, \quad \Phi_m(k) |w'_m(k) - w_m(k)| \leq K_F (s_1 + t_{k,m}),$$

and consequently $\|D_m(w'_m - w_m)\|_2^2 \leq K'_F (1 + \log_2(1/s_1)) s_1^2$, with K'_F depending only on K_F .

Proof. Fix m and drop it. Put $v_k := m|u_k(\hat{z})|/|R_m^{**} \hat{z}|$ and $v'_k := m|u_k(\hat{x}')|/|R_m \hat{x}'|$. By Lemma 1.7 (peaks are the k with $|\zeta(k)|/|\zeta| \geq \Phi_k^2 M/C$, and off P , $w(k) = C\zeta(k)/(\Phi_k^2 |\zeta|)$), we have the clamp formula $\Phi_k w(k) = \text{sign}(u_k(\hat{z})) \min(\Phi_k M, C v_k)$, and the same at x' . Squaring and summing, and using $M = 1 - C$, C is a root of

$$F(c; v) := \sum_k \min\left(\Phi_k \frac{1-c}{c}, v_k\right)^2 = 1,$$

and C' is a root of $F(\cdot; v') = 1$. Each $F(\cdot; v)$ is nonincreasing on $(0, 1)$. Since $\|\alpha\|_1 = 1$, there is $k_{\mathfrak{h}} \in P$ with $\alpha(k_{\mathfrak{h}}) \neq 0$, i.e. $Cv_{k_{\mathfrak{h}}} > \Phi_{k_{\mathfrak{h}}}M$; choose $r > 0$ with $[C-r, C+r] \subseteq (0, 1)$ and $v_{k_{\mathfrak{h}}} \geq \Phi_{k_{\mathfrak{h}}}(1-c)/c+r$ for $c \in [C-r, C+r]$. At late stages $|C' - C| \leq r$ and $|v'_{k_{\mathfrak{h}}} - v_{k_{\mathfrak{h}}}| \leq r$, so on $[C-r, C+r]$ the $k_{\mathfrak{h}}$ -term of $F(\cdot; v')$ equals $\Phi_{k_{\mathfrak{h}}}^2((1-c)/c)^2$, whose derivative is at most $-c_* := -2\Phi_{k_{\mathfrak{h}}}^2(1-C-r)/(C+r)^3$; the other terms are nonincreasing. Hence $c_*|C' - C| \leq |F(C; v') - F(C'; v')| = |F(C; v') - F(C; v)|$, and since $|\min(x, s)^2 - \min(x, s')^2| \leq 2x|s - s'|$,

$$|C' - C| \leq \frac{2(1-C)}{Cc_*} \sum_k \Phi_k |v'_k - v_k|.$$

By (E2), (E3) and $|u_k(\hat{z})| \leq 1$, at late stages $|v'_k - v_k| \leq K(s_1 + t_k + \mathfrak{t}(N''))$ with K depending only on f, g, ρ ; as $\sum_k \Phi_k \leq 1$ and $\sum_k \Phi_k t_k \leq \mathfrak{t}(N'') \leq s_1^2$, this gives $|C' - C| \leq K_F s_1$. If $\text{sign } u_k(\hat{x}') = \text{sign } u_k(\hat{z})$ or one of them vanishes, the clamp formula gives $|\Phi_k w'(k) - \Phi_k w(k)| \leq |C' - C|(\Phi_k + v_k) + C'|v'_k - v_k|$, and $v_k \leq m/|R^{**}\hat{z}|$. If the signs are opposite, both $|u_k(\hat{x}')|$ and $|u_k(\hat{z})|$ are at most $|u_k(\hat{x}') - u_k(\hat{z})|$, and $|\Phi_k w'(k)| + |\Phi_k w(k)| \leq C'v'_k + Cv_k$. In both cases the bound follows from (E2). Finally, using $\min(x+y, c)^2 \leq \min(x, c)^2 + 2cy$ and $\Phi_k \leq 2^{-k}$,

$$\begin{aligned} \|D(w' - w)\|_2^2 &\leq \sum_k \min(K_F s_1, 2\Phi_k)^2 + 4K_F \sum_k \Phi_k t_k \\ &\leq K_F^2 s_1^2 [\log_2(1/s_1)] + \frac{4}{3} s_1^2 + 4K_F s_1^2. \end{aligned} \quad \square$$

Lemma 7.15 (Anchor). *Fix a block and drop its index. Let $w, w' \in \ell_\infty$ with $N(w) = N(w') = 1$ be the norming functionals of nonzero block vectors, with data M, C, P and M', C', P' , and put $\gamma := C' - C$; assume $C \leq \frac{1}{2}$ and $|\gamma| \leq C/4$. Let $S_1 := \{k \in P \cap P' : \text{sign } w(k) = \text{sign } w'(k)\}$, $S_2 := \{k \notin P \cup P' : |2w'(k) - w(k)| \leq M' - |\gamma|\}$, $A := \mathbb{N} \setminus (S_1 \cup S_2)$ (the scrambled set), and*

$$y(k) := 2w'(k) - w(k) \quad (k \in S_1 \cup S_2), \quad y(k) := (1 - \gamma_+/M')w'(k) \quad (k \in A).$$

Then $N(y) \leq 1 + \mathcal{X}/(2(C+2\gamma))$ with $\mathcal{X} := 4\|D(w' - w)\|_2^2 + 4\sum_{k \in A} \Phi(k)^2$. Moreover $w' - w = (y - w') + Z_A$ with $Z_A := (w' - w)1_A + (\gamma_+/M')w'1_A$, $\|y - w'\|_\infty \leq 3$, and if w' norms Rx' with $\sigma' := |Rx'| > 0$, then

$$N(y) - \frac{\langle y, Rx' \rangle}{\sigma'} \leq \frac{\mathcal{X}}{2(C+2\gamma)} + \frac{1}{\sigma'} \sum_{k \in A} |(Rx')(k)|(|w'(k) - w(k)| + \gamma_+).$$

Proof. Sup norm. Note $M' - M = -\gamma$, $M \geq \frac{1}{2}$ and $|\gamma| \leq \frac{1}{8}$. On S_1 , $|y(k)| = 2M' - M = M' - \gamma$; on S_2 , $|y(k)| \leq M' - |\gamma|$; on A , $|y(k)| \leq M' - \gamma_+$. So $\|y\|_\infty \leq M' - \gamma$.

Hilbert part. Put $Y := y - w' = (w' - w)1_{S_1 \cup S_2} - (\gamma_+/M')w'1_A$ and $R_A := \langle Dw', D(w' - w)1_A \rangle + (\gamma_+/M')\|Dw'1_A\|^2$. Then $\langle Dw', DY \rangle = \langle Dw', D(w' - w) \rangle - R_A$ and $2\langle Dw', D(w' - w) \rangle = C'^2 - C^2 + \|D(w' - w)\|^2$, so

$$\begin{aligned} \|Dy\|^2 &= C'^2 + 2\langle Dw', DY \rangle + \|DY\|^2 = 2C'^2 - C^2 + \mathcal{X}_0, \\ \mathcal{X}_0 &:= \|D(w' - w)\|^2 + \|DY\|^2 - 2R_A. \end{aligned}$$

Since $1_{S_1 \cup S_2}$ and 1_A are disjoint and $\gamma_+/M' \leq 1$, $\|DY\|^2 \leq \|D(w' - w)\|^2 + \|Dw'1_A\|^2$ and $-2R_A \leq \|Dw'1_A\|^2 + \|D(w' - w)\|^2$; with $|w'| \leq 1$ this gives $\mathcal{X}_0 \leq \mathcal{X}$. As $2C'^2 - C^2 = (C+2\gamma)^2 - 2\gamma^2$ and $C+2\gamma \geq C/2 > 0$, $\|Dy\| \leq C+2\gamma + \mathcal{X}/(2(C+2\gamma))$. Adding, $N(y) \leq M' + C + \gamma + \mathcal{X}/(2(C+2\gamma)) = 1 + \mathcal{X}/(2(C+2\gamma))$.

The rest. The identity for $w' - w$ is immediate, and $\|y - w'\|_\infty \leq \|w' - w\|_\infty + 1 \leq 3$. Finally $\langle y, Rx' \rangle = \sigma' + \langle Y, Rx' \rangle$ and $\langle Y, Rx' \rangle = \langle w' - w, Rx' \rangle - \langle (w' - w)1_A, Rx' \rangle - (\gamma_+/M')\langle w'1_A, Rx' \rangle$, where the first term is ≥ 0 because w' norms Rx' and $N(w) = 1$. $\square$

Definition 7.16 (Scrambling condition). For a block m and $s > 0$ let

$$\text{Scr}_m(s) := \sum_{\substack{k \in P_m \\ \mu_k, m \leq s}} \min(\Phi_m(k), s) + \sum_{\substack{k \in Q_m \\ \Phi_m(k) \text{ gap}_m(k) \leq s \text{ or } \text{gap}_m(k) \leq s}} \min(\Phi_m(k), s).$$

A set $I_- \subseteq I$ of blocks satisfies the *scrambling condition* (SC) if for every $\Upsilon \geq 1$ there is a sequence $s_i \downarrow 0$ (common to all $m \in I_-$) with $\max_{m \in I_-} \text{Scr}_m(\Upsilon s_i)/s_i \rightarrow 0$.

The common sequence matters: each Scr_m may be $o(s)$ along a sequence of scales while the good scales of two blocks are disjoint. (SC) excludes degenerate peaks in the blocks of I_- (each contributes $\min(\Phi_m(k), \Upsilon s) = \Upsilon s$ for small s). If f satisfies (BT), then every block satisfies $\text{Scr}_m(\Upsilon s) = o(s)$ as $s \rightarrow 0$ for every Υ : for small s the finitely many strict non-peaks do not contribute, and the peak part is at most $\sum \{\Phi_m(k) : \mu_{k,m} < 2\Upsilon s\} = o(s)$ by (MS).

Lemma 7.17 (Scrambling estimate). *There is $K_* \geq 1$, depending only on f, g, ρ , with the following property. Let I_- satisfy (SC), let (s_i) be the sequence of (SC) for $\Upsilon := 2K_*$, and consider a construction sequence with $s_{1,i} = s_i$ and N_i'' so large that (11) and*

$$(12) \quad \sum_{m \in I_-} \sum_k \Phi_m(k) 1[t_{k,m} > s_{1,i}] \leq s_{1,i}^2$$

hold. Then for $m \in I_-$, with A_m the scrambled set of Lemma 7.15 for (w_m, w'_m) ,

$$\begin{aligned} \mathfrak{S}_m &:= \|D_m(w'_m - w_m)\|_2^2 \\ &+ \sum_{k \in A_m} \left(\Phi_m(k)^2 + \lambda_{k,m}(|w'_m(k) - w_m(k)| + (C'_m - C_m)_+) \right) = o(s_1). \end{aligned}$$

Condition (12) can be met by choosing N'' after s_1 : for fixed s_1 the left side tends to 0 as $N'' \rightarrow \infty$ by dominated convergence.

Proof. Fix $m \in I_-$ and drop it. The first term is $o(s_1)$ by Lemma 7.14. At late stages $|C' - C| \leq C/4$, so Lemma 7.15 applies ($C \leq \|\Phi\|_2 < \frac{1}{2}$).

Peaks. Put $\hat{v} := |R^{**}\hat{z}|M/(mC)$ and $\hat{v}' := |R\hat{x}'|M'/(mC')$, so that $\mu_k = q_0(|u_k(\hat{z})| - \hat{v}\Phi_k)$, and k is a peak of w' of sign $\text{sign } u_k(\hat{x}')$ as soon as $|u_k(\hat{x}')| > \hat{v}'\Phi_k$ (Lemma 1.7). By (E3) and Lemma 7.14, $|\hat{v}' - \hat{v}| \leq K_\vartheta s_1$ at late stages. If $k \in P$ and $\mu_k > K_*(s_1 + t_k)$ with $K_* \geq K_e + K_\vartheta + 1$, then by (E2) $|u_k(\hat{x}') - u_k(\hat{z})| < |u_k(\hat{z})|$ and $|u_k(\hat{x}')| - \hat{v}'\Phi_k > 0$; so $k \in P'$ with the same sign, i.e. $k \in S_1$.

Strict non-peaks. If $k \in Q$, $\Phi_k \text{ gap}_k > K_*(s_1 + t_k)$ and $\text{gap}_k > K_*s_1$ with $K_* \geq 4K_F$, then by Lemma 7.14 $|w'(k) - w(k)| < \text{gap}_k/4$ and $|\gamma| = |M' - M| < \text{gap}_k/4$. Hence $|w'(k)| < M'$, so $k \notin P'$, and $|2w'(k) - w(k)| \leq |w(k)| + 2|w'(k) - w(k)| < M - \text{gap}_k/2 \leq M' - |\gamma|$; i.e. $k \in S_2$.

Put $K_* := \max(K_e + K_\vartheta + 1, 4K_F)$. Let $A^{(1)} := \{k \in A : t_k \leq s_1\}$ and $A^{(2)} := A \setminus A^{(1)}$. By the two steps, every $k \in A^{(1)}$ is a peak with $\mu_k \leq 2K_*s_1$ or a strict non-peak with $\Phi_k \text{ gap}_k \leq 2K_*s_1$ or $\text{gap}_k \leq 2K_*s_1$; so $\sum_{A^{(1)}} \min(\Phi_k, 2K_*s_1) \leq \text{Scr}(2K_*s_1) = o(s_1)$. In particular, at late stages no $k \in A^{(1)}$ has $\Phi_k \geq 2K_*s_1$, and

$\sum_{A^{(1)}} \Phi_k = o(s_1)$. By (12), $\sum_{A^{(2)}} \Phi_k \leq s_1^2$. Since $\Phi_k \leq 1$, $|w' - w| \leq 2$, $\gamma_+ \leq 1$ and $\lambda_k = m\Phi_k$, the sum over A in $\mathfrak{S}$ is at most $4m \sum_A \Phi_k = o(s_1)$. $\square$

7.4. Engineering without steering.

Theorem 7.18 (Engineered recovery). *Let I be finite, let $f \in S_{p^*}$ have finite base support F , let $g \in \mathcal{C}(f)$ carry two-piece data $(b^\pm, \omega^\pm)$, and let $\rho \in (0, 1)$ with $\rho^2 \kappa_w < 1$. Assume that $I_- := \{m : \Delta d_m < 0\}$ satisfies the scrambling condition (SC) (no assumption if $I_- = \emptyset$). Then there are norm-attaining $f'_i \rightarrow f$ (engineered approximants) and $g'_i \in \mathcal{C}(f'_i)$ with $g'_i \rightarrow \rho g$; in particular $(f, \rho g) \in \text{clNA}((c_0, p), \ell_2^2)$. If moreover $\kappa_w \leq 1$, this holds for every $\rho < 1$, and $(f, g) \in \text{clNA}((c_0, p), \ell_2^2)$. No hypothesis is made on the contact set K , on the number of active blocks, on the sets Q_m (for $m \notin I_-$), or on rates of T .*

Proof. We use the notation of Definition 7.12; data of f' carry primes, $\sigma'_m = |R_m x'|_m$, $c' = q(x')$, and d'_m is as in (E4). Put $d_m^\diamond := d_m(\omega_m^\diamond)$ for $\diamond \in \{+, -, \theta\}$, so $d_m^\theta = \frac{1}{2}(d_m^+ + d_m^-)$.

Step 0: fixed data. Let $\delta := (1 - \rho^2 \kappa_w)/2$ and $\Gamma_\diamond := \Gamma_w(b^\diamond, \omega^\diamond)$; by convexity $\Gamma_\theta \leq \kappa_w$, so $\rho^2 \Gamma_\diamond \leq 1 - 2\delta$ for all three $\diamond$. Let $\gamma_0 > 0$ be the least gap $\text{gap}_m(k)$, $k \in \Omega_m$, $m \in I$, and $a_{\min} := \min_F |a_j|$. In every block fix $\gamma_m \in (M_m/2, M_m)$ with $|w_m(k)| \neq \gamma_m$ for all k . By Lemma 7.13, Lemma 7.14, Lemma 7.17 there are $T_1 \in (0, 1]$ and $K_\# \geq 1$, depending only on f, g, ρ , such that the conditions marked (T_1) below hold for $T_0 \leq T_1$, and such that the quantities of Step 3 satisfy, at late stages and for $0 < |\tau| \leq T_1$, $|\ell'_m| \leq K_\# |\tau| s_1$ and $\hat{G}_b, \hat{G}_m \leq K_\# \tau^2$ (for $\diamond = \pm$ we use $|\tau| > s_1$ to bound the terms $|\tau| o(s_1)$ by τ^2). Put

$$\eta_1 := \min \left(\frac{1}{4}, \frac{\delta}{192|I|K_\#} \right),$$

and choose transfer data $(\gamma_m, k_{*,m}^\varsigma, \Lambda_m^\varsigma)$ of f in every block by Lemma 7.4 with this η_1 . By Lemma 7.5 the transfer vectors $y_m'^\varsigma$ built at f' satisfy, at late stages, $e_{y,m}^\varsigma \geq \frac{1}{2}$, $\ell'_m \leq 2\eta_1$, $\|D_m y_m'^\varsigma\|_2 \leq K_y$ and $|Y_m'^\varsigma|/c' \leq K_Y$, where K_y, K_Y do not depend on η_1 (Lemma 7.4(b),(d)). Finally choose $T_0 \in (0, T_1]$ with $T_0^2 \leq 4\delta$ and so small that the finitely many conditions marked (T_0) below hold; they involve f, g, ρ, η_1 and the transfer data. For the reader's convenience we collect them here (the constants $K, K_W, K_y, K_Y, K_A, K'$ are introduced in Steps 2–5 and depend only on f, g, ρ , except K_A , which also depends on the transfer data). Conditions (T_1) : the number $2\rho T_0$ is at most half of the coordinatewise radius of every $\omega_m^\diamond$ at f ; at late stages $\rho T_0 \max_m |\Delta d_m| + 2T_0 K s_1 \leq \frac{1}{4}$; and $\rho T_0 (\|b^+\|_\infty + \|b^-\|_\infty) \leq a_{\min}/8$. Conditions (T_0) : for all m and ς ,

$$\begin{aligned} K_W T_0 &\leq \min((M_m - \gamma_m)/8, C_m/4), & 6K_\# T_0^2 (2 + \Lambda_m^\varsigma) &\leq \gamma_m/2, \\ 6K_\# T_0^2 &\leq (M_m - \gamma_m)/8, & 6|I|K_\# K_Y T_0^2 &\leq 1, & K' T_0 (1 + \rho^2 \kappa_w) &\leq \delta/8; \end{aligned}$$

and the last two terms of the bound for E in Step 4 are at most $\delta\tau^2/16$ for $|\tau| \leq T_0$. The order of choices is $T_1, K_\# \rightarrow \eta_1 \rightarrow \text{transfer data} \rightarrow T_0 \rightarrow (s_{1,i}) \rightarrow (N_i'')$, and each condition involves only quantities fixed before it.

Construction. If $I_- \neq \emptyset$, let (s_i) be the sequence given by (SC) for the multiplier $\Upsilon := 2K_*$ of Lemma 7.17; otherwise let $s_i \downarrow 0$ be arbitrary. Consider the engineered approximants with $N_{w,i} \rightarrow \infty$, $s_{1,i} := s_i$, and N_i'' chosen after $s_{1,i}$ so that (11),

(12) and

$$(13) \quad \rho V_{>N''} \leq \delta s_1/16, \quad V_{>N''} := \sum_{j \in K, j > N''} |v_j|,$$

hold. All statements below are at late stages. By Lemma 7.13(b), $w'_m \rightarrow w_m$ coordinatewise, $C'_m \rightarrow C_m$, $M'_m \rightarrow M_m$, $\sigma'_m \rightarrow \sigma_m$, $c' \rightarrow q_0$, $\nu' \rightarrow \nu$, $e' \rightarrow e$, and $R_m^* w'_m \rightarrow R_m^* w_m$ in ℓ_1 ; in particular every $k \in \Omega_m$ is a strict non-peak of w'_m with $\text{gap}'_m(k) \geq \gamma_0/2$, the coordinatewise radius of $\omega_m^\diamond$ at f' is at least half the one at f , and $d'_m(\omega) \rightarrow d_m(\omega)$, $H'_m(\omega) \rightarrow H_m(\omega)$ for $\omega \in \{\omega_m^\diamond\}$.

Step 1: the target. Let

$$g'' := b^\theta 1_{[1, N_w]} + \sum_m R_m^* (\omega_m^\theta - d'_m(\omega_m^\theta) w'_m), \quad c := g''(\hat{x}'), \quad g' := \rho(g'' - ca').$$

Then $g'(\hat{x}') = 0$. Since $(b^\theta, \omega^\theta)$ represents g , $g'' - g = -b^\theta 1_{(N_w, \infty)} + \sum_m R_m^* (d_m^\theta w_m - d'_m(\omega_m^\theta) w'_m) \rightarrow 0$, and $c = (g'' - g)(\hat{x}') + g(\hat{x}') \rightarrow g(\hat{z}) = 0$ because $\hat{x}' \rightarrow \hat{z}$ weak* boundedly. So $g' \rightarrow \rho g$.

Step 2: three exact decompositions. Let $\beta^\theta := b^\theta 1_{[1, N_w]}$, $\beta^\pm := b^\pm 1_{[1, N_w]} \pm \frac{1}{2} v 1_{(N_w, \infty)}$, $B^\diamond := \rho(\beta^\diamond - ca')$, and

$$\Omega_m^\theta := \rho(\omega_m^\theta - d'_m(\omega_m^\theta) w'_m), \quad \Omega_m^\pm := \rho(\omega_m^\pm - d'_m(\omega_m^\theta) w'_m - (d_m^\pm - d_m^\theta) w_m).$$

Since $\beta^\pm - \beta^\theta = \pm \frac{1}{2} v$, $\omega^\pm - \omega^\theta = \mp \frac{1}{2} (\omega^- - \omega^+)$, $d^\pm - d^\theta = \mp \frac{1}{2} \Delta d$ and $v = \sum_m R_m^* ((\omega_m^- - \omega_m^+) - \Delta d_m w_m)$, we get $g' = B^\diamond + \sum_m R_m^* \Omega_m^\diamond$ for $\diamond \in \{+, -, \theta\}$, and so $f' + \tau g' = (a' + \tau B^\diamond) + \sum_m R_m^* (w'_m + \tau \Omega_m^\diamond)$ for every τ . We use $\diamond = \theta$ for $0 < |\tau| \leq s_1$ and $\diamond = \text{sign } \tau$ for $s_1 < |\tau| \leq T_0$.

For $\diamond = \pm$ rewrite, with $\kappa_m^\diamond := \rho(d'_m - d_m)(\omega_m^\diamond - \omega_m^\theta)$,

$$\Omega_m^\diamond = \rho(\omega_m^\diamond - d'_m(\omega_m^\diamond) w'_m) + \kappa_m^\diamond w'_m + \rho(d_m^\diamond - d_m^\theta)(w'_m - w_m).$$

For $\diamond = \text{sign } \tau$ we have $\tau \rho(d_m^\diamond - d_m^\theta) = -r_m \text{sign}(\Delta d_m)$ with $r_m := \frac{1}{2} |\tau| \rho |\Delta d_m|$. Put $W_{0,m} := w'_m + \tau \rho(\omega_m^\diamond - d'_m(\omega_m^\diamond) w'_m) + \tau \kappa_m^\diamond w'_m$. If $\Delta d_m \geq 0$, let $V_m^{\text{an}} := w_m$ and $W_m^\natural := w'_m + \tau \Omega_m^\diamond = W_{0,m} - r_m w'_m + r_m V_m^{\text{an}}$. If $\Delta d_m < 0$, let $V_m^{\text{an}} := y_{A,m}$ and $Z_{A,m}$ be given by Lemma 7.15 for (w_m, w'_m) , and put $W_m^\natural := w'_m + \tau \Omega_m^\diamond - r_m Z_{A,m} = W_{0,m} - r_m w'_m + r_m V_m^{\text{an}}$; the remainder $r_m R_m^* Z_{A,m}$ is moved to the base. The decomposition

$$f' + \tau g' = A_\tau + \sum_m R_m^* W_m^\natural, \quad A_\tau := a' + \tau B^\diamond + \mathcal{Z}, \quad \mathcal{Z} := \sum_{m \in I_-} r_m R_m^* Z_{A,m},$$

is exact. For $\diamond = \theta$ put $W_m^\natural := w'_m + \tau \Omega_m^\theta$, $A_\tau := a' + \tau B^\theta$. In all cases $\|W_m^\natural - w'_m\|_\infty + \|D_m(W_m^\natural - w'_m)\|_2 \leq K_W |\tau|$ with K_W depending only on f, g, ρ , because $\|V_m^{\text{an}} - w'_m\|_\infty \leq 3$.

Step 3: first-order terms and excesses. Blocks. Since $\omega_m^\diamond$ vanishes on P'_m , $\ell'_m(w'_m + \tau \rho(\omega_m^\diamond - d'_m(\omega_m^\diamond) w'_m)) = 0$ (Lemma 3.2 at f'). Hence $\ell'_m = 0$ for $\diamond = \theta$, and for $\diamond = \pm$

$$\ell'_m(W_m^\natural) = \tau \kappa_m^\diamond + r_m \langle V_m^{\text{an}} - w'_m, R_m x' \rangle / \sigma'_m.$$

By (E4) and (11), $|\kappa_m^\diamond| \leq K s_1$. For $V_m^{\text{an}} = w_m$, $\langle V_m^{\text{an}} - w'_m, R_m x' \rangle = -c' B x_m$; for $V_m^{\text{an}} = y_{A,m}$ its modulus is at most $c' B x_m + c' \sum_{A_m} \lambda_{k,m} (|w'_m - w_m|(k) + (C'_m - C_m)_+)$ (proof of Lemma 7.15 and $|(R_m x')(k)| \leq \lambda_{k,m} c'$). By (E5) and Lemma 7.17 both are $o(s_1)$; so $|\ell'_m| \leq K_\sharp |\tau| s_1$. For the excess write $W_{0,m} - r_m w'_m = (1 - r_m + \tau \kappa_m^\diamond)(w'_m + t'(\omega_m^\diamond - d'_m(\omega_m^\diamond) w'_m))$ with $t' := \tau \rho / (1 - r_m + \tau \kappa_m^\diamond)$. By

convexity and homogeneity of N_m and Lemma 3.4(d) at f' (condition (T_1)): $2\rho T_0$ is at most half the radius of every $\omega_m^\diamond$ at f , and $\rho T_0 \max_m |\Delta d_m| + 2T_0 K s_1 \leq \frac{1}{4}$,

$$N_m(W_m^\natural) \leq (1 - r_m + \tau \kappa_m^\diamond) + \frac{\rho^2 \tau^2}{2} H'_m(\omega_m^\diamond)(1 + K_H T_0) + r_m N_m(V_m^{\text{an}}),$$

and therefore, using $\langle w'_m, R_m x' \rangle = \sigma'_m$,

$$G'_m(W_m^\natural) \leq \hat{G}_m := \frac{\rho^2 \tau^2}{2} H'_m(\omega_m^\diamond)(1 + K_H T_0) + r_m \mathfrak{g}_m,$$

$$\mathfrak{g}_m := N_m(V_m^{\text{an}}) - \frac{\langle V_m^{\text{an}}, R_m x' \rangle}{\sigma'_m}.$$

For $V_m^{\text{an}} = w_m$, $\mathfrak{g}_m = c' B x_m / \sigma'_m = o(s_1)$ by (E5); for $V_m^{\text{an}} = y_{A,m}$, $\mathfrak{g}_m \leq K \mathfrak{S}_m = o(s_1)$ by Lemmas 7.15 and 7.17. For $\diamond = \theta$ the same computation with $r_m = \kappa = 0$ gives $\hat{G}_m := \frac{\rho^2 \tau^2}{2} H'_m(\omega_m^\theta)(1 + K_H T_0)$.

Base. By Lemma 7.1(b) at f' , $G'_b(A_\tau) = \text{Exc}'(\tau B^\diamond + \mathcal{Z}) + \nu' \Psi'(U^*(\tau B^\diamond + \mathcal{Z})/\nu')$ and $\text{Exc}'(\tau B^\diamond + \mathcal{Z}) \leq \text{Exc}'(\tau B^\diamond) + 2\|\mathcal{Z}\|_1$. We claim $\text{Exc}'(\tau B^\diamond) \leq \frac{1}{2}\rho|\tau|V_{>N''}$, and $= 0$ for $\diamond = \theta$. On $\text{supp } a'$, $a'_j + \tau B_j^\diamond = (1 - \tau \rho c)a'_j + \tau \rho \beta_j^\diamond$ with $|\tau \rho c| \leq \frac{1}{2}$. On F , $|a'_j| \geq a_{\min}/2$ and $|\tau \rho \beta_j^\diamond| \leq a_{\min}/8$ (condition (T_1) : $\rho T_0(\|b^+\|_\infty + \|b^-\|_\infty) \leq a_{\min}/8$): no flip. On a window contact with mass: for $\diamond = \theta$, $|\tau \rho \beta_j^\theta| \leq s_1 \rho |b_j^\theta| = m_j/4 \leq |a'_j|/2$; for $\diamond = +$, $\tau > 0$ and $z_j b_j^+ \geq 0$, so both terms have the sign $z_j = \text{sign } a'_j$ or vanish; for $\diamond = -$, $\tau < 0$ and $z_j b_j^- \leq 0$, with the same conclusion. Off $\text{supp } a'$, $B_j^\diamond = \rho \beta_j^\diamond$, and the summand of Exc' is $|\tau \rho \beta_j^\diamond| - z'_j \tau \rho \beta_j^\diamond$: it vanishes off $F \cup K$ (where $b^\pm$ and v vanish), on window contacts without mass (there $\beta_j^\diamond = 0$ and $\tau \beta_j^\pm = \tau b_j^\pm$ has the sign $z_j = z'_j$ on the designated side), and on contacts in (N_w, N'') (there $\tau \beta_j^\pm = \pm \frac{1}{2} \tau v_j$ has the sign z_j because $z_j v_j \geq 0$, and $\beta_j^\theta = 0$); on contacts $j > N''$, $z'_j = 0$ and the summand is at most $\frac{1}{2}\rho|\tau||v_j|$, and it is 0 for $\diamond = \theta$. This proves the claim. Further $\|\mathcal{Z}\|_1 \leq \sum_{m \in I_-} r_m \sum_{A_m} \lambda_{k,m}(|w'_m - w_m|(k) + (C'_m - C_m)_+) = |\tau|o(s_1)$ by Lemma 7.17, and $P'^\perp U^* a' = 0$, so by Lemma 3.3

$$G'_b(A_\tau) \leq \hat{G}_b := \frac{\rho^2 \tau^2}{2} h'(\beta^\diamond)(1 + K_h T_0) + \frac{1}{2}\rho|\tau|V_{>N''} + |\tau|o(s_1),$$

where $h'(\beta) := \|P'^\perp U^* \beta\|^2 / \nu' \rightarrow h(b^\diamond)$ for $\beta = \beta^\diamond$ (as $\beta^\diamond - b^\diamond = -b^\theta 1_{(N_w, \infty)} \rightarrow 0$ in ℓ_1); for $\diamond = \theta$ the last two terms are absent.

Step 4: rebalancing. Apply Proposition 7.3 at f' with $h := \tau g'$ (so $h(x') = 0$), the decomposition of Step 2, the bounds $\hat{G}_b, \hat{G}_m$ of Step 3, and $y_m := y_m^{\downarrow}$ or $y_m^{\uparrow}$ according to the sign of $\mathfrak{l}'_m + \hat{G}_m - \hat{\Gamma}$. Then $|\varepsilon_m| \leq 2(|\mathfrak{l}'_m| + \hat{G}_m + \hat{\Gamma}) \leq 6K_\# \tau^2$ when $|\tau| > s_1$, and $\leq 4K_\# \tau^2$ for $\diamond = \theta$. Condition (9) follows from Lemma 7.2 with $\eta := K_W |\tau|$ (condition (T_0) : $K_W T_0 \leq \min_m (M_m - \gamma_m)/8$, $6K_\# T_0^2(2 + \Lambda_m^\varsigma) \leq \gamma_m/2$, $6K_\# T_0^2 \leq \min_m (M_m - \gamma_m)/8$ and $K_W T_0 \leq \min_m C_m/4$), and $|\lambda| \leq 6|I|K_\# K_Y T_0^2 \leq 1$ by condition (T_0) . The error E of Proposition 7.3 satisfies

$$E \leq 12|I|K_\# \eta_1 \tau^2 + 6|I|K_\# K_Y |\tau|^3 K_A + \frac{(48K_\# K_W K_Y |\tau|^3 + 36K_\#^2 K_Y^2 \tau^4)}{\min_m C_m/2},$$

where K_A bounds $(|q^*(A_\tau) - 1| + q^*(A_\tau - a'))/|\tau|$. By the choice of η_1 and condition (T_0) on the last two terms, $E \leq \delta \tau^2/8$.

Step 5: the levels. By Step 3, $\hat{\Gamma} = c' \hat{G}_b + \sum_m \sigma'_m \hat{G}_m \leq \frac{\rho^2 \tau^2}{2} (c' h'(\beta^\diamond) + \sum_m \sigma'_m H'_m(\omega_m^\diamond))(1 + K' T_0) + \frac{1}{2}\rho|\tau|V_{>N''} + |\tau|o(s_1)$. At late stages the bracket

is at most $\Gamma_\diamond + \delta/8$; by (13) and $|\tau| > s_1$ the last two terms are at most $\delta\tau^2/16$ (and they are absent for $\diamond = \theta$). With condition (T_0) : $K'T_0(1 + \rho^2\kappa_w) \leq \delta/8$,

$$p^*(f' + \tau g') \leq 1 + \hat{\Gamma} + E \leq 1 + \frac{\tau^2}{2} \left(\rho^2\kappa_w + \frac{\delta}{8} + \frac{\delta}{8} + \frac{\delta}{8} + \frac{\delta}{4} \right) \leq 1 + \frac{\tau^2}{2}(1 - \delta)$$

for $0 < |\tau| \leq T_0$, since $\rho^2\kappa_w = 1 - 2\delta$.

Step 6: conclusion. At late stages $p^*(f' - f) \leq (1 - \rho^2)T_0^2/6$ and $p^*(g' - \rho g) \leq (1 - \rho^2)T_0/6$. Lemma 7.6 gives $g' \in \mathcal{C}(f')$, and f' attains its norm. If $\kappa_w \leq 1$, then $\rho^2\kappa_w < 1$ for every $\rho < 1$, and $(f, g) = \lim_{\rho \rightarrow 1}(f, \rho g) \in \text{cl NA}$. $\square$

Remark 7.19 (On the proof). (a) Earlier engineered constructions had to make the two side decompositions represent the *same* functional at the new normer (“steering”): far contacts with flipped signs and negative masses, a tuning mass fixed by the intermediate value theorem, a spanning hypothesis for several blocks. None of this is needed here. The linear mismatches between the pieces are of order $|\tau|s_1$, and Proposition 7.3 removes them at the cost $|\tau|s_1\iota$, the inefficiency ι being fixed before s_1 ; the first-order identity of Lemma 7.1(a) guarantees that a linear mismatch is a pure shift of levels. (b) The only genuinely nonlinear first-order costs are the kinks beyond the cut-off, the Bregman terms Bx_m (always $o(s_1)$) and, in blocks with $\Delta d_m < 0$, the anchor costs controlled by (SC). (c) The approximants are not the canonical truncations; this is consistent with Proposition 6.8.

7.5. Consequences.

Corollary 7.20 (Weighted engineered recovery). *Let I be finite, $f \in S_{p^*}$ with F finite, and let $g \in \mathcal{C}(f)$ carry d -neutral two-piece data with any number of active blocks. If $\rho^2\kappa_w < 1$, then $(f, \rho g) \in \text{cl NA}((c_0, p), \ell_2^2)$; if $\kappa_w \leq 1$, then $(f, g) \in \text{cl NA}((c_0, p), \ell_2^2)$.*

Proof. $I_- = \emptyset$; apply Theorem 7.18. $\square$

Since $q_0 + \sum_m \sigma_m = 1$, the coefficient κ_w is at most the max-form coefficient

$$\kappa_{\max} := \max_{\pm} \max (h(b^\pm), \max_m H_m(\omega_m^\pm)),$$

and it can be much smaller. Corollary 7.20 therefore applies under the weighted coefficient $\kappa_w \leq 1$ rather than $\kappa_{\max} \leq 1$, and needs no spanning hypothesis when several blocks are active.

Corollary 7.21 (Non-negative d -mismatch). *Let I be finite, $f \in S_{p^*}$ with F finite, and let $g \in \mathcal{C}(f)$ carry two-piece data with $\Delta d_m \geq 0$ for every m and $\kappa_w \leq 1$. Then $(f, g) \in \text{cl NA}((c_0, p), \ell_2^2)$. No hypothesis is made on Q_m (infinitely many strict non-peaks, near-threshold coordinates and degenerate peaks are allowed), on K , or on T .*

Proof. $I_- = \emptyset$; apply Theorem 7.18. The fine structure of the blocks enters the proof only through (E4), (E5) and Lemma 7.14, which hold at every f with F finite. $\square$

Corollary 7.22 (Block-tame points are recoverable). *Let I be finite. Every $f \in S_{p^*}$ satisfying (BT) belongs to $\mathcal{R}$; that is, $(f, g) \in \text{cl NA}((c_0, p), \ell_2^2)$ for every $g \in \mathcal{C}(f)$, whatever the contact set K .*

Proof. By Theorem 7.10(c), every $g \in \mathcal{C}(f)$ carries two-piece data with $\kappa_w \leq 1$. By the remark after Definition 7.16, (BT) implies $\text{Scr}_m(\Upsilon s) = o(s)$ as $s \rightarrow 0$ for every block and every Υ , so every set of blocks satisfies (SC). Theorem 7.18 gives $(f, \rho g) \in \text{cl NA}$ for every $\rho < 1$, hence $(f, g) \in \text{cl NA}$. $\square$

Corollary 7.22 contains the conclusion $f \in \mathcal{R}$ of Theorem 5.13 (which needed K finite), and at (BT) points the recoverability of the mates in $\mathcal{K}(f)$ given by Proposition 6.1(c). It does not contain the stronger statements of these results that the recovery takes place along the canonical truncations.

Corollary 7.23 (The example of Section 6). *Let $T, I \ni 1$ and f be as in Lemma 6.4. Then f satisfies (BT), and:*

- (a) $f \in \mathcal{R}$: every mate of f , in particular every element of the defect $\text{Def}(f)$ of Theorem 6.6, is recovered along engineered norm-attaining approximants;
- (b) every $g \in \mathcal{C}(f)$ is supported in $\{1\} \cup K_0$, and $\mu^+ \leq g_j/u_{2,1}(j) \leq \mu^-$ for $j \in K_0$, where $(\mu^\pm/\lambda_0)e_2$ (in block 1) are the block parts of optimal two-piece data of g .

Proof. (BT): $F = \{1\}$, $Q_1 = \{2\}$ and $Q_m = \emptyset$ for $m \neq 1$ (Lemma 6.4). By the proof of Lemma 6.4, $\vartheta_m \Phi_m(k) \leq q_0 \varpi_{k,m}$ for $k \geq 2$ and $\vartheta_m \Phi_m(1) \leq q_0/2$. With (P2), (P3) this gives $\mu_{1,m} \geq q_0(\frac{7}{9} - \frac{1}{2}) \geq q_0/4$, $\mu_{k,m} \geq q_0(\frac{1}{2} - \varpi_{k,m}) \geq q_0/4$ for exceptional (k, m) , and $\mu_{k,m} \geq q_0(\frac{3}{2}r_l - r_l^2) \geq q_0 r_l$ ($l = \ell(k, m)$, $r_l \leq \frac{1}{2}$) for the other peaks; so there are no degenerate peaks. For $s < q_0/4$, $\mu_{k,m} < s$ forces a non-exceptional (k, m) with $q_0 r_l < s$, i.e. $2\Phi_m(k) \leq \varpi_{k,m} < s^2/q_0^2$. Since $\Phi_m(k+1) \leq \Phi_m(k)/2$, the sum of $\Phi_m(k)$ over these k is at most $s^2/q_0^2 = o(s)$: (MS) holds. (a) is Corollary 7.22. (b) By Theorem 7.10(c), $g = b^\pm + R_1^*(\omega_1^\pm - d_1(\omega_1^\pm)w_1)$ with $\omega_1^\pm = (\mu^\pm/\lambda_0)e_2$; since $w_1(2) = 0$, $d_1(\omega_1^\pm) = 0$ and $g = b^\pm + \mu^\pm u_{2,1}$. The vectors $b^\pm$ are supported in $F \cup K = \{1\} \cup K_0$, with $b_j^+ \geq 0 \geq b_j^-$ on K_0 (where $z_j = 1$), and $\text{supp } u_{2,1} = \{1\} \cup K_0$ with $u_{2,1} > 0$ on K_0 (P1). $\square$

Part (b) is the converse statement of Remark 6.7(b), and (a) shows that the defect of Theorem 6.6 is a defect of the intrinsic mechanisms of Sections 3–5 only. The explicit mates g_{K_1} are also covered directly by Corollary 7.20, through their d -neutral data described after Definition 7.7 (for ρ with $\rho^2 \kappa_w < 1$).

Remark 7.24 (Negative d -mismatch without (SC)). (a) Without (SC), the proof of Theorem 7.18 still gives $(f, \rho g) \in \text{cl NA}$ whenever

$$\rho^2 \kappa_w + \rho \sum_{m \in I_-} |\Delta d_m| K_m^{\text{scr}} < 1, \quad K_m^{\text{scr}} := \limsup_i \frac{\mathfrak{g}_m + 2\|R_m^* Z_{A,m}\|_1}{s_{1,i}},$$

along a construction sequence for which all K_m^{scr} are finite: in Step 5 the anchor costs then contribute at most $\frac{\tau^2}{2} \rho \sum_{I_-} |\Delta d_m| K_m^{\text{scr}} (1 + o(1))$ for $|\tau| > s_1$. Finiteness of the K_m^{scr} is a *hypothesis* of this implication, and it is not automatic. (*Sketch, not proved here:* for suitable admissible T , built after $\hat{z}$, the truncated mass of near-threshold peaks is of order $s \log(1/s)$.) A sufficient condition is that the blocks of I_- have no degenerate peaks, that (MS) holds in them, and that for $m \in I_-$, as $s \rightarrow 0$,

$$\sum \{\Phi_m(k) : k \in Q_m, \Phi_m(k) \text{gap}_m(k) \leq s \text{ or } \text{gap}_m(k) \leq s\} = O(s);$$

indeed, the proof of Lemma 7.17 then gives $\mathfrak{S}_m = O(s_1)$ instead of $o(s_1)$, and hence $K_m^{\text{scr}} < \infty$. Gaps of the strict non-peaks bounded below off a finite set do *not* suffice

for this in general (the bound $\Phi_m(k) \leq 2^{-m-k}$ only gives $O(s \log(1/s))$ for the sum above); they do suffice if moreover $\Phi_m(k+1) \leq \beta \Phi_m(k)$ for some $\beta < 1$, since then $\sum \{\Phi_m(k) : \Phi_m(k) \leq x\} \leq x/(1-\beta)$. So without (SC) the possible failure of recovery for $\Delta d < 0$ is confined to ρ close to 1 only in such cases. (b) *Sketch, not used.* One can construct admissible operators for which (SC) fails at a first row (prescribe block vectors in $\hat{z}^\perp$ at a positive proportion of the indices of one block, so that these become strict non-peaks with $w_m(k) = 0$ at all fine scales). Then the estimate of Theorem 7.18 does not apply to mates with $\Delta d < 0$ in that block for ρ close to 1. Whether these mates are recovered along other approximants is open; that the approximants of Definition 7.12 fail to recover them is *not* proved.

8. A DESIGNED ADMISSIBLE OPERATOR

Martín's construction works with every admissible operator T (Remark 1.2). In this section we *design* T and prove that, for this T and every finite block set, every first row in a large class $\mathcal{R}_0$ is recoverable; consequently density for this T is equivalent to a single statement about approximating mates by mates at first rows in $\mathcal{R}_0$ (Lemma Z, Problem 8.21). The mechanism is *signature pinning*: every block vector carries a private signature on its own set of coordinates, so that at a first row whose signature sets keep some room, the block coefficients of any two admissible decompositions of a mate are fixed by the mate itself, up to $O(t)$.

Throughout this section $N \geq 1$ is fixed, $I = \{1, \dots, N\}$ and $p = p_N$. The symbols $S_l, h_l, \delta_l, c_l, n_l, y_l, u_l$ below are local to this section and are unrelated to the construction of Section 6; so are the ladder bijection j , the room constant ϖ_0 , the block coefficients $\theta_l^\pm$, the pinning weights $\pi_{l',l}$, the base coefficient $\mathfrak{a}_t$ and the local-validity constant c_b .

8.1. The signature-ladder design.

Definition 8.1 (Signature-ladder design). (D0) Fix a bijection $j : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ with $j(k, m) < j(k', m)$ for $k < k'$, and write $(k(l), m(l)) := j^{-1}(l)$. Fix a coordinate j_0 , pairwise disjoint infinite sets $S_l \subseteq \mathbb{N} \setminus \{j_0\}$ ($l \in \mathbb{N}$), and put

$$h_l := \sum_{s \in S_l} 2^{-s} e_s^* (\geq 0), \quad \delta_l := \min \{2^{-l}, (4(1 + \|U\|)\|h_l\|_1)^{-1}\},$$

so that $q^*(\delta_l h_l) \leq (1 + \|U\|)\delta_l \|h_l\|_1 \leq \frac{1}{4}$. Fix *targets* $y^{(i)} \in c_{00} \cap S_{q^*}$ ($i \geq 1$), dense in S_{q^*} , with $y^{(1)} := e_{j_0}^*/q^*(e_{j_0}^*)$, and a sequence $(i_r)_{r \geq 1}$ in which every positive integer occurs infinitely often.

(D1) Recursively in $l = 1, 2, \dots$: given $c_l \in (0, 1]$ ($c_1 := 1$), call $y \in c_{00}$ *allowed at l* if

- (a) $\text{supp } y \cap S_{l'} = \emptyset$ for every $l' \geq l$, and
- (b) $2c_l \leq 2^{-2s} c_{l'} \delta_{l'}$ for every $l' < l$ and every $s \in \text{supp } y \cap S_{l'}$.

Let $y_l := y^{(i_{k(l)})}$ if this target is allowed at l , and $y_l := y^{(1)}$ otherwise ($y^{(1)}$ is allowed at every l). Put

$$n_l := q^*(y_l + \delta_l h_l), \quad u_l := (y_l + \delta_l h_l)/n_l, \quad \delta_l^\circ := \delta_l \|h_l\|_1/n_l,$$

$$\Lambda^\circ(l) := \prod_{l'' \leq l} \left(1 + \frac{2}{\delta_{l''}^\circ}\right), \quad n_l^w := \lceil l 2^{l^3} \Lambda^\circ(l) \rceil,$$

$$T_{\text{hi}}(l) := \min \left\{ T_{\text{lo}}(l-1), \frac{2^{-l^3}}{l \Lambda^\circ(l)} \right\} \quad (T_{\text{lo}}(0) := 1),$$

$$T_{\text{lo}}(l) := 2^{-n_l^w} T_{\text{hi}}(l), \quad c_{l+1} := \min\{c_l/4, T_{\text{lo}}(l)^3\}.$$

(D2) $Te_{k,m} := c_{j(k,m)} u_{j(k,m)}$, extended linearly and continuously to $\ell_1(\mathbb{N} \times \mathbb{N})$.

The interval $\mathcal{W}(l) := [T_{\text{lo}}(l), T_{\text{hi}}(l)]$ is the l -th window; it contains the n_l^w dyadic scales $T_{\text{hi}}(l)2^{1-i}$, $1 \leq i \leq n_l^w$.

Theorem 8.2 (The design is admissible). *The operator T of Definition 8.1 is admissible. For $l = j(k, m)$ we have $u_{k,m} = u_l$, $\Phi_m(k) = 2^{-m-k} c_l$ and $\lambda_l := \lambda_{k,m} \leq c_l/4$, and:*

(P1) $y_l \in c_{00}$, $\|y_l\|_1 \leq 1$, $n_l \in [\frac{3}{4}, \frac{5}{4}]$ and $\text{supp } y_l \cap S_{l'} = \emptyset$ for $l' \geq l$. Consequently, for $s \in S_l$: $u_l(s) = \delta_l 2^{-s}/n_l$, $u_{l'}(s) = 0$ for $l' < l$, and $u_{l'}(s) = y_{l'}(s)/n_{l'}$ for $l' > l$.

(P2) $\sum_{l' > l} c_{l'} \leq 2c_{l+1} \leq 2T_{\text{lo}}(l)^3$ and $\sum_{l' > l} \lambda_{l'} \leq T_{\text{lo}}(l)^3$.

(P3) $T_{\text{hi}}(l)2^{l^3} \Lambda^\circ(l) \leq 1/l$, $n_l^w \geq l 2^{l^3} \Lambda^\circ(l)$, and $T_{\text{hi}}(l+1) \leq T_{\text{lo}}(l)$.

Proof. (P1): $q^*(y_l) = 1$, $\|y_l\|_1 \leq q^*(y_l)$ and $q^*(\delta_l h_l) \leq \frac{1}{4}$ give the bounds; the support statement is (a); for $s \in S_l$ the signatures $h_{l'}$, $l' \neq l$, vanish at s (disjointness), and the targets $y_{l'}$ with $l' \leq l$ vanish on S_l by (a). (P2): $c_{l'+1} \leq c_l/4$ and $\lambda_{l'} \leq c_{l'}/4$. (P3) is immediate from (D1).

(T-a). $q^*(Te_{k,m}) = c_l \leq 1 = c_1$.

(T-d). Fix m and i . The finite set $\text{supp } y^{(i)}$ meets only finitely many $S_{l'}$, and $c_l \rightarrow 0$, so $y^{(i)}$ is allowed at every large l . Since i occurs infinitely often in (i_r) , $y_l = y^{(i)}$ for infinitely many $l = j(k, m)$, and along these $\delta_l \rightarrow 0$ and $n_l \rightarrow 1$, so $q^*(u_l - y^{(i)}) \rightarrow 0$. Hence every $y^{(i)}$ is a limit of $u_{k,m}$, $k \rightarrow \infty$, and $\{u_{k,m} : k \in \mathbb{N}\}$ is dense in S_{q^*} .

(T-b), (T-c). Let $x \in \ell_1(\mathbb{N} \times \mathbb{N})$, indexed by l , with $\mathcal{Z} := Tx = \sum_l x_l c_l u_l \in c_{00}$, and suppose $x_{l'} \neq 0$. For $s \in S_{l'}$, by (P1),

$$\mathcal{Z}(s) = \frac{x_{l'} c_{l'} \delta_{l'} 2^{-s}}{n_{l'}} + \sum_{l > l', s \in \text{supp } y_l} \frac{x_l c_l y_l(s)}{n_l}.$$

If the last sum is not empty, let $L(s)$ be its least index; by (b) at $L(s)$, $2c_{L(s)} \leq 2^{-2s} c_{l'} \delta_{l'}$, and the sum is at most

$$\frac{4}{3} \|x\|_\infty \sum_{l \geq L(s)} c_l \leq \frac{16}{9} \|x\|_\infty c_{L(s)} \leq \frac{8}{9} \|x\|_\infty 2^{-2s} c_{l'} \delta_{l'}.$$

Hence

$$|\mathcal{Z}(s)| \geq c_{l'} \delta_{l'} 2^{-s} \left(\frac{|x_{l'}|}{n_{l'}} - \frac{8}{9} \|x\|_\infty 2^{-s} \right) > 0$$

for all large $s \in S_{l'}$, contradicting $\mathcal{Z} \in c_{00}$. So $x = 0$: T is injective and $T(\ell_1(\mathbb{N} \times \mathbb{N})) \cap c_{00} = \{0\}$. $\square$

Only (T-a)–(T-d), (P1), (P2) and (P3) are used below. No rate of approximation of the targets and no independence property of the targets is used; the same target sequence serves all blocks, so Martín’s cross-block near duplicates are compatible with the design.

8.2. The class $\mathcal{R}_0$. Let $\mathcal{L}_N := \{l : m(l) \leq N\}$, the ladder indices of the blocks present in p_N . For $f \in S_{p^*}$ and $\gamma \in (0, 1)$ let $J_\gamma := \{j \notin F : |z_j| \leq 1 - \gamma\}$ (coordinates with room $\geq \gamma$).

Definition 8.3 (Signature room). $f \in S_{p^*}$ belongs to $\mathcal{R}_0$ if F is finite and

(SR) there are $\gamma \in (0, 1)$ and $\varpi_0 \in (0, 1]$ with $\|h_l 1_{S_l \cap J_\gamma}\|_1 \geq \varpi_0^l \|h_l\|_1$ for every $l \in \mathcal{L}_N$.

For such f put $\delta'_l := \delta_l \|h_l 1_{S_l \cap J_\gamma}\|_1 / n_l \geq \varpi_0^l \delta_l^\circ$, $\Lambda_f(l) := \prod_{l'' \leq l, l'' \in \mathcal{L}_N} (1 + 2/\delta'_{l''})$.

Since $1 + 2/\delta'_l \leq \varpi_0^{-l} (1 + 2/\delta_l^\circ)$ and $\sum_{l'' \leq l} l'' \leq l^2$, we have $\Lambda_f(l) \leq \varpi_0^{-l^2} \Lambda^\circ(l)$.

Lemma 8.4. (a) Every $f' \in \text{NA}((X, p), \mathbb{R}) \cap S_{p^*}$ belongs to $\mathcal{R}_0$. (b) Every base-tame f (F and K finite and $\sup_{j \notin F \cup K} |z_j| < 1$) belongs to $\mathcal{R}_0$. (c) $\mathcal{R}_0$ contains first rows with infinite contact sets and with near-contacts; (SR) imposes no condition on the block data.

Proof. (a) By Proposition 1.10(c), $F' = \text{supp } a'$ is finite and $z' \in c_0$, so $\{j \notin F' : |z'_j| > \frac{1}{2}\}$ is finite. Choose $\gamma \in (0, \frac{1}{2}]$ with $\gamma \leq 1 - |z'_j|$ for each of these j with $|z'_j| < 1$. Then $\mathbb{N} \setminus J_\gamma = F' \cup K'$ is finite. Every S_l is infinite, so $\|h_l 1_{S_l \cap J_\gamma}\|_1 > 0$ for all l , and the ratio is 1 for all but finitely many l ; take ϖ_0 as the minimum of 1 and of the l -th roots of the finitely many other ratios. (b) The same argument with $\gamma := 1 - \sup_{j \notin F \cup K} |z_j|$. (c) Since $\sum_{s > s_0} 2^{-s} = 2^{-s_0}$, the least element of S_l carries at least half of the mass of h_l . Hence every f with F finite and disjoint from $\bigcup_l S_l$ (e.g. $F = \{j_0\}$), $|z_j| \leq \frac{1}{2}$ at $j = \min S_l$ for all l , and arbitrary z elsewhere (for instance $|z_j| = 1$ on the infinite set $\bigcup_l (S_l \setminus \{\min S_l\})$) satisfies (SR) with $\gamma = \frac{1}{2}$ and $\varpi_0 = \frac{1}{2}$. First rows with prescribed (a, z) exist by Remark 8.22(c). $\square$

The first row of Section 6, transplanted to this design (its contact set being a signature set), is the typical first row outside $\mathcal{R}_0$ (*signature-resonant*).

8.3. Two-sided decompositions (any admissible T). In this subsection T is an arbitrary admissible operator, I is finite, $f \in S_{p^*}$ and $g \in \mathcal{C}(f)$. For $\Theta \in \ell_\infty$ put $H_m(\Theta) := \|P_m^\perp D_m \Theta\|_2^2 / C_m$ (this agrees with Definition 3.1 on vectors vanishing on P_m , and $H_m(\Theta + c w_m) = H_m(\Theta)$), and for $B \in \ell_1$, $\Theta = (\Theta_m) \in \prod_m \ell_\infty$ put $\Gamma_w(B, \Theta) := q_0 h(B) + \sum_m \sigma_m H_m(\Theta_m)$. Then $\sqrt{\Gamma_w}$ is a seminorm.

Lemma 8.5 (Two-sided decompositions). *For every $t > 0$ there are $B_\pm \in \ell_1$ and $\Theta_\pm \in \prod_m \ell_\infty$ with $g = B_+ + L^* \Theta_+ = B_- + L^* \Theta_-$ and*

$$q^*(a + t B_+), N_m(w_m + t \Theta_{+,m}), q^*(a - t B_-), N_m(w_m - t \Theta_{-,m}) \leq s(t) \quad (m \in I).$$

Proof. By Lemma 1.6, $f + t g = A + L^* W$ with $\max(q^*(A), \|W\|_{V^*}) = p^*(f + t g) \leq s(t)$; put $B_+ := (A - a)/t$ and $\Theta_+ := (W - w)/t$. The same with $-t$ gives B_-, Θ_- . $\square$

We call $(B_\pm, \Theta_\pm)$ a *two-sided decomposition at scale t* .

Lemma 8.6 (Uniform smallness). *For every $\eta > 0$ there is $t_\eta > 0$ such that every two-sided decomposition at a scale $t \in (0, t_\eta]$ satisfies, for both signs and every m , $\|tB_\pm\|_1 \leq \eta$, $\|D_m(t\Theta_{\pm,m})\|_2 \leq \eta$ and $|\|w_m \pm t\Theta_{\pm,m}\|_\infty - M_m| \leq \eta$.*

Proof. Otherwise there are $t_n \rightarrow 0$ and decompositions violating one of the inequalities, say on the $+$ side. Put $A_n := a + t_n B_{+,n}$ and $W_n := w + t_n \Theta_{+,n}$; they are bounded, so after passing to a subsequence $A_n \rightarrow A$ weak* in ℓ_1 and $W_{n,m} \rightarrow W_m$ weak* in ℓ_∞ . As L^* is weak*-continuous, $A + L^*W = f$ with $q^*(A) \leq 1$ and $\|W\| \leq 1$, so $A = a$ and $W = w$ by Proposition 1.9(b). Since U is compact, $\|U^*A_n\| \rightarrow \nu$, hence $\limsup \|A_n\|_1 \leq \lim(s(t_n) - \|U^*A_n\|) = \|a\|_1$; for bounded coordinatewise convergent sequences this gives $\|A_n - a\|_1 \rightarrow 0$. D_m is weak*-to-norm continuous on bounded sets, so $\|D_m(W_{n,m} - w_m)\|_2 \rightarrow 0$ and $\|D_m W_{n,m}\|_2 \rightarrow C_m$; then $\limsup \|W_{n,m}\|_\infty \leq 1 - C_m = M_m \leq \liminf \|W_{n,m}\|_\infty$. This contradicts the choice of the decompositions. $\square$

Lemma 8.7 (First-order terms, budget and one-sided bounds). *Let $\eta \leq \eta_* := \frac{1}{2} \min(\nu/(2\|U\| + 2), \min_m C_m, \frac{1}{4})$ and $0 < t \leq \min(t_\eta, 1)$, and let $(B_\pm, \Theta_\pm)$ be a two-sided decomposition at scale t . Then for both signs:*

- (a) $|q_0 B_\pm(\hat{z})| \leq t/2$ and $|\langle \Theta_{\pm,m}, \zeta_m \rangle| \leq t/2$ for every m ;
- (b) the one-sided base bounds

$$\sum_{j \notin F} (|B_+(j)| - z_j B_+(j)) \leq \frac{t}{2q_0}, \quad \sum_{j \notin F} (|B_-(j)| + z_j B_-(j)) \leq \frac{t}{2q_0}$$

hold; consequently $\sum_{j \in J_\gamma} |B_\pm(j)| \leq t/(2\gamma q_0)$ for $\gamma \in (0, 1)$, and

$$\|B_+ 1_{F^c}\|_1 + \|B_- 1_{F^c}\|_1 \leq \|(B_+ - B_-) 1_{F^c}\|_1 + t/q_0;$$

- (c) $\sum_{k \in P_m} |\alpha_m(k)| (\|W\|_\infty - \text{sign}(w_m(k))W(k)) \leq t^2/(2\sigma_m)$ for $W = w_m + t\Theta_{+,m}$ and for $W = w_m - t\Theta_{-,m}$;
- (d) $\Gamma_w(B_\pm, \Theta_\pm) \leq 1 + \eta_\Gamma(\eta)$, where $\eta_\Gamma(\eta) := (1 + \|U\|\eta/\nu) \max_m (1 + \eta/C_m) - 1 \rightarrow 0$ as $\eta \rightarrow 0$.

Proof. Consider the $+$ side; the $-$ side is the same with $-t$, $-B_-$, $-\Theta_-$. Let $A := a + tB_+$, $W_m := w_m + t\Theta_{+,m}$. By Lemma 7.1(a) with $h = tg$ and $g(\xi) = 0$,

$$(14) \quad q_0 G_b(A) + \sum_m \sigma_m G_m(W_m) = q_0 q^*(A) + \sum_m \sigma_m N_m(W_m) - 1 \leq s(t) - 1 \leq \frac{t^2}{2}.$$

(a) $q^*(A) \geq A(\hat{z})$ gives $tB_+(\hat{z}) \leq s(t) - 1 \leq t^2/2$, and $N_m(W_m) \geq \langle W_m, \zeta_m \rangle / \sigma_m$ gives $\langle \Theta_{+,m}, \zeta_m \rangle \leq \sigma_m t/2$. Since $q_0 B_+(\hat{z}) + \sum_m \langle \Theta_{+,m}, \zeta_m \rangle = g(\xi) = 0$ and $q_0 + \sum_m \sigma_m = 1$, each of these terms is also $\geq -t/2$. (b) By Lemma 7.1(b), $G_b(A) \geq \sum_{j \notin F} (|tB_+(j)| - z_j tB_+(j))$; use (14). On J_γ , $|x| \mp z_j x \geq \gamma|x|$. The last inequality follows by summing over $j \notin F$ the inequality $|x| + |y| \leq |x - y| + (|x| - zx) + (|y| + zy)$ ($|z| \leq 1$), whose two sides differ by $|x - y| - z(x - y) \geq 0$. (c) Lemma 7.1(c) and (14). (d) By Lemma 3.3, $G_b(A) \geq \nu \Psi(tU^*B_+/\nu) \geq \frac{t^2}{2} h(B_+)/ (1 + \|U\|\eta/\nu)$, since $t\|U^*B_+\| \leq \|U\|\eta \leq \nu/2$. By Lemma 7.1(c) and Lemma 8.6,

$$\begin{aligned} G_m(W_m) &\geq \|D_m W_m\| - \frac{\langle D_m W_m, D_m w_m \rangle}{C_m} = \frac{\|P_m^\perp D_m W_m\|^2}{\|D_m W_m\| + \langle D_m W_m, D_m w_m \rangle / C_m} \\ &\geq \frac{t^2}{2} \frac{H_m(\Theta_{+,m}) C_m}{C_m + \eta}. \end{aligned}$$

Insert both bounds in (14). $\square$

Lemma 8.8 (Sup-level parametrization). *In the setting of Lemma 8.7, fix m and drop it. Define $d_\pm$ by $\|w + t\Theta_+\|_\infty = (1 - d_+t)M$ and $\|w - t\Theta_-\|_\infty = (1 + d_-t)M$, and put $\omega_\pm := \Theta_\pm + d_\pm w$. Then:*

- (a) $|d_\pm|t \leq \eta/M \leq \frac{1}{2}$;
- (b) (box) $|(1 - d_+t)w(k) + t\omega_+(k)| \leq (1 - d_+t)M$ and $|(1 + d_-t)w(k) - t\omega_-(k)| \leq (1 + d_-t)M$ for every k ;
- (c) (peaks) for $k \in P$ and $\varsigma_k := \text{sign } w(k)$, $\varsigma_k \omega_+(k) \leq 0 \leq \varsigma_k \omega_-(k)$, and $\sum_{k \in P} |\alpha(k)| |\omega_\pm(k)| \leq t/(2\sigma)$;
- (d) $|d(\omega_\pm) - d_\pm| \leq t/\sigma$, where $d(\omega) := \langle Dw, Dw \rangle / C$;
- (e) $|t\Theta_\pm(k)| \leq 3$ for all k , and $|d_\pm|M \leq 3\|\Phi\|_2/t + t/\sigma$;
- (f) (non-peaks) for $k \in Q$ and $\varsigma_k := \text{sign } w(k)$ (any sign if $w(k) = 0$), $\varsigma_k \omega_+(k) \leq (1 - d_+t) \text{gap}(k)/t$ and $\varsigma_k \omega_-(k) \geq -(1 + d_-t) \text{gap}(k)/t$.

Proof. (a) is Lemma 8.6 ($M \geq \frac{1}{2}$, $\eta \leq \frac{1}{8}$). (b): $w + t\Theta_+ = (1 - d_+t)w + t\omega_+$ and $w - t\Theta_- = (1 + d_-t)w - t\omega_-$. (c): at a peak, $|(1 - d_+t)\varsigma_k M + t\omega_+(k)| \leq (1 - d_+t)M$ forces $\varsigma_k t\omega_+(k) \leq 0$, and then $\|w + t\Theta_+\|_\infty - \varsigma_k(w + t\Theta_+)(k) = t|\omega_+(k)|$; insert in Lemma 8.7(c). The $-$ side is symmetric. (d): by Lemma 1.7, $\langle \Theta_+, \zeta \rangle / \sigma = \langle \omega_+ - d_+w, \alpha \rangle + \langle D(\omega_+ - d_+w), Dw \rangle / C = -\sum_P |\alpha(k)| |\omega_+(k)| + d(\omega_+) - d_+$, using (c) and $M + C = 1$; similarly $\langle \Theta_-, \zeta \rangle / \sigma = \sum_P |\alpha(k)| |\omega_-(k)| + d(\omega_-) - d_-$. Use Lemma 8.7(a) and (c). (e): $|t\Theta_\pm(k)| \leq N(w \pm t\Theta_\pm) + |w(k)| \leq s(1) + 1$; and $d(\omega_\pm) - d_\pm = \langle Dw, D\Theta_\pm \rangle / C - d_\pm M$, so $|d_\pm|M \leq \|D\Theta_\pm\|_2 + t/\sigma$. (f): multiply the box inequality (b) by ς_k and use $|w(k)| = M - \text{gap}(k)$. $\square$

Lemma 8.9 (Base part on a finite support). *If F is finite, there is $C_F < \infty$ with $\|\beta\|_1 \leq C_F(\|P^\perp U^* \beta\| + |\beta(\hat{z})|)$ for every β supported in F .*

Proof. On the finite-dimensional space $\ell_1(F)$ the map $\beta \mapsto (P^\perp U^* \beta, \beta(\hat{z}))$ is injective: if $P^\perp U^* \beta = 0$, then $U^* \beta = \mu U^* a / \nu$, so $\beta = (\mu/\nu)a$ (U^* injective) and $\beta(\hat{z}) = \mu/\nu$. $\square$

8.4. Signature pinning. From now on T is the operator of Definition 8.1, $f \in \mathcal{R}_0$ with constants $\gamma, \varpi_0, g \in \mathcal{C}(f)$, $\eta \leq \eta_*$ with $\eta_\Gamma(\eta) \leq 1$, $0 < t \leq \min(t_\eta, 1)$, and $(B_\pm, \Theta_\pm)$ is a two-sided decomposition of g at scale t . For $l \in \mathcal{L}_N$ put $\theta_l^\pm := \lambda_l \Theta_{\pm, m(l)}(k(l))$ (and $\theta_l^\pm := 0$ for $l \notin \mathcal{L}_N$), so that $L^* \Theta_\pm = \sum_l \theta_l^\pm u_l$, and

$$\Delta \theta_l := \theta_l^+ - \theta_l^-, \quad \Delta B := B_+ - B_- = - \sum_l \Delta \theta_l u_l.$$

For a *window index* l_* the indices $l \leq l_*$ are *coarse* and the indices $l > l_*$ are *fine*.

Lemma 8.10 (Box bound). $|\theta_l^\pm| \leq 3\lambda_l/t$ for every l .

Proof. Lemma 8.8(e). $\square$

Lemma 8.11 (Pinning inequality). *For every $l \in \mathcal{L}_N$,*

$$\delta'_l |\Delta \theta_l| \leq E_l + \sum_{l' > l} \pi_{l', l} |\Delta \theta_{l'}|, \quad E_l := \|\Delta B 1_{S_l \cap J_\gamma}\|_1, \quad \pi_{l', l} := \|y_{l'} 1_{S_l}\|_1 / n_{l'},$$

and $\sum_l \pi_{l', l} \leq \frac{4}{3}$ for every l' .

Proof. By (P1), for $s \in S_l$, $-\Delta B(s) = \Delta \theta_l \delta_l 2^{-s} / n_l + \sum_{l' > l} \Delta \theta_{l'} y_{l'}(s) / n_{l'}$. Take the ℓ_1 -norm over $s \in S_l \cap J_\gamma$: the first term has norm $|\Delta \theta_l| \delta'_l$. The sets S_l are disjoint and $\|y_{l'}\|_1 / n_{l'} \leq \frac{4}{3}$. $\square$

Lemma 8.12 (Triangular solution). *For every l_* , $\sum_{l \leq l_*} |\Delta \theta_l| \leq \Lambda_f(l_*) \left(\sum_{l \leq l_*} E_l + \frac{4}{3} \sum_{l > l_*} |\Delta \theta_l| \right)$.*

Proof. For coarse $l \in \mathcal{L}_N$ put $D_l := |\Delta \theta_l|$, $X_l := E_l + \sum_{l' > l_*} \pi_{l',l} D_{l'}$ and $\Sigma_l := \sum_{l \leq l' \leq l_*} D_{l'}$ ($\Sigma_{l_*+1} := 0$; $D_l := 0$ for $l \notin \mathcal{L}_N$). By Lemma 8.11, $D_l \leq (X_l + \frac{4}{3} \Sigma_{l+1})/\delta'_l$, so $\Sigma_l \leq X_l/\delta'_l + (1 + \frac{4}{3\delta'_l}) \Sigma_{l+1}$. Unrolling from l_* downwards (products over $\mathcal{L}_N$),

$$\Sigma_1 \leq \sum_{l \leq l_*} \frac{X_l}{\delta'_l} \prod_{l'' < l} \left(1 + \frac{4}{3\delta'_{l''}} \right) \leq \Lambda_f(l_*) \sum_{l \leq l_*} X_l,$$

and $\sum_{l \leq l_*} X_l \leq \sum_{l \leq l_*} E_l + \frac{4}{3} \sum_{l' > l_*} D_{l'}$ by Lemma 8.11. $\square$

Proposition 8.13 (Pinning on a window). *If $t \in \mathcal{W}(l_*)$, then with $K := K_1 \Lambda_f(l_*)$, $K_1 := 1/(\gamma q_0) + 14$, and $K_d := \max_m(1/(mC_m) + 2/(\sigma_m M_m))$:*

- (a) $\sum_l |\Delta \theta_l| \leq Kt$ and $\|\Delta B\|_1 \leq Kt$;
- (b) $\|B_+ 1_{F^c}\|_1 + \|B_- 1_{F^c}\|_1 \leq Kt + t/q_0$; in particular the total base mass of both decompositions on contacts and near-contacts is $O(Kt)$;
- (c) $|d_{+,m} - d_{-,m}| \leq K_d Kt$ for every m , with $d_{\pm,m}$ as in Lemma 8.8.

Proof. (a) By Lemma 8.7(b) and disjointness of the S_l , $\sum_l E_l \leq \sum_{j \in J_\gamma} (|B_+(j)| + |B_-(j)|) \leq t/(\gamma q_0)$. By Lemma 8.10 and (P2), $\sum_{l > l_*} |\Delta \theta_l| \leq 6 \sum_{l > l_*} \lambda_l/t \leq 6T_{\text{io}}(l_*)^3/t \leq 6t^2$. By Lemma 8.12, $\sum_l |\Delta \theta_l| \leq \Lambda_f(l_*)(t/(\gamma q_0) + 8t^2) + 6t^2 \leq Kt$. And $\|u_l\|_1 \leq q^*(u_l) = 1$. (b) Lemma 8.7(b) and (a). (c) Drop m ; put $\Delta \Theta := \Theta_+ - \Theta_-$ and $\Delta d := d_+ - d_-$. By Lemma 8.8(d), $\Delta d = d(\omega_+ - \omega_-) + r$ with $|r| \leq 2t/\sigma$, and $\omega_+ - \omega_- = \Delta \Theta + \Delta d w$, so $\Delta d M = \langle Dw, D\Delta \Theta \rangle/C + r$. Since $\lambda_k = m\Phi_m(k)$, $\langle Dw, D\Delta \Theta \rangle = \frac{1}{m} \sum_k \Phi_m(k) w(k) \Delta \theta_{j(k,m)}$, whose modulus is at most $\frac{M}{m} \sum_l |\Delta \theta_l|$. Divide by M and use (a) and $K \geq 1$. $\square$

Thus every one-sided resource used by the two decompositions (contacts and near-contacts on the base side; on the block side peaks, near-threshold coordinates and coordinates pushed beyond their gap, as the next proof shows) is controlled by the two-sided differences $\Delta \theta_l$, ΔB , which are pinned by the signatures. At the first row of Section 6 this fails: there the signature set of the resonant carrier is the contact set.

8.5. The window certificate.

Definition 8.14 (Window certificate). For $t \in \mathcal{W}(l_*)$ let $\mathbf{a}_t := (B_+ 1_F)(\hat{z})$, $b_t := B_+ 1_F - \mathbf{a}_t a$, and in block m let

$$\omega_m^c(k) := \max \left\{ -\frac{2 \text{gap}_m(k)}{t}, \min \left\{ \omega_{+,m}(k), \frac{2 \text{gap}_m(k)}{t} \right\} \right\}$$

if $k \in Q_m$, $j(k,m) \leq l_*$ and $\text{gap}_m(k) \geq t^2$, and $\omega_m^c(k) := 0$ otherwise. Put $c_t := (b_t, (\omega_m^c)_m)$.

Proposition 8.15 (Window certificate). *Let $t \in \mathcal{W}(l_*)$ with $Kt \leq 1$. There are constants K_b, K_2, K_4 , depending only on f, γ and the design, such that:*

- (a) c_t is a balanced finite certificate (Definition 5.5); on $\text{supp } \omega_m^c$, $\text{gap}_m(k) \geq t^2$ and $|\omega_m^c(k)| \leq 2 \text{gap}_m(k)/t$; $\|D_m \omega_m^c\|_2 \leq 1/t$ and $|d_m(\omega_m^c)| \leq 1/t$;

- (b) $\|b_t\|_1 \leq K_b$;
- (c) $\|g - g_{c_t}\|_1 \leq K_2 \Lambda_f(l_*)t$;
- (d) $\sqrt{\Gamma_w(c_t)} \leq \sqrt{1 + \eta_\Gamma(\eta)} + K_4 \Lambda_f(l_*)t$.

Proof. (a) $b_t(\hat{z}) = \mathbf{a}_t - \mathbf{a}_t a(\hat{z}) = 0$; the coarse set is finite; and $\|D_m \omega_m^c\|_2 \leq (2/t)\|\Phi_m\|_2 \leq 1/t$, $|d_m(\omega)| \leq \|D_m \omega\|_2$.

(b) By Lemma 8.7(a) and Proposition 8.13(b),

$$|\mathbf{a}_t| \leq |B_+(\hat{z})| + \|\hat{z}\|_\infty \|B_+ 1_{F^c}\|_1 \leq \frac{t}{2q_0} + (1 + \|U\|) \left(Kt + \frac{t}{q_0} \right).$$

By Lemma 8.9,

$$\|B_+ 1_F\|_1 \leq C_F (\|P^\perp U^* B_+\| + \|U\| \|B_+ 1_{F^c}\|_1 + |\mathbf{a}_t|),$$

and $\|P^\perp U^* B_+\|^2 = \nu h(B_+) \leq 2\nu/q_0$ by Lemma 8.7(d). Use $Kt \leq 1$ and $t \leq 1$.

(c) $g - g_{c_t} = (B_+ - b_t) + \sum_m R_m^* X_m$ with $X_m := \Theta_{+,m} - \omega_m^c + d_m(\omega_m^c)w_m$, and $\|R_m^* X\|_1 \leq \sum_k \lambda_{k,m} |X(k)|$. First, $B_+ - b_t = B_+ 1_{F^c} + \mathbf{a}_t a$ has norm $\leq (2 + \|U\|)(K + 2/q_0)t$. Fix m and drop it; with $\Theta_+ = \omega_+ - d_+ w$,

$$X = [(\omega_+ - \omega^c) - (d_+ - d(\omega^c))w] 1_{\text{coarse}} + [\Theta_+ + d(\omega^c)w] 1_{\text{fine}}.$$

Fine part. $\sum_{\text{fine}} \lambda_k |\Theta_+(k)| = \sum_{l>l_*} |\theta_l^+| \leq 3t^2$ and $|d(\omega^c)| \sum_{\text{fine}} \lambda_k \leq t^2$, by Lemma 8.10, (P2) and $t \geq T_{\text{io}}(l_*)$.

Coarse part. Let $\varrho_k := |\omega_+(k) - \omega^c(k)|$. We claim $\lambda_k \varrho_k \leq |\Delta \theta_k| + \lambda_k |\Delta d| M + 2t \lambda_k$, where $\Delta \theta_k := \lambda_k (\Theta_+(k) - \Theta_-(k))$, so that $\lambda_k |\omega_+(k) - \omega_-(k)| \leq |\Delta \theta_k| + \lambda_k |\Delta d| M$. At a peak $\omega^c(k) = 0$, and $\omega_+(k), \omega_-(k)$ have opposite signs in the weak sense (Lemma 8.8(c)), so $\varrho_k \leq |\omega_+(k) - \omega_-(k)|$. At $k \in Q$ with $|\omega_+(k)| \leq 2 \text{gap}(k)/t$: $\varrho_k = 0$ if $\text{gap}(k) \geq t^2$, and $\varrho_k \leq 2 \text{gap}(k)/t < 2t$ otherwise. At $k \in Q$ with $|\omega_+(k)| > 2 \text{gap}(k)/t$: by Lemma 8.8(f) and $|d_\pm t| \leq \frac{1}{2}$, $\varsigma_k \omega_+(k) \leq \frac{3}{2} \text{gap}(k)/t$, hence (choosing $\varsigma_k := -\text{sign } \omega_+(k)$ if $w(k) = 0$) $\varsigma_k \omega_+(k) < -2 \text{gap}(k)/t$, and $e_k := -\varsigma_k \omega_+(k) - 2 \text{gap}(k)/t > 0$; also $\varsigma_k \omega_-(k) \geq -2 \text{gap}(k)/t$, so $e_k \leq \varsigma_k (\omega_-(k) - \omega_+(k))$: the part of $\omega_+(k)$ beyond the clamp is one-sided and bounded by the two-sided difference. Then $\varrho_k = e_k$ if $\text{gap}(k) \geq t^2$, and $\varrho_k = 2 \text{gap}(k)/t + e_k \leq 2t + |\omega_+(k) - \omega_-(k)|$ otherwise. This proves the claim. Summing, with $\sum_k \lambda_k \leq 1$ and Proposition 8.13, $\sum_{\text{coarse}} \lambda_k \varrho_k \leq (1 + K_d) Kt + 2t$. Next, $|d_+ - d(\omega^c)| \leq |d_+ - d(\omega_+)| + \|D(\omega_+ - \omega^c)\|_2 \leq t/\sigma + \sum_k \Phi_k |\omega_+(k) - \omega^c(k)|$, and

$$\begin{aligned} \sum_k \Phi_k |\omega_+(k) - \omega^c(k)| &\leq \frac{1}{m} \sum_{\text{coarse}} \lambda_k \varrho_k + \sum_{\text{fine}} \Phi_k (|\Theta_+(k)| + |d_+| M) \\ &\leq \frac{1}{m} \sum_{\text{coarse}} \lambda_k \varrho_k + 3t^2 + \left(\frac{3}{t} + \frac{t}{\sigma} \right) t^3, \end{aligned}$$

by Lemma 8.8(e) and (P2). Multiplying by $\sum_{\text{coarse}} \lambda_k |w(k)| \leq 1$ and collecting terms (with $t \leq 1$, $K \geq 1$, $\Lambda_f \geq 1$) gives (c).

(d) The pair of c_t is

$$(b_t, (\omega_m^c - d_m(\omega_m^c)w_m)_m) = (B_+, \Theta_+) - (B_+ - b_t, (X_m)_m),$$

and H_m vanishes on multiples of w_m . As $\sqrt{\Gamma_w}$ is a seminorm,

$$\sqrt{\Gamma_w(c_t)} \leq \sqrt{\Gamma_w(B_+, \Theta_+)} + \sqrt{\Gamma_w(B_+ - b_t, X)}.$$

The first term is at most $\sqrt{1 + \eta_\Gamma(\eta)}$ (Lemma 8.7(d)). For the second, $q_0 h(x) \leq \|U\|^2 \|x\|_1^2 / \nu$ and $\sigma_m H_m(X_m) \leq \|D_m X_m\|_2^2 / C_m \leq (\sum_k \lambda_{k,m} |X_m(k)|)^2 / C_m$, and both were bounded in (c) by a constant times $\Lambda_f(l_*)t$. $\square$

8.6. Uniform transfer expansion and windowed averaging.

Lemma 8.16 (Uniform transfer expansion). *Let T be admissible, I finite and $f \in S_{p^*}$ with F finite. For every $\varepsilon_{\text{tr}} > 0$ and $A_0 \geq 1$ there are $c_b \in (0, \frac{1}{8}]$ and $t_1 > 0$ such that for every $t \in (0, t_1]$ and every balanced finite certificate $c = (b, \omega)$ with*

(C-a) $\|b\|_1 \leq A_0$ and $\Gamma_w(c) \leq 2$, and

(C-b) $\text{supp } \omega_m \subseteq \{k \in Q_m : \text{gap}_m(k) \geq t^2\}$ and $|\omega_m(k)| \leq 2 \text{gap}_m(k)/t$ for all m, k ,

we have $p^*(f + rg_c) \leq 1 + \frac{r^2}{2}(\Gamma_w(c) + \varepsilon_{\text{tr}})$ for all $|r| \leq c_b t$.

Proof. Fix $\gamma_m \in (M_m/2, M_m)$ with $|w_m(k)| \neq \gamma_m$ for all k . Let $|r| \leq c_b t$, $t \leq t_1$, and consider $f + rg_c = A + \sum_m R_m^* W_m$ with $A := a + rb$, $W_m := w_m + r(\omega_m - d_m w_m)$, $d_m := d_m(\omega_m)$; the first-order terms vanish (Lemma 3.2, $b(\xi) = 0$). *Base.* If $c_b t_1 A_0 \leq \min(a_{\min}, \nu/(2\|U\| + 2))$, there are no flips on F and $\text{supp } b \subseteq F$, so by Lemmas 7.1(b) and 3.3, $G_b(A) \leq \hat{G}_b := \frac{r^2}{2} h(b)(1 + K_A |r|)$, $K_A := 2\|U\|A_0/\nu$. *Blocks.* $|d_m| \leq \|D_m \omega_m\|_2 \leq (2/t)\|\Phi_m\|_2 \leq 1/t$, so the coordinatewise radius of ω_m is at least $\min(t/4, C_m t/2) \geq c_b t$ if $c_b \leq \min_m C_m/2$, and Lemma 3.4(d) gives $G_m(W_m) \leq \hat{G}_m := \frac{r^2}{2} H_m(\omega_m)(1 + 2c_b/C_{\min})$, $C_{\min} := \min_m C_m$. Hence $\hat{\Gamma} \leq \frac{r^2}{2} \Gamma_w(c)(1 + K_A |r| + 2c_b/C_{\min})$. *Rebalancing.* Since $h(b) \leq 2/q_0$ and $H_m(\omega_m) \leq 2/\sigma_m$, $|\varepsilon_m| := |\hat{G}_m - \hat{\Gamma}|/e_{y,m} \leq K_3 r^2$ with K_3 depending only on f (as $e_{y,m} \geq \frac{1}{2}$). Choose $\eta_1 := \min(\frac{1}{4}, \varepsilon_{\text{tr}}/(8|I|K_3))$ and transfer data (Lemma 7.4); $\|D_m y_m\|_2 \leq K_y$ and $|Y_m|/q_0 \leq K_Y$ with K_y, K_Y independent of η_1 . Now $\|W_m - w_m\|_\infty \leq |r|(2/t + 1/t) \leq 3c_b$ and $\|D_m(W_m - w_m)\|_2 \leq 2c_b$; so if $c_b \leq \min_m \min((M_m - \gamma_m)/12, C_m/4)$ and t_1 is so small that $K_3 c_b^2 t_1^2 (2 + \Lambda_m^c) \leq \gamma_m/2$ and $K_3 c_b^2 t_1^2 \leq (M_m - \gamma_m)/4$, Lemma 7.2 gives (9). Proposition 7.3 yields

$$p^*(f + rg_c) \leq 1 + \hat{\Gamma} + |I|K_3 \eta_1 r^2 + 2|I|K_3 K_Y (1 + \|U\|) A_0 |r|^3 + \frac{8K_3 K_y c_b r^2 + K_3^2 K_y^2 r^4}{C_{\min}}.$$

Choose first η_1 as above, then c_b with $(4 + 16K_3 K_y) c_b / C_{\min} \leq \varepsilon_{\text{tr}}/4$, then t_1 so small that the terms of order $|r|^3, r^4$ (with $|r| \leq c_b t_1$) and $\Gamma_w(c) K_A |r|$ contribute at most $\frac{r^2}{2} \cdot \frac{\varepsilon_{\text{tr}}}{4}$, besides the conditions above. Since $\Gamma_w(c) \leq 2$, the bound is at most $1 + \frac{r^2}{2}(\Gamma_w(c) + \varepsilon_{\text{tr}})$. $\square$

Theorem 8.17 (Windowed averaging). *Let I be finite, $f \in S_{p^*}$ with F finite, $g \in \mathcal{C}(f)$, $\rho \in (0, 1)$ and $\eta_0 > 0$ with $\rho^2(1 + \eta_0) \leq (1 + \rho^2)/2$. Suppose there are $c_b \in (0, 1]$ and triples $(t^{(j)}, n_j, K_j)$ with $t^{(j)} \rightarrow 0$, $n_j \geq 24\rho^2 K_j / (c_b(1 - \rho^2))$ and $K_j t^{(j)} / n_j \rightarrow 0$, such that for every j and every $t \in \{t^{(j)} 2^{1-i} : 1 \leq i \leq n_j\}$ there is a balanced finite certificate c_t at f with*

- (i) $p^*(f + rg_{c_t}) \leq 1 + \frac{r^2}{2}(1 + \eta_0)$ for $|r| \leq c_b t$;
- (ii) $p^*(g - g_{c_t}) \leq K_j t$;
- (iii) $\Gamma_w(c_t) \leq 1 + \eta_0$.

Then $(f, \rho g) \in \text{clNA}((c_0, p), \ell_2^2)$. More precisely, the averaged certificates c below satisfy $\rho g_c \in \mathcal{C}(f)$, $\Gamma_w(\rho c) \leq 1$ and $\rho g_c \rightarrow \rho g$.

Proof. Fix $r_0 \in (0, 1]$ with $r_0^2 \leq 1 - \rho^2$, and j so large that $2\rho K_j t^{(j)} / n_j \leq (1 - \rho^2)r_0/3$. Write $n := n_j$, $K := K_j$, $t_i := t^{(j)} 2^{1-i}$ and $c := \frac{1}{n} \sum_i c_{t_i}$, a balanced finite certificate with $g_c = \frac{1}{n} \sum_i g_{c_{t_i}}$. For each i and real r : (A) if $\rho|r| \leq c_b t_i$, then

$p^*(f + r\rho g_{c_{t_i}}) \leq 1 + \frac{\rho^2 r^2}{2}(1 + \eta_0)$; (B) always $p^*(f + r\rho g_{c_{t_i}}) \leq p^*(f + r\rho g) + \rho|r|Kt_i \leq s(\rho r) + \rho|r|Kt_i$. By convexity, $p^*(f + r\rho g_c) \leq \frac{1}{n} \sum_i p^*(f + r\rho g_{c_{t_i}})$.

Case $|r| \leq r_0$. Let $I_r := \{i : c_b t_i < \rho|r|\}$; as the t_i are dyadic, $\sum_{I_r} t_i < 2\rho|r|/c_b$. Using (A) off I_r and (B) on I_r , $p^*(f + r\rho g_c) \leq \max\{1 + \frac{\rho^2 r^2}{2}(1 + \eta_0), s(\rho r)\} + Qr^2$ with $Q := 2\rho^2 K/(c_b n) \leq (1 - \rho^2)/12$. Now $1 + \frac{\rho^2 r^2}{2}(1 + \eta_0) + Qr^2 \leq 1 + \frac{r^2}{2}(1 - \frac{1 - \rho^2}{3}) \leq 1 + \frac{r^2}{2} - \frac{r^4}{8} \leq s(r)$ because $r^2 \leq 1 - \rho^2$, and $s(\rho r) + Qr^2 \leq s(r)$ by Lemma 3.6(a).

Case $|r| \geq r_0$. Using (B) for all i , $p^*(f + r\rho g_c) \leq s(\rho r) + 2\rho K t^{(j)}|r|/n \leq s(\rho r) + (1 - \rho^2)r_0|r|/3 \leq s(r)$, by Lemma 3.6(a) and $\min(r^2, |r|) \geq r_0|r|$.

Hence $\rho g_c \in \mathcal{C}(f)$. By convexity of Γ_w in the data, $\Gamma_w(\rho c) \leq \rho^2(1 + \eta_0) \leq 1$, and $p^*(\rho g_c - \rho g) \leq 2\rho K t^{(j)}/n \rightarrow 0$. Since $a \in c_{00}$, Theorem 5.8 gives $(f, \rho' \rho g_c) \in \text{cl NA}$ for all $\rho' < 1$, hence $(f, \rho g_c) \in \text{cl NA}$; let $j \rightarrow \infty$. $\square$

8.7. Recovery on $\mathcal{R}_0$ and the reduction to Lemma Z.

Theorem 8.18 (Recovery on $\mathcal{R}_0$). *Let T be the operator of Definition 8.1 and $N \geq 1$. Then $\mathcal{R}_0 \subseteq \mathcal{R}$: for every $f \in \mathcal{R}_0$ and every $g \in \mathcal{C}(f)$, $(f, g) \in \text{cl NA}((c_0, p_N), \ell_2^2)$. No condition on the blocks of f is needed (infinitely many strict non-peaks, near-threshold coordinates, weak or degenerate peaks, failure of (MS), several active blocks and infinite contact sets are allowed).*

Proof. Fix $f \in \mathcal{R}_0$ (constants γ, ϖ_0), $g \in \mathcal{C}(f)$ and $\rho \in (0, 1)$. Choose $\eta_0 \in (0, 2]$ with $\rho^2(1 + \eta_0) \leq (1 + \rho^2)/2$ (the bound $\eta_0 \leq 2$ is used below for (C-a)) and put

$$\kappa_0 := \frac{1}{2}(\sqrt{1 + \eta_0/2} - 1), \quad \varepsilon_{\text{tr}} := \eta_0/2, \quad A_0 := K_b + 1,$$

with K_b from Proposition 8.15; let c_b, t_1 be given by Lemma 8.16. Choose $\eta \leq \eta_*$ with $\eta_{\Gamma}(\eta) \leq 1$ and $\sqrt{1 + \eta_{\Gamma}(\eta)} \leq 1 + \kappa_0$, and put $t_* := \min(t_\eta, t_1, 1)$. For a window index l put $t^{(l)} := T_{\text{hi}}(l)$ and $K_l := (1 + \|U\|)K_2\Lambda_f(l)$. The scales $t_i = T_{\text{hi}}(l)2^{1-i}$, $1 \leq i \leq n_l^w$, lie in $\mathcal{W}(l)$. By (P3) and $\Lambda_f(l) \leq \varpi_0^{-l^2}\Lambda^\circ(l)$,

$$\Lambda_f(l)T_{\text{hi}}(l) \leq \varpi_0^{-l^2}2^{-l^3}/l \rightarrow 0 \quad (l \rightarrow \infty).$$

Hence for all large l and every $t \in \mathcal{W}(l)$: $t \leq t_*$, $K_1\Lambda_f(l)t \leq 1$ and $K_4\Lambda_f(l)t \leq \kappa_0$. Take a two-sided decomposition at scale t (Lemma 8.5) and its window certificate c_t . By Proposition 8.15, $\|b_t\|_1 \leq A_0$, $\sqrt{\Gamma_w(c_t)} \leq 1 + 2\kappa_0 = \sqrt{1 + \eta_0/2}$, i.e. $\Gamma_w(c_t) \leq 1 + \eta_0/2 \leq 2$, and (C-b) holds; by Lemma 8.16, $p^*(f + r g_{c_t}) \leq 1 + \frac{r^2}{2}(1 + \eta_0)$ for $|r| \leq c_b t$; and $p^*(g - g_{c_t}) \leq q^*(g - g_{c_t}) \leq (1 + \|U\|)\|g - g_{c_t}\|_1 \leq K_l t$. Finally, by (P3),

$$\frac{n_l^w}{K_l} \geq \frac{l2^{l^3}\varpi_0^{l^2}}{(1 + \|U\|)K_2} \rightarrow \infty, \quad \frac{K_l t^{(l)}}{n_l^w} \leq K_l T_{\text{hi}}(l) \leq (1 + \|U\|)K_2 \frac{\varpi_0^{-l^2}2^{-l^3}}{l} \rightarrow 0,$$

so the hypotheses of Theorem 8.17 hold along large l , and $(f, \rho g) \in \text{cl NA}$. As $\rho < 1$ was arbitrary, $(f, g) \in \text{cl NA}$. $\square$

The super-exponential factor 2^{l^3} of the design absorbs the f -dependent constant $\varpi_0^{-l^2}$ and every constant depending on f, g and ρ ; this is the only place where the order of quantifiers (T before f) matters.

Corollary 8.19 (No intrinsic defect on $\mathcal{R}_0$). *For the operator of Definition 8.1 and $f \in \mathcal{R}_0$, $\mathcal{C}(f)$ is the norm closure of the set of $g_c \in \mathcal{C}(f)$ with c a balanced finite certificate and $\Gamma_w(c) \leq 1$. In particular $\text{Def}(f) = \emptyset$.*

Proof. By the proof of Theorem 8.18 and Theorem 8.17, every ρg ($g \in \mathcal{C}(f)$, $\rho < 1$) is a limit of such g_c ; let $\rho \rightarrow 1$. These g_c lie in $\mathfrak{B}(f) \subseteq \mathcal{K}(f)$. $\square$

Theorem 8.20 (Reduction). *For the operator of Definition 8.1 and $N \geq 1$, the set $\text{NA}((c_0, p_N), \ell_2^2)$ is dense if and only if the following statement (Lemma Z) holds: for every $f \in S_{p_N}^*$, $g \in \mathcal{C}(f)$, $\rho < 1$ and $\varepsilon > 0$ there is $f' \in \mathcal{R}_0$ with $p_N^*(f' - f) < \varepsilon$ and $\text{dist}(\rho g, \mathcal{C}(f')) < \varepsilon$.*

Proof. If Lemma Z holds, take $f' \in \mathcal{R}_0$ and $g' \in \mathcal{C}(f')$ with $p^*(f' - f) < \varepsilon$ and $p^*(g' - \rho g) < \varepsilon$. By Theorem 8.18, $(f', g') \in \text{clNA}$, and $\|(f', g') - (f, \rho g)\| < 2\varepsilon$. So $(f, \rho g) \in \text{clNA}$ for all f, g, ρ , and density follows from Proposition 1.13(b). Conversely, if $\text{NA}((c_0, p_N), \ell_2^2)$ is dense, Lemma 1.16 gives norm-attaining $f_n \rightarrow f$ with $\rho g \in \text{Li}_n \mathcal{C}(f_n)$, and $f_n \in \mathcal{R}_0$ by Lemma 8.4(a). $\square$

By Lemma 1.5, Lemma Z for infinitely many N would give the density of $\text{NA}((c_0, p), \ell_2^2)$ for Martín's norm p built with the designed operator.

Problem 8.21 (Lemma Z). Let T be the operator of Definition 8.1 and $N \geq 1$. Is it true that for every $f \in S_{p_N}^*$, every $g \in \mathcal{C}(f)$, every $\rho < 1$ and every $\varepsilon > 0$ there is $f' \in \mathcal{R}_0$ (not necessarily norm attaining) with $p_N^*(f' - f) < \varepsilon$ and $\text{dist}_{p_N^*}(\rho g, \mathcal{C}(f')) < \varepsilon$?

Remark 8.22 (What is known about Lemma Z). (a) Lemma Z holds at every $f \in \mathcal{R}_0$ (take $f' = f$), and for every mate covered by Theorem 7.18 or Corollary 7.22 (the engineered approximants are norm attaining, hence in $\mathcal{R}_0$); in particular at every (BT) point.

(b) $f \notin \mathcal{R}_0$ iff F is infinite or, for all $\gamma \in (0, 1)$ and $\varpi_0 \in (0, 1]$, some signature set S_l , $l \in \mathcal{L}_N$, satisfies

$$\|h_l 1_{S_l \cap J_\gamma}\|_1 < \varpi_0^l \|h_l\|_1 :$$

the one-sided base resources (support, contacts, near-contacts) swallow the signature of some carrier, or of a fast sequence of carriers (*signature-resonant* first rows). If the signature of the carrier l_0 is swallowed (i.e. $\delta'_{l_0} = 0$ for every γ), then $\Delta\theta_{l_0}$ is not pinned, and every coarser carrier $l < l_0$ whose signature set is met by the target y_{l_0} becomes slaved to the free coefficient $\Delta\theta_{l_0}$ through the pinning inequality; so more than the swallowed carriers may lose their pinning.

(c) The approximants in Lemma Z need not attain their norms. Every pair (a', z') with $q^*(a') = 1$, $z' \in B_{\ell_\infty}$ and $z' = \text{sign } a'$ on $\text{supp } a'$ is the forced data of some $f' \in S_{p^*}$: put $e' := U^* a' / \|U^* a'\|$, $\hat{z}' := z' + U e'$, $q'_0 := (1 + \sum_m |R_m^{**} \hat{z}'|_m)^{-1}$, $\xi' := q'_0 \hat{z}'$ and $f' := a' + L^* J_V(L^{**} \xi')$. Indeed $q^{**}(\hat{z}') = a'(\hat{z}') = 1$, so $f'(\xi') = 1 = p^{**}(\xi')$ and $p^*(f') \leq 1$ by Lemma 1.6; Proposition 1.9 identifies the forced data. If $a' \rightarrow a$ in ℓ_1 and $z' \rightarrow z$ coordinatewise, then $f' \rightarrow f$ (as in the proof of Proposition 1.12). So one may lower $|z|$ on far parts of the swallowed signature sets; whether the resulting f' satisfy the mate condition of Lemma Z is exactly the open question.

(d) *Heuristic, not used.* If only finitely many carriers are unpinned, the arguments of this section suggest that every two-sided decomposition is a pinned part of size $O(Kt)$, plus finitely many free (and slaved) coefficients with their base counterparts, i.e. an exact two-piece structure up to $O(Kt)$ on whole windows; combining windowed averaging with the engineering of Section 7 is a natural route to Lemma Z in that case. Lowering $|z|$ on far parts of the swallowed sets makes the corresponding pinning constants tiny; above that scale f' copies the mate structure of f , below

it the windows of f' take over, and the transition band would have to be handled by the ρ -slack. None of this is proved.

(e) *Sketch, not used.* For F infinite with (SR), the proof of Theorem 8.18 appears to go through after clamping B_+ on F at $2|a_j|/t$ (the excess beyond the no-flip capacity is one-sided and bounded by $|\Delta B(j)|$ plus flip costs), truncating the clamped base part to a finite set, adapting Lemma 8.16 to $|b_j| \leq 2|a_j|/t$, and using a version of Theorem 5.8 for $a \notin c_{00}$; we have not written out the details.

9. STATUS OF THE DENSITY PROBLEM

9.1. What is proved. Let T be an arbitrary admissible operator. Density of $\text{NA}((c_0, p), \ell_2^2)$ is equivalent to $\mathcal{R} = S_{p^*}$ (Proposition 1.13), and by Lemma 1.5 it suffices to prove $\mathcal{R} = S_{p_N^*}$ for infinitely many N . A first row f belongs to $\mathcal{R}$ under each of the following sufficient conditions:

- (S1) f attains its norm (Proposition 1.13(d));
- (S2) f belongs to the residual set $\mathcal{O}$ of Hausdorff continuity points of the fibres (residual uniform recovery theorem of [7]);
- (S3) $\mathcal{C}(f) = \text{cl Cert}^{\text{sh}}(f)$; then every mate is recovered along every sequence $f_n \rightarrow f$ (Theorem 3.13); more generally $\mathcal{C}(f) = \mathcal{K}(f)$ when $a \in c_{00}$ (Proposition 6.1);
- (S4) I is finite and f satisfies (BT): finite base support, finitely many strict non-peaks, no degenerate peaks and (MS), with an *arbitrary* contact set (Corollary 7.22; this contains the conclusion $f \in \mathcal{R}$ of Theorem 5.13, though not its recovery along the canonical truncations, and covers the first row of Section 6);
- (S5) f is a limit of points of $\mathcal{R}$ along which the support functions $\tilde{r}$ are asymptotically at least those of f (Theorem 2.4);
- (S6) T is the signature-ladder operator of Definition 8.1, $I = \{1, \dots, N\}$, and $f \in \mathcal{R}_0$, i.e. F is finite and the signature sets keep room (Theorem 8.18); no condition on the blocks of f is needed, and $\mathcal{R}_0$ contains all norm-attaining and all base-tame first rows.

For an individual mate $g \in \mathcal{C}(f)$, $(f, g) \in \text{cl NA}$ whenever g lies in $\text{cl Cert}^{\text{sh}}(f)$; this covers finite and shifted certificates (Theorems 3.8, 3.13), weighted directions with $O(\sigma)$ box tails (Theorem 4.6), locally split mates and split-contractive operators when I is finite (Theorem 4.10, Corollary 4.11), every mate that is a certificate of radius $\gtrsim t$ up to an $O(t)$ remainder at all small scales (Corollary 4.2), and carrier mates with non-summable gaps (Theorem 4.12). When $a \in c_{00}$, also every balanced finite certificate with $\Gamma_w \leq 1$ is recovered (Theorem 5.8). When I and F are finite, every mate with two-piece data (one-sided linear decompositions, possibly through infinitely many contacts) with $\kappa_w \leq 1$ is recovered along engineered norm-attaining approximants if $\Delta d_m \geq 0$ in every block (Corollary 7.21), and if the blocks with $\Delta d_m < 0$ satisfy the scrambling condition (SC) (Theorem 7.18); no hypothesis on the number of active blocks, on the contact set or on rates of T is needed. At (BT) points the exact one-sided second-order coefficients are known (Theorem 7.10). For ℓ_2^{d+1} we only prove the joint recovery of tuples of finite certificates under the hypothesis of Theorem 3.15.

On the negative side, the intrinsic mechanisms of Sections 3–5 do not suffice in general: for some admissible T there is a non-attaining f at which the defect $\mathcal{C}(f) \setminus \mathcal{K}(f)$ spans an infinite-dimensional space (Theorem 6.6), and its explicit

elements are not recovered along the canonical truncations (Proposition 6.8). This is a defect of the intrinsic mechanisms only: that first row satisfies (BT), and every one of its mates is recovered along engineered norm-attaining approximants (Corollary 7.23). At every point where $a \in c_{00}$ and K is finite (in particular at every norm-attaining point) exact resonances do not exist (Proposition 6.10). Nothing proved here points to a counterexample; a counterexample would produce a two-dimensional Hilbert space without Lindenstrauss' property B, answering Question 6 of Johnson and Wolfe [2] negatively.

9.2. What remains. By Lemma 1.16 and Corollary 3.9(c), the density problem for p_N is equivalent to the following statement, in which only the defect matters.

Problem 9.1. Let $N \geq 1$, let T be admissible, and let $f \in S_{p_N}^*$. Is it true that for every $g \in \text{Def}(f)$ and every $\rho < 1$ there are norm-attaining $f_n \rightarrow f$ with $\rho g \in \text{Li}_n \mathcal{C}(f_n)$? Equivalently, is every pair $(f, \rho g)$ with $g \in \mathcal{C}(f) \setminus \text{cl Cert}^{\text{sh}}(f)$ a limit of norm-attaining operators into ℓ_2^2 ?

By Lemma 1.5, a positive answer for infinitely many N gives density for Martín's norm; a negative answer for one N and one admissible T gives a two-dimensional range without property B. By Proposition 6.8, the approximants in Problem 9.1 cannot in general be the canonical truncations.

For the designed operator. For the operator T of Definition 8.1 and every N , Problem 9.1 is equivalent to Lemma Z (Problem 8.21 and Theorem 8.20): every mate of every first row must be approximated by mates at first rows in $\mathcal{R}_0$, which need not attain their norms. What remains is base-side engineering at signature-resonant first rows (contacts or near-contacts swallowing signature sets, as at the first row of Section 6) and at first rows with infinite base support (Remark 8.22).

For an arbitrary admissible T . The following classes are open.

Problem 9.2 (Negative d -mismatch). Let I be finite, $f \in S_{p^*}$ with F finite, and let $g \in \mathcal{C}(f)$ carry two-piece data with $\kappa_w \leq 1$ such that the blocks with $\Delta d_m < 0$ do not satisfy (SC). Is $(f, g) \in \text{cl NA}$?

Theorem 7.18 settles $\Delta d \geq 0$ (for any number of active blocks and without steering) and $\Delta d < 0$ under (SC); Remark 7.24(a) gives a quantitative partial answer, and Remark 7.24(b) indicates that (SC) can fail for admissible T (sketch). The explicit switching mates of Theorem 6.6 are d -neutral and are recovered.

Problem 9.3 (Generic supports). Let F be finite and let some Q_m be infinite, or (MS) fail, or degenerate peaks exist. Is every $g \in \mathcal{C}(f)$ a limit of mates $g_n \in \mathcal{C}(f_n)$, $f_n \rightarrow f$, carrying two-piece data to which Theorem 7.18 applies?

At such f the lower bound of Theorem 7.10 is not available (the block expansion is multiscale), and mates need not have two-piece data; mixed mates combining deep weighted parts with switching parts are not covered either.

Problem 9.4 (Second-order defect). Is there f with $a \in c_{00}$ and a balanced finite certificate $g \in \mathcal{C}(f) \cap S(f)$ with $\Gamma_w(g) > 1$? By Proposition 5.12 such an f must have degenerate peaks, infinitely many strict non-peaks, failure of (MS), or infinitely many contacts; in the last case a finite-dimensional analogue suggests that the second-order coefficient can drop below Γ_w (Remark 5.14). At (BT) points the question does not affect recoverability: Corollary 7.22 recovers every mate, and the relevant coefficients are the one-sided $\gamma^\pm \leq \Gamma_w$ (Theorem 7.10).

Problem 9.5 (Approximate resonances). Do switching mates carried, on both sides of $t = 0$, by near-threshold strict non-peaks with summable gaps or by weak peaks belong to $\text{cl Cert}(f)$, or at least to the closure of the mates recovered in Section 7? There is no linear obstruction for them (Remark 4.13). One way to prove non-membership in $\text{cl Cert}(f)$ is to exhibit a sequence $f_n \rightarrow f$ with $g \notin \text{Li}_n \mathcal{C}(f_n)$ (Corollary 3.9).

Such scale-dependent switching is the main remaining mechanism for an arbitrary admissible T . One can design operators T whose near-threshold peak carriers are one-sided with bounded conversion capacity (we do not include this construction). For them no recovery argument is known: the available arguments leave a Hilbert (room) capacity requirement on the converted carriers that is independent of ρ and of the error terms. No counterexample is known either.

Problem 9.6 (Infinite base support). Extend Theorems 7.10, 7.18 and 8.18 to first rows with $a \notin c_{00}$. All results of Sections 7 and 8 use that F is finite (no flips on F below a fixed scale).

Problem 9.7 (Universality of the defect). Does *every* admissible T (in particular every T produced by the proof of Martín’s Lemma B) admit a first row with nonempty defect? The linear obstruction requires a first row with few strict non-peaks, which an adversarial T may prevent; for the designed operator of Section 8 the defect is empty at every first row in $\mathcal{R}_0$ (Corollary 8.19).

Finally we note a gap between necessary and sufficient conditions in the reduction itself. If $g_1, \dots, g_d \in \mathcal{C}(f)$, then $(g_1, \dots, g_d)/\sqrt{d} \in \mathcal{C}_d(f)$, and density into ℓ_2^{d+1} implies that such tuples are recovered along a common norm-attaining sequence (by the argument of Lemma 1.16, with rotations of ℓ_2^{d+1}). But we do not know whether density into ℓ_2^2 alone forces the existence, for each f , of one norm-attaining sequence along which the whole fibre $\mathcal{C}(f)$ is recovered (condition (a) of Corollary 2.3); only the sufficiency of that condition is used.

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